Co-ordination Compounds

Question1

The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.

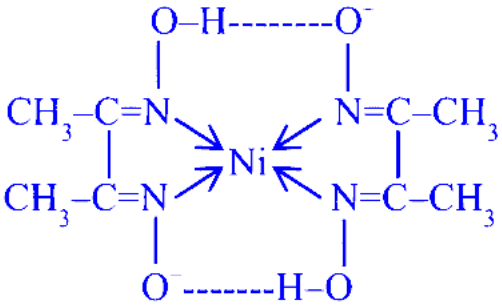
JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 14

Solution:

Ni^{2+} with (dmg) forms $[Ni(dmg)_2]^{2+}$ having 2 H -Bonds as shown :

The no. of H atoms = 14



Question2

Total number of molecules/species from following which will be paramagnetic is _____.

$O_2, O_2^+, O_2^-, NO, NO_2, CO, K_2[NiCl_4], [Co(NH_3)_6]Cl_3, K_2[Ni(CN)_4]$

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 6

Solution:

To determine which molecules or species are paramagnetic, we need to identify those with unpaired electrons. Here's the analysis:

O_2: This molecule has 2 unpaired electrons as predicted by the Molecular Orbital Theory (MOT), making it paramagnetic.

O_2^+: This ion has 1 unpaired electron according to MOT, which also makes it paramagnetic.

O_2^-: This ion contains 1 unpaired electron as determined by MOT, indicating it is paramagnetic.

NO: As a species with an odd number of electrons, NO has unpaired electrons, making it paramagnetic.

NO_2: This molecule is also an odd-electron species, which means it has unpaired electrons and is paramagnetic.

$K_2[NiCl_4]$: In this compound, Ni^{2+} (nickel with a +2 oxidation state) has an electron configuration of $3d^8$. Given the presence of weak field ligands and a coordination number of 4, this forms a tetrahedral complex. As a result, it is paramagnetic with 2 unpaired electrons.

Thus, the paramagnetic species in the provided list are O_2, O_2^+, O_2^-, NO, NO_2, and $K_2[NiCl_4]$.

Question3

The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among $\mathrm{Mn_2O_3}, \mathrm{TiO}$ and VO is _____ B.M. (Nearest integer).

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 5

Solution:

The compound with the strongest oxidizing power among Mn_2O_3, TiO, and VO is Mn_2O_3. This conclusion is based on the standard electrode potential (E°) value:

$E^\circ_{Mn^{+3}/Mn^{+2}} = +1.57\,\text{V}$

For Mn^{3+}, the electronic configuration is d^4. The spin-only magnetic moment (μ) is calculated as follows:

$\mu = \sqrt{n(n+2)}\,\text{BM}$

Where n is the number of unpaired electrons. Here, $n = 4$:

$\mu = \sqrt{4(4+2)} = \sqrt{24}\,\text{BM}$

This calculates to approximately 4.89 BM, which rounds to an integer value of 5 BM.

Question4

Consider the following low-spin complexes

$$\mathrm{K_3\,[Co(NO_2)_6]}, \mathrm{K_4\,[Fe(CN)_6]}, \mathrm{K_3\,[Fe(CN)_6]}, \mathrm{Cu_2\,[Fe(CN)_6]} \text{ and } \mathrm{Zn_2\,[Fe(CN)_6]}$$

The sum of the spin-only magnetic moment values of complexes having yellow colour is _____ B.M. (answer in nearest integer)

JEE Main 2025 (Online) 29th January Evening Shift

Answer: 0

Solution:

Given low-spin complexes:

$K_3\,[Co(NO_2)_6]$ – Yellow colour

$K_4\,[Fe(CN)_6]$ – light yellow colour

$K_3\,[Fe(CN)_6]$ – bright red colour not yellow.

$Cu_2\,[Fe(CN)_6]$ – reddish brown color not yellow

$Zn_2\,[Fe(NO_2)_6]$ – bluish-white colour not yellow

$K_3\,[Co(NO_2)_6]$

Oxidation state of Co : $+3$

Configuration:

$Co\,d^7\,s^2$

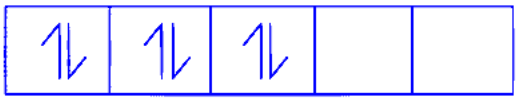
$Co^{3+}\,d^6$

$3(+1) + x + 6(-1) = 0$

$3 + x - 6 = 0$

$x = 6 - 3 = +3$

For low-spin d^6 configuration, no unpaired electrons are present :



spin-only magnetic moment, $\mu = \sqrt{n(n+2)}$

$n = 0$, $\mu = 0$

This complex shows yellow colour (bright yellow)

$K_4\,[Fe(CN)_6]$

Oxidation state of Fe : $+2$

$4(+1) + x + 6(-1) = 0$

$4 + x - 6 = 0$

$x = 6 - 4 = +2$

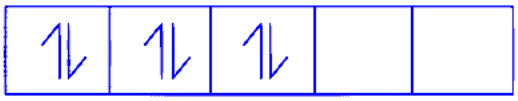
Configuration:

$Fe\,d^6\,s^2$

$Fe^{+2}\,d^6$

$\mu = 0$

For low-spin d^6 configuration, no unpaired electrons are present :



This complex shows yellow colour (light yellow)

$K_3\,[Fe(CN)_6]$

Oxidation state of Fe : $+3$

$3(+1) + x + 6(-1) = 0$

$3 + x - 6 = 0$

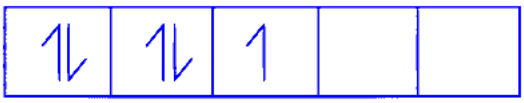
$x = 6 - 3 = +3$

Configuration:

$Fe\,d^6\,s^2$

$Fe^{+2}\,d^5$

low spin d^5



One unpaired electron is present.

Spin-only magnetic moment, $\mu = \sqrt{n(n+2)}$

n (number of unpaired electrons) = 1

So, $\mu = \sqrt{n(n+2)}$

$= \sqrt{1 \times 3}$

$= \sqrt{3}$

= 1.732 BM

Colour of the complex is not yellow.

$Cu_2\,[Fe(CN)_6]$, Cu^{2+} and $[Fe(CN)_6]^{4-}$

$x + 6(-1) = -4$

$x = -4 + 6$

$= +2$

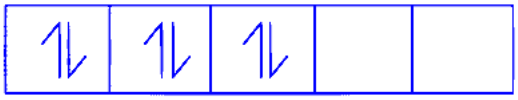
Oxidation state of Fe : $+2$

Configuration:

$Fe\,d^6\,s^2$

$Fe^{2+}\,d^6$

For low-spin d^6 configuration, no unpaired electrons are present



$n = 0, \mu = \sqrt{n(n+2)} = 0$

Colour of the complex is not yellow.

$Zn_2\,[Fe(CN)_6]$

$[Fe(CN)_6]^{4-}$ and Zn^{2+}

$x + 6(-1) = -4$

$x = -4 + 6 = +2$

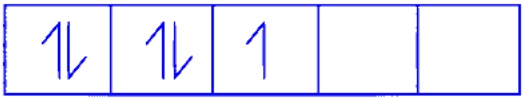
Oxidation state of Fe : $+2$

Configuration:

$Fe\,d^6\,s^2$

$Fe^{2+}\,d^6$

For low-spin d^6 configuration, no unpaired electrons are present.



$n = 0,\ \mu = \sqrt{n(n+2)} = 0$

Colour of the complex is not yellow

The complexes having yellow colour are $K_3\,[Co(NO_2)_6]$ and $K_4\,[Fe(CN)_6]$

The sum of the spin-only magnetic moment values of complexes having yellow colour is 0 (zero).

Question5

A transition metal (M) among $\mathrm{Mn, Cr, Co}$ and Fe has the highest standard electrode potential $\left(\mathrm{M^{3+}/M^{2+}}\right)$. It forms a metal complex of the type $\mathrm{[M(CN)_6]^{4-}}$. The number of electrons present in the $\mathrm{e_g}$ orbital of the complex is _____.

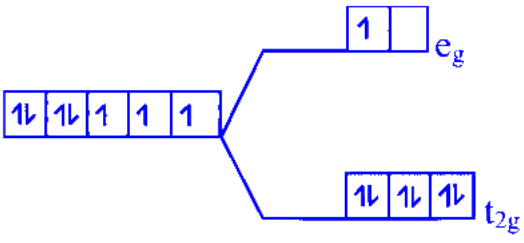
JEE Main 2025 (Online) 2nd April Morning Shift

Answer: 1

Solution:

Co has highest standard electrode potential $\left(\mathrm{M^{+3}/M^{+2}}\right)$ among Mn, Cr, Co, Fe

$\therefore$ Complex is $\mathrm{[Co(CN)_6]^4}$ and its splitting is as follows.



$\therefore$ electron in $\mathrm{e_g}$ orbital is one.

Question6

The spin-only magnetic moment value of M^{n+} ion formed among Ni, Zn, Mn and Cu that has the least enthalpy of atomisation is_____ . (in nearest integer) Here n is equal to the number of diamagnetic complexes among $K_2\,[NiCl_4]$, $[Zn(H_2O)_6]Cl_2$, $K_3\,[Mn(CN)_6]$ and $[Cu(PPh_3)_3I]$

JEE Main 2025 (Online) 2nd April Evening Shift

Answer: 0

Solution:

$K_2\,[NiCl_4] \Rightarrow sp^3$, Paramagnetic

$[Zn(H_2O)_6]Cl_2 \Rightarrow sp^3\,d^2$, Diamagnetic

$K_3\,[Mn(CN)_6] \Rightarrow d^2sp^3$, Paramagnetic

$[Cu(PPh_3)_3I] \Rightarrow sp^3$, Diamagnetic

Hence the value of n is 2

Least value of enthalpy of atomisation among Ni,

Zn, Mn and Cu is of Zn

$Zn^{+2} : -[Ar]3\,d^{10}$
$\mu = 0$

Question7

The number of optical isomers exhibited by the iron complex (A) obtained from the following reaction is_____.

$$FeCl_3 + KOH + H_2C_2O_4 \rightarrow \ A$$

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 2

Solution:

$$FeCl_3 + KOH + H_2SO_4 \rightarrow \underset{(A)}{K_3[Fe(C_2O_4)_3]}$$

$\Rightarrow [Fe(C_2O_4)_3]^{3-}$ is $[M(AA)_3]$ type complex.

So total optical isomers $= 2$

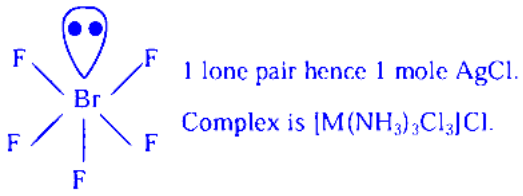
Question8

A metal complex with a formula $MCl_4 \cdot 3NH_3$ is involved in $sp^3\,d^2$ hybridisation. It upon reaction with excess of $AgNO_3$ solution gives ' x ' moles of AgCl . Consider ' x ' is equal to the number of lone pairs of electron present in central atom of BrF_5. Then the number of geometrical isomers exhibited by the complex is _____.

JEE Main 2025 (Online) 4th April Evening Shift

Answer: 2

Solution:



It shows 2 geometrical isomers (Ma_3b_3 type)

facial (fac) & meridional (Mer)

Question9

The number of paramagnetic complexes among $[FeF_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Mn(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}$, and $[CoF_6]^{3-}$, which involved d^2sp^3 hybridization is _____.

JEE Main 2025 (Online) 7th April Morning Shift

Answer: 2

Solution:

$[FeF_6]^{3-}$	Paramagnetic	sp^3d^2
$[Fe(CN)_6]^{3-}$	Paramagnetic	d^2sp^3
$[Mn(CN)_6]^{3-}$	Paramagnetic	d^2sp^3
$[Co(C_2O_4)_3]^{3-}$	Diamagnetic	d^2sp^3
$[MnCl_6]^{3-}$	Paramagnetic	sp^3d^2
$[CoF_6]^{3-}$	Paramagnetic	sp^3d^2

Only $[Fe(CN)_6]^{3-}$ and $[Mn(CN)_6]^{3-}$ are paramagnetic and d^2sp^3 hybridisation of metal.

Question10

The number of paramagnetic metal complex species among $[Co(NH_3)_6]^{3+}$, $[Co(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}$, $[Mn(CN)_6]^{3-}$, $[CoF_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ with same number of unpaired electrons is _____.

JEE Main 2025 (Online) 7th April Evening Shift

Answer: 2

Solution:

To determine which metal complexes are paramagnetic and have the same number of unpaired electrons, let's analyze each one:

$[Co(NH_3)_6]^{3+}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $\mathrm{t}_{2g}^{2,2,2}\ \mathrm{e}_g^{0,0}$

Magnetic property: Diamagnetic (unpaired electrons = 0)

$[\mathrm{Co}(\mathrm{C}_2\mathrm{O}_4)_3]^{3-}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $\mathrm{t}_{2g}^{2,2,2}\ \mathrm{e}_g^{0,0}$

Magnetic property: Diamagnetic (unpaired electrons = 0)

$[\mathrm{MnCl}_6]^{3-}$:

Manganese ion: Mn^{3+}

Electron configuration: $3d^4$

Orbital occupancy: $\mathrm{t}_{2g}^{1,1,1}\ \mathrm{e}_g^{1,0}$

Magnetic property: Paramagnetic (unpaired electrons = 4)

$[\mathrm{Mn}(\mathrm{CN})_6]^{3-}$:

Manganese ion: Mn^{3+}

Electron configuration: $3d^4$

Orbital occupancy: $\mathrm{t}_{2g}^{2,1,1}\ \mathrm{e}_g^{0,0}$

Magnetic property: Paramagnetic (unpaired electrons = 2)

$[\mathrm{CoF}_6]^{3-}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $\mathrm{t}_{2g}^{2,1,1}\ \mathrm{e}_g^{1,1}$

Magnetic property: Paramagnetic (unpaired electrons = 4)

$[\mathrm{Fe}(\mathrm{CN})_6]^{3-}$:

Iron ion: Fe^{3+}

Electron configuration: $3d^5$

Orbital occupancy: $\mathrm{t}_{2g}^{2,2,1}\ \mathrm{e}_g^{0,0}$

Magnetic property: Paramagnetic (unpaired electrons = 1)

$[\mathrm{FeF}_6]^{3-}$:

Iron ion: Fe^{3+}

Electron configuration: $3d^5$

Orbital occupancy: $\mathrm{t}_{2g}^{1,1,1}\ \mathrm{e}_g^{1,1}$

Magnetic property: Paramagnetic (unpaired electrons = 5)

From this analysis, the paramagnetic complexes having the same number of unpaired electrons (4 unpaired electrons) are:

$[\mathrm{MnCl}_6]^{3-}$

$[\mathrm{CoF}_6]^{3-}$

These two complexes show the same paramagnetic property, each with 4 unpaired electrons.

Question11

In which of the following complexes the CFSE, Δ_o will be equal to zero?

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $K_3\,[Fe(SCN)_6]$

B. $[Fe(en)_3]Cl_3$

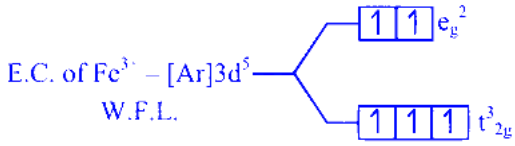
C. $[Fe(NH_3)_6]Br_2$

D. $K_4\,[Fe(CN)_6]$

Answer: A

Solution:

For complex $K_3\,[Fe(SCN)_6]$



Calculation of CFSE

$= (-0.4 \times 3 + 0.6 \times 2)\Delta_0$

$= 0\Delta_0$

Question12

From the magnetic behaviour of $[NiCl_4]^{2-}$ (paramagnetic) and $[Ni(CO)_4]$ (diamagnetic), choose the correct geometry and oxidation state.

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $[NiCl_4]^{2-}$: Ni(0), tetrahedral$[Ni(CO)_4]$: Ni(0), square planar

B. $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral $[Ni(CO)_4]$: Ni(0), tetrahedral

C. $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral $[Ni(CO)_4]$: Ni^{II}, square planar

D. $[NiCl_4]^{2-}$: Ni^{II}, square planar $[Ni(CO)_4]$: Ni(0), square planar

Answer: B

Solution:

The magnetic behavior of complexes can provide insights into their geometry and oxidation states. Given two complexes, $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$, we need to analyze the magnetic properties to deduce these characteristics.

$[NiCl_4]^{2-}$

Oxidation State: In $[NiCl_4]^{2-}$, nickel is in the +2 oxidation state (Ni^{2+}). The electron configuration for Ni^{2+} is $[Ar]\,3d^8\,4s^0$.

Geometry and Hybridization: The complex shows paramagnetic properties, indicating the presence of unpaired electrons. The hybridization in this case is sp^3, which corresponds to a tetrahedral geometry. Thus, there are two unpaired electrons leading to its paramagnetism.

$[Ni(CO)_4]$

Oxidation State: In this complex, nickel is in the zero oxidation state (Ni(0)). The electron configuration is $[Ar]\,3d^{10}\,4s^0$ after rearrangement.

Geometry and Hybridization: The compound is diamagnetic, indicating all electrons are paired. Its hybridization is also sp^3, meaning the geometry is tetrahedral, resulting in no unpaired electrons and confirming its diamagnetic nature.

In summary, $[NiCl_4]^{2-}$ is a tetrahedral complex with nickel in a +2 oxidation state and is paramagnetic. In contrast, $[Ni(CO)_4]$ is a tetrahedral complex with nickel in a 0 oxidation state and is diamagnetic.

Question13

Identify the homoleptic complex(es) that is/are low spin.

(A) $[Fe(CN)_5NO]^{2-}$

(B) $[CoF_6]^{3-}$

(C) $[Fe(CN)_6]^{4-}$

(D) $[Co(NH_3)_6]^{3+}$

(E) $[Cr(H_2O)_6]^{2+}$

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. (C) and (D) only

B. (C) only

C. (A) and (C) only

D. (B) and (E) only

Answer: A

Solution:

(A) $[Fe(CN)_5NO]^{2-}$: This is a heteroleptic complex with iron in the Fe^{2+} oxidation state, resulting in a $3d^6$ electronic configuration. It fo spin complex with electron configuration $t_{2g}^6e_g^0$ due to the strong field ligands (SFL).

(B) $[CoF_6]^{3-}$: This homoleptic complex contains cobalt in the Co^{3+} oxidation state, leading to a $3d^6$ electron configuration. It is a high complex with an sp^3d^2 hybridization due to weak field ligands (WFL).

(C) $[Fe(CN)_6]^{4-}$: This homoleptic complex has iron in the Fe^{2+} state, giving a $3d^6$ configuration. It forms a low spin complex with hybr d^2sp^3 and electron configuration $t_{2g}^6e_g^0$, due to the strong field nature of the cyanide ligands.

(D) $[Co(NH_3)_6]^{3+}$: This homoleptic complex contains cobalt in the Co^{3+} state, resulting in a $3d^6$ electron configuration. It forms a low complex with hybridization d^2sp^3 and configuration $t_{2g}^6e_g^0$ due to the strong field ligands.

(E) $[Cr(H_2O)_6]^{2+}$: This homoleptic complex with chromium in the Cr^{2+} state results in a $3d^4$ electronic configuration. It is a high spin with hybridization d^2sp^3 and configuration $t_{2g}^3e_g^1$ due to weak field ligands.

In summary, the homoleptic complexes that are low spin are:

(C) $[Fe(CN)_6]^{4-}$

(D) $[Co(NH_3)_6]^{3+}$

Question14

The correct order of the following complexes in terms of their crystal field stabilization energies is :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$

B. $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$

C. $[Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_4]^{2+} < [Co(en)_3]^{3+}$

D. $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_4]^{2+}$

Answer: A

Solution:

The crystal field stabilization energy (CFSE) of a complex depends on two main factors:

Charge/Oxidation State of the Central Metal Atom: Higher oxidation states generally lead to greater crystal field splitting energy (Δ).

Field Strength of the Ligand: Stronger field ligands also result in greater splitting. Additionally, chelating ligands typically increase the field strength.

For octahedral complexes, the CFSE is calculated as:

CFSE $= [-0.4\, t_{2g} + 0.6\, e_g]\, \Delta_\circ$

Considering the complexes:

$[Co(en)_3]^{3+}$:

Co exists as Co^{3+} with electronic configuration $t_{2g}^6 e_g^0$.

CFSE = $-2.4\, (\Delta_0)_1$.

$[Co(NH_3)_6]^{3+}$:

Co also as Co^{3+} with $t_{2g}^6 e_g^0$.

CFSE = $-2.4\, (\Delta_0)_2$.

$[Co(NH_3)_6]^{2+}$:

Co with Co^{2+} configuration $t_{2g}^5 e_g^2$.

CFSE = $-0.8\, (\Delta_0)_3$.

$[Co(NH_3)_4]^{2+}$:

Co as Co^{2+} with different configuration $e^4 t_2^3$.

CFSE = $-1.2\, \Delta_t$, where $\Delta_t = \frac{4}{9}(\Delta_0)_3$.

The order of crystal field splitting energies, based on our criteria, is:

$\Delta_t < (\Delta_0)_3 < (\Delta_0)_2 < (\Delta_0)_1$

Therefore, the CFSEs of these complexes compare as follows:

$[Co(en)_3]^{3+}$ has the highest CFSE due to the Δ_0 of the strongest field ligand and oxidation state.

$[Co(NH_3)_6]^{3+}$ follows with similarly high CFSE due to Co^{3+}.

$[Co(NH_3)_6]^{2+}$ has a moderate CFSE.

$[Co(NH_3)_4]^{2+}$ has the lowest CFSE with Δ_t.

This hierarchy of CFSE is determined by both the oxidation state of the cobalt metal and the field strength of the ligands involved.

Question15

The complex that shows Facial - Meridional isomerism is :

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $[Co(en)_2Cl_2]^+$

B. $[Co(NH_3)_3Cl_3]$

C. $[Co(NH_3)_4Cl_2]^+$

D. $[Co(en)_3]^{3+}$

Answer: B

Solution:

$Ma_3\ b_3$ type complexes show Facial - Meridional isomerism

(i) $[Co(NH_3)_3Cl_3] \Rightarrow Ma_3\ b_3$

(ii) $[Co(NH_3)_4Cl_2]^+ \Rightarrow Ma_4\ b_2$

(iii) $[Co(en)_3]^{3+} \quad \Rightarrow M(AA)_3$

(iv) $[Co(en)_2Cl_2]^+ \Rightarrow M(AA)_2\ b_2$

$a, b, = NH_3, Cl^-$

$AA = en$

Question16

The d-electronic configuration of an octahedral Co (II) complex having magnetic moment of 3.95 BM is:

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $e^4t_2^3$

B. $t_{2g}^6e_g^1$

C. $t_{2g}^5e_g^2$

D. $t_{2g^3}^3e_g^0$

Answer: C

Solution:

$$Co^{+2} = (Ar)_{18}\ 3d^7\ 4s^0$$

$$\underline{1}\quad \underline{1}\quad e_g^2$$

$$\underline{\uparrow\downarrow}\quad \underline{\uparrow\downarrow}\quad \underline{1}\ t_{2g}^5$$

Question17

$CrCl_3 \cdot xNH_3$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of $0.558^\circ C$. Assuming 100% ionisation of this complex and coordination number of Cr is 6 , the complex will be (Given $K_f = 1.86\ K\ kg\ mol^{-1}$)

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $[Cr(NH_3)_5Cl]Cl_2$

B. $[Cr(NH_3)_4Cl_2]Cl$

C. $[Cr(NH_3)_6]Cl_3$

D. $[Cr(NH_3)_3Cl_3]$

Answer: A

Solution:

Given the problem, we have the following information:

Depression in freezing point, $\Delta T_f = 0.558^\circ C$

Freezing point depression constant, $k_f = 1.86\,K\,kg\,mol^{-1}$

Molal concentration of the solution, $0.1\,mol\,kg^{-1}$

Coordination number of Cr is 6.

Using the formula for freezing point depression:

$\Delta T_f = i \times k_f \times m$

where i is the van 't Hoff factor, k_f is the freezing point depression constant, and m is the molality of the solution.

Substituting the known values:

$0.558 = i \times 1.86 \times 0.1$

Solving for i:

$i = \frac{0.558}{1.86 \times 0.1} = 3$

The van 't Hoff factor $i = 3$ indicates that the complex ionizes into three particles in solution. Given the coordination number of Cr is 6, we analyze the possible complexes:

$[Cr(NH_3)_5Cl]Cl_2$

$[Cr(NH_3)_4Cl_2]Cl$

$[Cr(NH_3)_6]Cl_3$

$[Cr(NH_3)_3Cl_3]$

The complex that gives three ions upon dissociation could be $[Cr(NH_3)_5Cl]Cl_2$. This is because the complex will dissociate as:

$[Cr(NH_3)_5Cl]^{2+} + 2\,Cl^-$

This results in three ions total, matching the van 't Hoff factor $i = 3$.

Therefore, the complex is $[Cr(NH_3)_5Cl]Cl_2$.

Question18

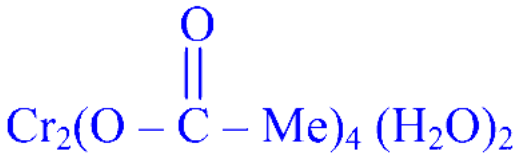
Identify the coordination complexes in which the central metal ion has d^4 configuration.

(A) $[FeO_4]^{2-}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$

(D)



(E) $[NiF_6]^{2-}$

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. (B), (C) and (D) only

B. (C) and (E) only

C. (A), (B) and (E) only

D. (B) and (D) only

Answer: D

Solution:

$Fe^{+6} = [Ar]3\,d^2$
$Mn^{+3} = [Ar]3\,d^4$
$Fe^{+3} = [Ar]3\,d^5$
$Cr^{+2} = [Ar]3\,d^4$
$Ni^{+4} = [Ar]3\,d^6$

Question19

One mole of the octahedral complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structure of the complex is:

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. $[Co(NH_3)_5Cl]Cl_2$

B. $[Co(NH_3)_4Cl] \cdot Cl_2 \cdot NH_3$

C. $[Co(NH_3)_3Cl_3] \cdot 2NH_3$

D. $[Co(NH_3)_4Cl_2] \cdot Cl \cdot NH_3$

Answer: A

Solution:

$$[Co(NH_3)_5Cl]Cl_2 \rightarrow \underbrace{[Co(NH_3)_5Cl]^{2+}(\text{ aq. }) + 2Cl^{-}(\text{aq.})}_{\text{3 ions in water}}$$

$$[Co(NH_3)_5Cl]Cl_2\,(\text{aq.})\,) + 2AgNO_3\,(\text{aq.}) \rightarrow [Co(NH_3)_5Cl](NO_3)_2(\text{ aq. }) + 2AgCl(\text{ s})$$

Question20

When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be:

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. Pale Blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet

B. Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green

C. Green $\rightarrow$ Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet

D. Violet $\rightarrow$ Blue $\rightarrow$ Pale Blue $\rightarrow$ Green

Answer: C

Solution:

$$[Ni(H_2O)_6]^{+2}_{(aq)} + en_{(aq)} \rightarrow [Ni(H_2O)_4(en)]^{+2}_{(aq)} + 2H_2O$$

Green

$$[Ni(H_2O)_4(en)]^{+2}_{(aq)} + en_{(aq)} \rightarrow [Ni(H_2O)_2(en)_2]^{+2}_{(aq)} + 2H_2O$$

Pale Blue

$$[Ni(H_2O)_2(en)]^{+2}_{(aq)} + en_{(aq)} \rightarrow [Ni(en)_3]^{+2}_{(aq)} + 2H_2O$$

Blue / purple

Question21

The conditions and consequence that favours the $t_{2g}^3 e_g^1$ configuration in a metal complex are :

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. strong field ligand, low spin complex

B. strong field ligand, high spin complex

C. weak field ligand, high spin complex

D. weak field ligand, low spin complex

Answer: C

Solution:

For $3d^4$

If ligand is SFL : $t_{2g}^4 e_g^0$ (Low spin)

If ligand is WFL : $t_{2g}^3 e_g^1$ (High spin)

Question22

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation of central metal ion)
(A) $[CoF_6]^{3-}$	(I) d^2sp^3
(B) $[NiCl_4]^{2-}$	(II) sp^3
(C) $[Co(NH_3)_6]^{3+}$	(III) sp^3d^2
(D) $[Ni(CN)_4]^{2-}$	(IV) dsp^2

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

B.

(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

C.

(A)-(I), (B)-(IV), (C)-(III), (D)-(II)

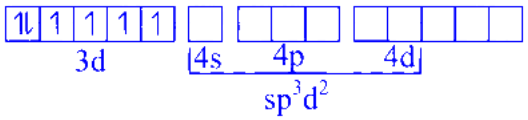
D.

(A)-(III), (B)-(II), (C)-(I), (D)-(IV)

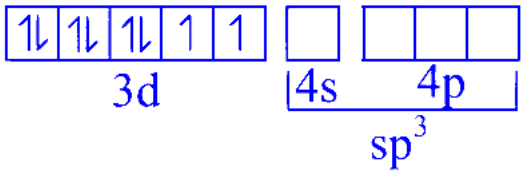
Answer: D

Solution:

(A) $[CoF_6]^{-3}$
$Co^{3+} \rightarrow 3\,d^6$

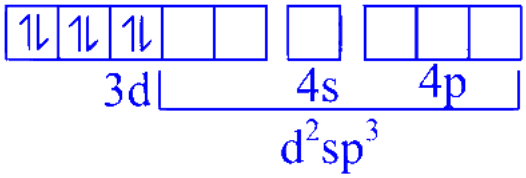


(B) $[NiCl_4]^{2-}$
$Ni^{2+} \rightarrow 3\,d^8$



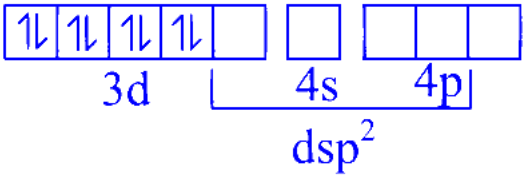
(C) $[Co(NH_3)_6]^{3+}$

$Co^{3+} \rightarrow 3\ d^6$



(D) $[Ni(CN)_4]^{2-}$

$Ni^{2+} \rightarrow 3\ d^8$



Question23

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation & Magnetic characters)
(A) $[MnBr_4]^{2-}$	(I) d^2sp^3 & diamagnetic
(B) $[FeF_6]^{3-}$	(II) sp^3d^2 & paramagnetic
(C) $[Co(C_2O_4)_3]^{3-}$	(III) sp^3 & diamagnetic
(D) $[Ni(CO)_4]$	(IV) sp^3 & paramagnetic

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

(A)-(III), (B)-(I), (C)-(I), (D)-(IV)

B.

(A)-(III), (B)-(II), (C)-(III), (D)-(IV)

C.

(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

D.

(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

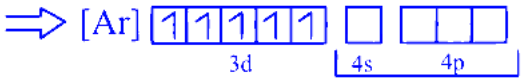
Answer: C

Solution:

(A) $[MnBr_4]^{2-}$

$Mn^{+2} \Rightarrow [Ar]3\,d^5$

In presence of ligand field

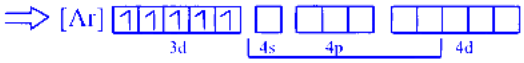


$\Rightarrow sp^3$ hybridization, paramagnetic in nature

(B) $[FeF_6]^{3-}$

$Fe^{+3} \Rightarrow [Ar]3\,d^5$

In presence of ligand field



$\Rightarrow sp^3\,d^2$ hybridization, paramagnetic in nature

(C) $[Co(C_2O_4)_3]^{3-}$

$Co^{+3} \Rightarrow [Ar]3\,d^6$

In presence of ligand field



$\Rightarrow d^2sp^3$ hybridization, diamagnetic in nature

(D) $[Ni(CO)_4]$

$Ni^0 \Rightarrow [Ar]3\,d^8 4\,s^2$

In presence of ligand field



$\Rightarrow sp^3$ hybridization, diamagnetic in nature

Question24

The correct increasing order of stability of the complexes based on $\triangle_0$ value is :

I. $[Mn(CN)_6]^{3-}$

II. $[Co(CN)_6]^{4-}$

III. $[Fe(CN)_6]^{4-}$

IV. $[Fe(CN)_6]^{3-}$

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

I < II < IV < III

B.

IV < III < II < I

C.

III < II < IV < I

D.

II < III < I < IV

Answer: A

Solution:

Δ_0 value of the octahedral complexes can be calculated using the formula

CASE $= (-0,4\Delta_0) \times$ number of electrons in f_{2g} orbitals + $(+0.6\Delta_0) \times$ number of electrons in eg orbitals.

The ligand CN^- is a strong field ligand and all the complexes form low-spin complex (pairing occurs).

I) $[M_N(CN)_6]^{3-}$

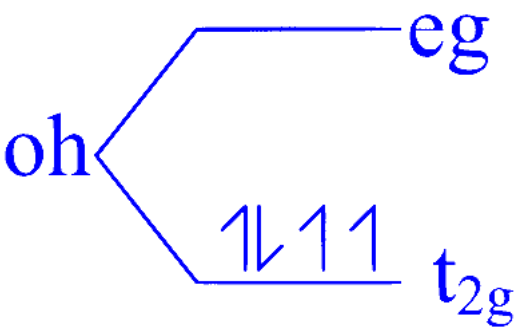
Oxidation state of M_n : $+3$

Mn configuration $\rightarrow d^5s^2$

Mn^{+3} configuration $\rightarrow d^4$

$$x + 6(-1) = -3$$
$$x = -3 + b = +3$$

$$CFSE = -0.4\Delta_0 \times 4 + (0.6\Delta_0) \times 0$$
$$= -1.6\Delta_0$$



II) $[Co(CN)_6]^{4-}$

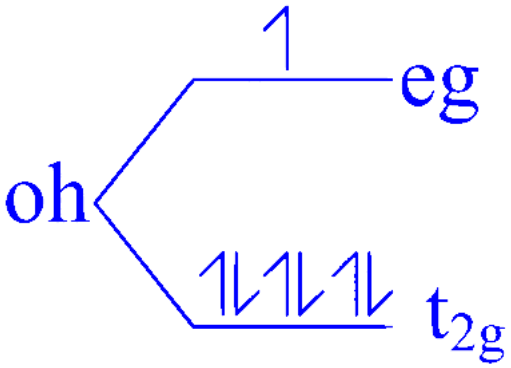
Oxidation state of C_0 : $+2$

C_0 configuration $\rightarrow d^7s^2$

Co^{+2} configuration $\rightarrow d^7$

$$CFSE = -0.4\Delta_0 \times 6 + (0.6\Delta_0) \times 1$$
$$= -2.4\Delta_0 + 0.6\Delta_0 = -1.8\Delta_0$$

$$x + 6(-1) = -4$$
$$x = -4 + 6 = +2$$



$$CFSE = -0.4\Delta_0 \times 6 + (0.6\Delta_{10}) \times 1$$
$$= -2.4\Delta_0 + 0.6\Delta_0 = -1.8\Delta_0$$

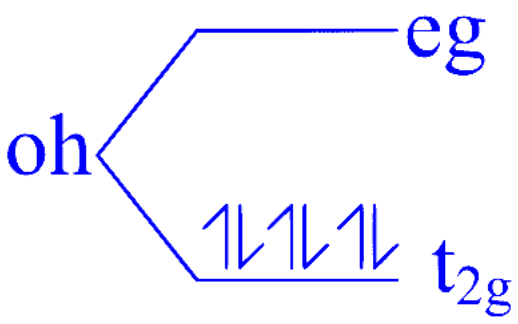
III) $[Fe(CN)_6]^{4-}$

Oxidation state of Fe : $+2$

Fe configuration $\rightarrow d^6s^2$

Fe^{+2} configuration $\rightarrow d^6$

$$x + (-1) \times 6 = -4$$
$$x = -4 + 6$$
$$= +2$$



$$CFSE = -0.4\Delta_0 \times 6 + 0.6\Delta_0 \times 0$$
$$= -2.4\Delta_0$$

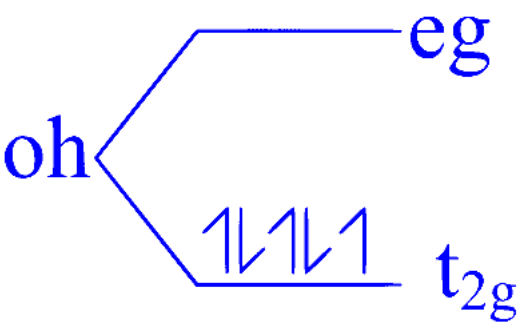
IV) $[\mathrm{Fe(CN)_6}]^{3-}$

Oxidation state of Fe : $+3$

Fe configuration $\rightarrow d^6s^2$

Fe^{+3} configuration $\rightarrow \mathrm{d}^5$

$$x + 6(-1) = -3$$
$$x = -3 + 6$$
$$= +3$$



I) $[\mathrm{Mn}_n(\mathrm{CN})_6]^{3-} \rightarrow -1.6\Delta_0$

II) $[\mathrm{Co(CN)_6}]^{4-} \rightarrow -1.8\Delta_0$

III) $[\mathrm{Fe(CN)_6}]^{4-} \rightarrow -2.4\Delta_0$

IV) $\left[\mathrm{Fe((N)_6}\right]^{3-} \rightarrow -2.0\Delta_0$

Stability of complexes based on Δ_0 value, greater the CFSE value, more stable the complex.

The more negative the CFSE value, the move stable the complex.

So, complex with CFSE -2.4Δ is most stable and complex with CFSE $-1.6\Delta_0$, is least stable. Stability order:

$-1.6\Delta_0 < -1.8\Delta_0 < -2.0\Delta_0 < -2.4\Delta_0$
$[Mn(\mathrm{CN})_6]^{3-} < [\mathrm{C(CN)_6}]^{4-} < [\mathrm{Fe(CN)_6}]^{3-} < [\mathrm{Fe(CN)_6}]^{4-}$
$I < II < IV < III$
Answer: Option (1)

Question25

The calculated spin-only magnetic moments of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are :

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

4.90 and 4.90 B.M.

B.

4.90 and 5.92 B.M.

C.

5.92 and 4.90 B.M.

D.

3.87 and 4.90 B.M.

Answer: C

Solution:

$K_3\left[Fe(OH)_6\right]$

$K_4\left[Fe(OH)_6\right]$

Spin-only magnetic moment formula is $\sqrt{n(n+2)}$, where n is the number of unpaired electrons in the central metal $K_3\left[Fe(OH)_6\right]$

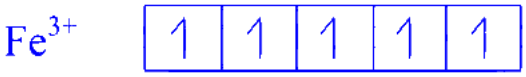
Also written as $[Fe(OH)_6]^{3-}$

Oxidation state of $Fe \rightarrow +3$

Configuration of $Fe \rightarrow d^6s^2$

Configuration of $Fe^{3+} \rightarrow d^5$

Since OH^- is a weak ligand, no pairing of electrons occurs.



Unpaired electrons, $n = 5$

So, spin-only magnetic moment

$= \sqrt{n(n+2)}$

$= \sqrt{5(5+2)}$

$= \sqrt{5 \times 7}$

$= \sqrt{35}$

$= 5.92$ BM

$3(+1) + x + 6(-1) = 0$

$3 + x - 6 = 0$

$x - 3 = 0$

$x = +3$

$K_4\left[Fe(OH)_6\right]$

$4(+1) + x + 6(-1) = 0$

$4 + x - 6 = 0$

$x = 6 - 4 = +2$

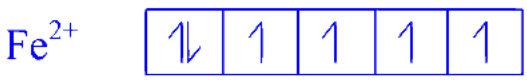
Also written as $[Fe(OH)_6]^{4-}$

Oxidation state of $Fe \rightarrow +2$

Configuration of $Fe \rightarrow d^6s^2$

Configuration of $Fe^{+2} \rightarrow d^6$

Since OH^- is a weak field ligand, no pairing of electrons occurs.



Unpaired electrons = 4

Spin-only magnetic moment

$= \sqrt{n(n+2)}$

$= \sqrt{4(4+2)}$

$= \sqrt{4 \times 6}$

$= \sqrt{24}$

= 4.90 BM

Spin-only magnetic moments of $K_3\left[Fe(OH)_6\right]$ and $K_4\left[Fe(OH)_6\right]$ respectively are 5.92 and 4.90 B.M.

Correct answer is option (3) 5.92 and 4.90 B.M.

Question26

Identify the homoleptic complexes with odd number of d electrons in the central metal :

(A) $\left[\mathrm{FeO_4}\right]^{2-}$

(B) $\left[\mathrm{Fe(CN)_6}\right]^{3-}$

(C) $\left[\mathrm{Fe(CN)_5NO}\right]^{2-}$

(D) $\left[\mathrm{CoCl_4}\right]^{2-}$

(E) $\left[\mathrm{Co(H_2O)_3\ F_3}\right]$

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

(C) and (E) only

B.

(A), (B) and (D) only

C.

(B) and (D) only

D.

(A), (C) and (E) only

Answer: C

Solution:

Homoleptic complexes are complexes with identical ligands attached to the central metal atom.

(A) $[FeO_4]^{2-}$ - Homdeptic

Central metal : Fe

IN this complex, Fe is in +6 oxidation state.

{oxidation state : $x + 4(-2) = -2$

$x - 8 = -2$

$x = -2 + 6 = +6$

Configuration of $Fe - d^6s^2$

Configuration of $Fe(+6) - cl^2$

Iron has even number of d electrons. So, it is not correct.

(B) $[Fe(N)_6]^{3-}$ - Homoleptic

Central metal : Fe

In this complex, the oxidation state of Fe is +3

Oxidation state :

$x + 6(-1) - 3$

$x - 6 = -3$

$x = -3 + 6 = +3$

Configuration of $Fe - d^6s^2$

Configuration of $Fe(+3) - d^5$

Iron has odd number of d electrons.

So, it is correct.

(C) $[Fe(CN)_5NO]^{2-}$ - Not homoleptic

This complex has different type of ligands.

$[CoCl_4]^{2-}$ - Homoleptic

Central metal - Co

In this complex, the oxidation state of Co is +2

Oxidation state:

$x + 4(-1) = -2$

$x - 4 = -2$

$x = -2 + 4 = +2$

Configuration of $Co - d^7s^2$

Configuration of $Co(+2) - d^7$

Cobalt has odd number of d electrons.

So, it is correct.

(E) $[Co(H_2O)_3F_3]$ - Not homoleptic,

It has different type of ligands.

Complexes B and D are homoleptic complexes with odd number of d electrons in the central metal. So, correct answer is

(B) and (D) only.

Question27

Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_0 < \mathrm{P}$ high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of $\Delta_t

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. Both Statement I and Statement II are incorrect

B. Statement I is correct but Statement II is incorrect

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are correct

Answer: D

Solution:

In octahedral complexes, which have a coordination number (CN) of 6, the formation of high spin or low spin complexes depends on the relative magnitude of the crystal field splitting energy, denoted as Δ_o, and the pairing energy (P.E.):

If Δ_o is less than the pairing energy (Δ_o < P.E.), high spin complexes are formed.

If Δ_o is greater than the pairing energy (Δ_o > P.E.), low spin complexes are formed.

In tetrahedral complexes, where the coordination number is 4:

The crystal field splitting energy (Δ_t) is typically less than the pairing energy (Δ_t < P.E.), resulting in the formation of mainly high spin complexes. Low spin complexes are rarely observed in tetrahedral coordination due to this energy consideration.

Question28

The d-orbital electronic configuration of the complex among $[Co(en)_3]^{3+}$, $[CoF_6]^{3-}$, $[Mn(H_2O)_6]^{2+}$ and $[Zn(H_2O)_6]^{2+}$ that has the highest CFSE is :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. ${t_{2g}}^6 e_g^0$

B. ${t_{2g}}^3 e_g^2$

C. ${t_{2g}}^4 {e_g}^2$

D. ${t_{2g}}^6 e_g^4$

Answer: A

Solution:

Identify the oxidation state and d-electron count for each complex:

For $[Co(en)_3]^{3+}$:

Ethylenediamine (en) is neutral.

Oxidation state of cobalt is +3.

Cobalt (atomic number 27) in the +3 state gives a d^6 configuration.

With strong/moderate field ligands like en, the complex is low spin, leading to a configuration of $t_{2g}^6 e_g^0$.

For $[CoF_6]^{3-}$:

Each fluoride ion is F^- so, with six of them, Co must be in the +3 state again.

However, since fluoride is a weak field ligand, this complex adopts a high spin configuration for d^6, resulting in $t_{2g}^4 e_g^2$.

For $[Mn(H_2O)_6]^{2+}$:

Water is neutral.

Manganese in the +2 state gives a d^5 configuration.

With weak field water, the d^5 configuration is high spin: $t_{2g}^3 e_g^2$.

For $[Zn(H_2O)_6]^{2+}$:

Zinc in the +2 state has a d^{10} configuration.

In an octahedral field, all 10 electrons fill the orbitals as $t_{2g}^6 e_g^4$.

Calculate the crystal field stabilization energy (CFSE) for each case:

The CFSE in an octahedral field can be estimated as:

$\text{CFSE} = (n_{t_{2g}} \times -0.4\Delta_o) + (n_{e_g} \times 0.6\Delta_o)$

For $t_{2g}^6 e_g^0$ (low-spin d^6 in $[\text{Co(en)}_3]^{3+}$):

$\text{CFSE} = 6(-0.4\Delta_o) + 0(0.6\Delta_o) = -2.4\Delta_o$

For $t_{2g}^4 e_g^2$ (high-spin d^6 in $[\text{CoF}_6]^{3-}$):

$\text{CFSE} = 4(-0.4\Delta_o) + 2(0.6\Delta_o) = -1.6\Delta_o + 1.2\Delta_o = -0.4\Delta_o$

For $t_{2g}^3 e_g^2$ (high-spin d^5 in $[\text{Mn(H}_2\text{O)}_6]^{2+}$):

$\text{CFSE} = 3(-0.4\Delta_o) + 2(0.6\Delta_o) = -1.2\Delta_o + 1.2\Delta_o = 0$

For $t_{2g}^6 e_g^4$ (for d^{10} in $[\text{Zn(H}_2\text{O)}_6]^{2+}$):

$\text{CFSE} = 6(-0.4\Delta_o) + 4(0.6\Delta_o) = -2.4\Delta_o + 2.4\Delta_o = 0$

Compare the CFSE values:

$t_{2g}^6 e_g^0$ gives a CFSE of $-2.4\Delta_o$ (most negative, hence highest stabilization).

The others result in less stabilization (or net zero).

Conclusion:

The complex with the highest CFSE is the one with the configuration $t_{2g}^6 e_g^0$, which corresponds to $[\text{Co(en)}_3]^{3+}$.

Thus, the answer is:

Option A: $t_{2g}^6 e_g^0$.

Question29

The type of hybridization and the magnetic property of $[\text{MnCl}_6]^{3-}$ are,

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. sp^3d^2, paramagnetic with four unpaired electrons.

B. d^2sp^3, paramagnetic with four unpaired electrons.

C. $\text{sp}^3\ \text{d}^2$, paramagnetic with two unpaired electrons.

D. d^2sp^3, paramagnetic with two unpaired electrons.

Answer: A

Solution:

Determine the oxidation state of manganese in $[\text{MnCl}_6]^{3-}$:

Chloride (Cl^-) has a charge of -1.

With six chloride ions, the total charge contributed by the ligands is $6(-1) = -6$.

Let the oxidation state of Mn be x. Then:

$x - 6 = -3 \quad \Longrightarrow \quad x = +3.$

So, manganese is in the +3 oxidation state.

Find the d-electron count for Mn(III):

The neutral manganese atom has the electronic configuration $[\text{Ar}]\, 3d^5 4s^2$.

Removing three electrons (first from the 4s, then from the 3d) gives:

Mn^{3+} : $[\text{Ar}]\, 3d^4$.

Therefore, Mn(III) is a d^4 system.

Predict the spin state in an octahedral complex:

Chloride (Cl^-) is a weak-field ligand, meaning it produces a relatively small crystal field splitting (Δ_o).

When Δ_o is small, the pairing energy is higher than Δ_o, so electrons prefer to remain unpaired.

For a d^4 configuration in a high-spin octahedral complex, the electrons will be distributed as:

Three electrons in the three t_{2g} orbitals (one in each)

One electron in one of the e_g orbitals

This results in a total of 4 unpaired electrons, making the complex paramagnetic.

Determine the type of hybridization:

In octahedral complexes, the two common hybridizations are:

d^2sp^3 (inner orbital complex), typically seen in low-spin complexes where inner 3d orbitals are available because the electrons are paired.

sp^3d^2 (outer orbital complex), seen in high-spin complexes where the inner $3d$ orbitals are occupied by unpaired electrons.

Since $[MnCl_6]^{3-}$ is high-spin (because of Cl^- being a weak-field ligand), the inner 3d orbitals are not available for hybridization.

Therefore, the complex uses the outer orbitals (the 4s, 4p, and 4d orbitals), leading to an sp^3d^2 hybridization.

Conclusion:

The hybridization of $[MnCl_6]^{3-}$ is sp^3d^2.

It is paramagnetic with four unpaired electrons.

Thus, the correct answer is:

Option A: sp^3d^2, paramagnetic with four unpaired electrons.

Question30

Match the LIST-I with LIST-II

LIST-I (Molecules/ion)		LIST-II (Hybridisation of central atom)	
A.	PF_5	I	dsp^2
B	SF_6	II	$sp^3\ d$
C	$Ni(CO)_4$	III	$sp^3\ d^2$
D	$[PtCl_4]^{2-}$	IV	sp^3

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. A-IV, B-I, C-II, D-III

B. A-III, B-I, C-IV, D-II

C. A-II, B-III, C-IV, D-I

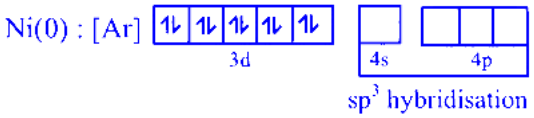
D. A-I, B-II, C-III, D-IV

Answer: C

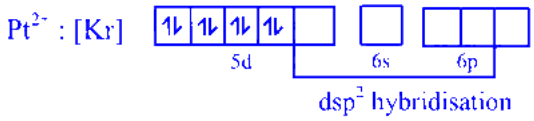
Solution:

$PF_5 : 5\sigma + 0\ell p \rightarrow sp^3\ d$
$SF_6 : 6\sigma + 0\ell p \rightarrow sp^3\ d^2$
$Ni(CO)_4 : Ni \rightarrow 0$

In presence of ligand field :-



$[PtCl_4]^{2-}$: Pt $\rightarrow +2$
In presence of ligand field :-



Question31

The correct order of the complexes $[Co(NH_3)_5(H_2O)]^{3+}(A)$, $[Co(NH_3)_6]^{3+}(B)$, $[Co(CN)_6]^{3-}(C)$ and $[CoCl(NH_3)_5]^{2+}(D)$ in terms of wavelength of light absorbed is

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. D > C > B > A

B. C > B > A > D

C. D > A > B > C

D. C > B > D > A

Answer: C

Solution:

The energy of absorbed light by a complex can be expressed as $E = hv = \frac{hC}{\lambda}$, where E is the energy, h is Planck's constant, v is the frequency, C is the speed of light, and λ is the wavelength of the absorbed light. This formula demonstrates that energy (E) is inversely proportional to the wavelength (λ).

In this context, all cobalt complexes are in the +3 oxidation state. The Crystal Field Stabilization Energy (CFSE) is influenced by the strength of the ligands surrounding the metal ion. As the ligand field strength increases, the CFSE also increases.

The order of field strength for the ligands in this case is:

$CN^- > NH_3 > H_2O > Cl^-$

This implies the following order for CFSE in the given complexes:

C > B > A > D

Because energy is inversely proportional to wavelength, complexes absorbing at higher energies will absorb at shorter wavelengths. Therefore, the order of the complexes in terms of the wavelength of light absorbed, from longest to shortest wavelength, is:

D > A > B > C

Question32

Identify the diamagnetic octahedral complex ions from below ;

A. $[Mn(CN)_6]^{3-}$

B. $[Co(NH_3)_6]^{3+}$

C. $[Fe(CN)_6]^{4-}$

D. $[Co(H_2O)_3F_3]$

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. B and C Only

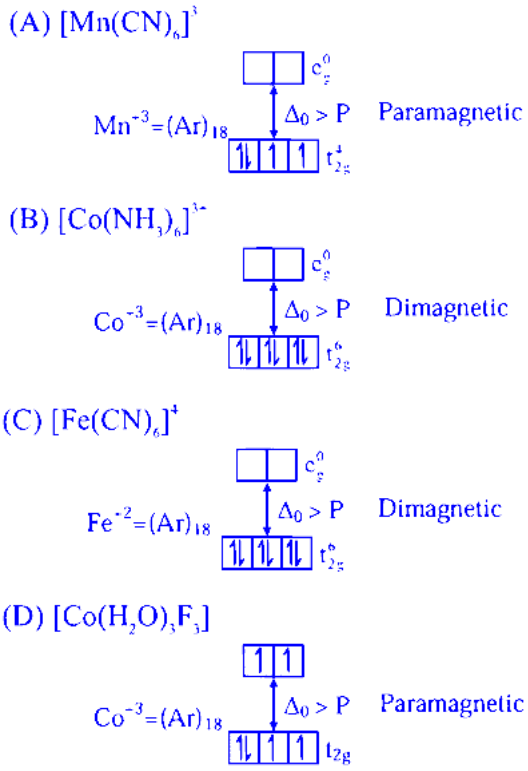
B. A and C Only

C. B and D Only

D. A and D Only

Answer: A

Solution:



Question33

Number of stereoisomers possible for the complexes, $[CrCl_3(py)_3]$ **and** $[CrCl_2(ox)_2]^{3-}$ **are respectively** $(py =$ **pyridine,**$ox =$ **oxalate** $)$

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. 3 & 3

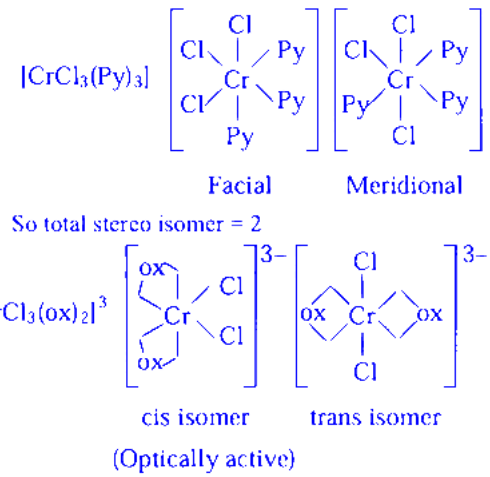
B. 2 & 3

C. 1 & 2

D. 2 & 2

Answer: B

Solution:



Geometrical isomer = 2(1 cis +1 trans)

Optical isomer = 3 (2 optically active +1 optically inactive)

Stereoisomer = 3

Question34

Which one of the following complexes will have $\Delta_o = 0$ and $\mu = 5.96$ B.M?

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $[FeF_6]^{4-}$

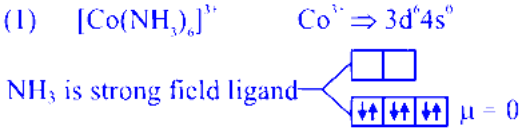
B. $[Mn(SCN)_6]^{4-}$

C. $[Co(NH_3)_6]^{3+}$

D. $[Fe(CN)_6]^{4-}$

Answer: B

Solution:



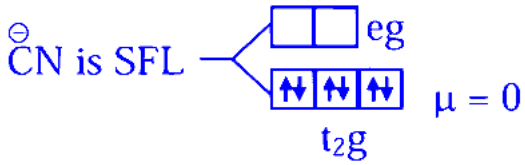
$= [-0.4 \times 6 + 0.6 \times (0)]\Delta_0 = -2.4\Delta_0$

(2) $[Mn(SCN)_6]^4$
$Mn^{2+} \Rightarrow 3\,d^5 4\,s^0$

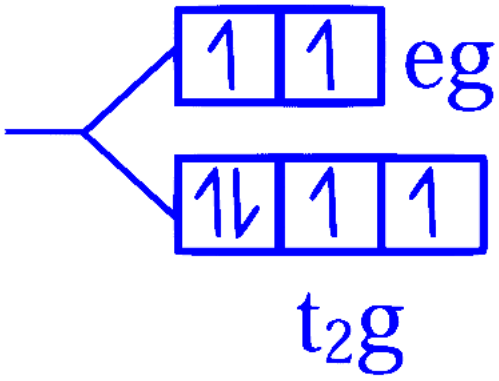


$\mu = \sqrt{35}$ B.M. $= 5.96$ B.M.
CFSE $= (-0.4 \times 3 + 0.6 \times 2)\Delta_0$
So $\Delta_0 = 0$

(3) $[Fe(CN)_6]^{-4}$ $Fe^{2+} \Rightarrow 3\,d^6 4\,s^0$



CFSE $= -2.4\Delta_0$
(4) $[FeF_6]^4 \quad Fe^{2+} \Rightarrow 3\,d^6 4\,s^0$



$\mu = \sqrt{24}$ B.M. $= 4.89$ B.M.
CFSE $= (-0.4 \times 4 + 0.6 \times 2)\Delta_0 = -1.2\Delta_0$

Question35

The correct order of $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ complex species based on the number of unpaired electrons present is:

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. $[CoF_6]^{3-} > [FeF_6]^{3-} > [Ni(CO)_4] > [Ni(CN)_4]^{2-}$

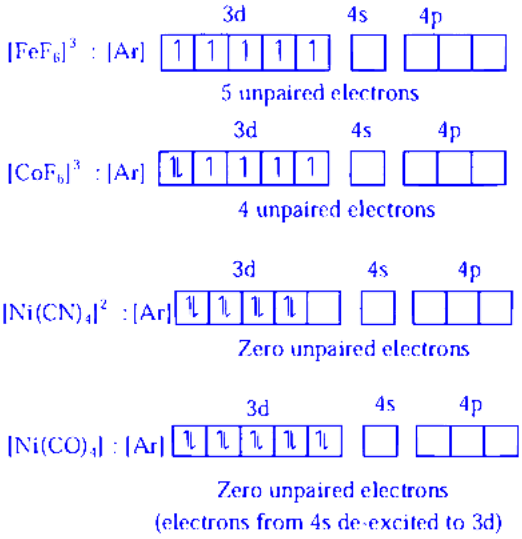
B. $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} > [Ni(CO)_4]$

C. $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} = [Ni(CO)_4]$

D. $[Ni(CN)_4]^{2-} > [FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CO)_4]$

Answer: C

Solution:



Question36

' X ' is the number of electrons in t_{2g} orbitals of the most stable complex ion among $[\mathrm{Fe(NH_3)_6}]^{3+}, [\mathrm{FeCl_6}]^{3-}, \quad [\mathrm{Fe(C_2O_4)_3}]^{3-}$ and $[\mathrm{Fe(H_2O)_6}]^{3+}$. The nature of oxide of vanadium of the type $\mathrm{V_2O_X}$ is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. Amphoteric

B. Acidic

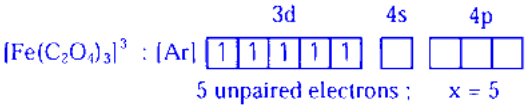
C. Basic

D. Neutral

Answer: A

Solution:

Most stable is $[\mathrm{Fe(C_2O_4)_3}]^{3-}$ due to Chelation effect.



V_2O_5 is amphoteric.

Question37

An octahedral complex having molecular composition $\mathrm{Co \cdot 5NH_3 \cdot Cl^2 . SO_4}$ has two isomers A and B. The solution of A gives a white precipitate with $\mathrm{AgNO_3}$ solution and the solution of B gives white precipitate with $\mathrm{BaCl_2}$ solution. The type of isomerism exhibited by the complex is,

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. Co-ordinate isomerism

B. Ionisation isomerism

C. Geometrical isomerism

D. Linkage isomerism

Answer: B

Solution:

In this octahedral complex with the molecular composition $\mathrm{Co \cdot 5NH_3 \cdot Cl^2 \cdot SO_4}$, two isomers, A and B, are present.

Isomer A is represented as $\mathrm{[Co(NH_3)_5\,(SO_4)]Cl}$. When this isomer is dissolved in solution, it reacts with $\mathrm{AgNO_3}$, forming a white precipitate of silver chloride (AgCl).

Isomer B is written as $\mathrm{[Co(NH_3)_5Cl]SO_4}$. In solution, it reacts with $\mathrm{BaCl_2}$, forming a white precipitate of barium sulfate ($\mathrm{BaSO_4}$).

These two isomers demonstrate ionisation isomerism. This type of isomerism occurs when the exchange of ions inside and outside the coordination sphere of a complex leads to the formation of isomers with different chemical behaviors in solution.

Question38

'X' is the number of acidic oxides among VO_2, V_2O_3, CrO_3, V_2O_5 and Mn_2O_7. The primary valency of cobalt in $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$ is Y. The value of X + Y is _____.

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

3

B.

4

C.

2

D.

5

Answer: D

Solution:

In the given compounds, we evaluate which are acidic oxides:

CrO_3 is acidic.

Mn_2O_7 is acidic.

Therefore, the number of acidic oxides, X, is 2.

Next, for the complex $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$, we consider its dissociation:

$$\overset{III}{[Co}(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3 \rightleftharpoons 2\overset{III}{[Co}(H_2NCH_2CH_2NH_2)_3]^{3+} + 3SO_4^{2-}$$

The primary valency of cobalt (Y) in this context is 3.

Thus, the value of $X + Y$ is:

$X + Y = 2 + 3 = 5$

Question39

The number of unpaired electrons responsible for the paramagnetic nature of the following complex species are respectively :

$[Fe(CN)_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

1, 5, 4, 2

B.

1, 4, 4, 2

C.

1, 5, 5, 2

D.

1, 1, 4, 2

Answer: A

Solution:

The number of unpaired electrons responsible for the paramagnetic nature of each complex species is calculated as follows:

Complex: $[Fe(CN)_6]^{3-}$

Ion: Fe^{3+}

Electronic Configuration: $3d^5$

Orbital Population:

$t_{2g}^{2,2,1}$

$e_g^{0,0}$

Unpaired Electrons: 1

Complex: $[FeF_6]^{3-}$

Ion: Fe^{3+}

Electronic Configuration: $3d^5$

Orbital Population:

$t_{2g}^{1,1,1}$

$e_g^{1,1}$

Unpaired Electrons: 5

Complex: $[CoF_6]^{3-}$

Ion: Co^{3+}

Electronic Configuration: $3d^6$

Orbital Population:

$t_{2g}^{2,1,1}$

$e_g^{1,0}$

Unpaired Electrons: 4

Complex: $[Mn(CN)_6]^{3-}$

Ion: Mn^{3+}

Electronic Configuration: $3d^4$

Orbital Population:

$t_{2g}^{2,1,1}$

$e_g^{0,0}$

Unpaired Electrons: 2

Each complex's paramagnetic nature is due to the specific number of unpaired electrons as indicated above.

Question40

Match List - I with List - II.

List - I (Complex)	List - II (Primary valency and Secondary valency)
(A) $[Co(en)_2Cl_2]Cl$	(I) 3, 6
(B) $[Pt(NH_3)_2Cl(NO_2)]$	(II) 3, 4
(C) Hg $[Co(SCN)_4]$	(III) 2, 6
(D) $[Mg\ (EDTA)]^{2-}$	(IV) 2, 4

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

(A)-(I), (B)-(III), (C)-(II), (D)-(IV)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(I), (B)-(IV), (C)-(II), (D)-(III)

D.

(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Answer: C

Solution:

Primary valency = Oxidation state

Secondary valency = Co-ordination number

	Complex	Primary valency	Secondary
(A)	$[Co(en)_2Cl_2]Cl$	3	6
(B)	$[Pt(NH_3)_2Cl\,(NO_2)]$	2	4
(C)	$Hg\,[Co(SCN)_4]$	3	4
(D)	$[Mg(EDTA)]^{2-}$	2	6

Question41

Given below are two statements:

Statement I: A homoleptic octahedral complex, formed using monodentate ligands, will not show stereoisomerism.

Statement II: cis- and trans- platin are heteroleptic complexes of Pd.

In the light of the above statements, choose the correct answer from the options given below:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are true

C.

Both Statement I and Statement II are false

D.

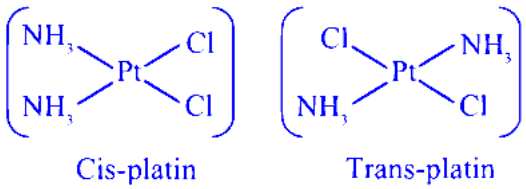
Statement I is false but Statement II is true

Answer: A

Solution:

Homoleptic complex of type $[Ma_6]$ (Where a $\Rightarrow$ monodentate ligand) cannot show geometrical as well as optical isomerism.

Cis-platin and trans-platin has formula $[Pt(NH_3)_2Cl_2]$ which is a heteroleptic complex of platinum.



Cis-platin Trans-platin

Question42

The number of species from the following that are involved in sp^3d^2 hybridization is :

$[Co(NH_3)_6]^{3+}$, SF_6, $[CrF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$, and $[MnCl_6]^{3-}$

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

4

B.

3

C.

5

D.

6

Answer: B

Solution:

In $[Co(NH_3)_6]^{3+}$, Co^{+3} : $[Ar]3\ d^6$, NH_3 is S.F.L

Hybridisation state of Co^{3+} is d^2sp^3

In SF_6, Hybridisation state of sulphur is $sp^3\ d^2$

In $[CrF_6]^{3-}$, Cr^{+3} : $[Ar]3\ d^3$

Hybridisation state of Cr^{3+} is d^2sp^3

$[CoF_6]^{3-}$, Co^{+3} : $[Ar]3\ d^6$ $F^-$is W.F.L

Hybridisation state of Co^{3+} is $sp^3\ d^2$

$[Mn(CN)_6]^{3-}$, Mn^{+3} : $[Ar]3\ d^4CN^-$is S.F.L

Hybridisation state of Mn^{3+} is d^2sp^3

$[MnCl_6]^{3-}$, Mn^{+3} : $[Ar]3\ d^4Cl^-$is W.F.L

Hybridisation state of $Cl^-$is $sp^3\ d^2$

Total number of $sp^3\ d^2$ hybridized molecules is 3

Question43

Match the LIST-I with LIST-II

LIST-I (Complex/ Species)	LIST-II (Shape & magnetic moment)
A. $[Ni(CO)_4]$	I. Tetrahedral, 2.8 BM
B. $[Ni(CN)_4]^{2-}$	II. Square planar, 0 BM
C. $[NiCl_4]^{2-}$	III. Tetrahedral, 0 BM
D. $[MnBr_4]^{2-}$	IV. Tetrahedral, 5.9 BM

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

A-I, B-II, C-III, D-IV

B.

A-IV, B-I, C-III, D-II

C.

A-III, B-II, C-I, D-IV

D.

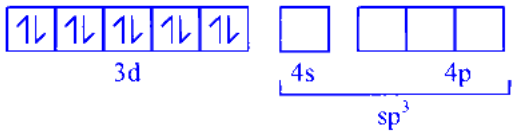
A-III, B-IV, C-II, D-I

Answer: C

Solution:

(A) $[Ni(CO)_4], Ni^0 : [Ar]3\ d^8 4\ s^2$

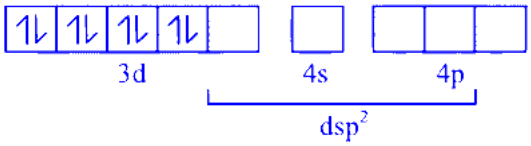
Valence orbitals of Ni^0 in pre-hybridisation state :



Tetrahedral, Diamagnetic, $\mu = 0$ B.M.

(B) $[Ni(CN)_4]^{2-}, Ni^{+2} : [Ar]3\ d^8 4\ s^0$

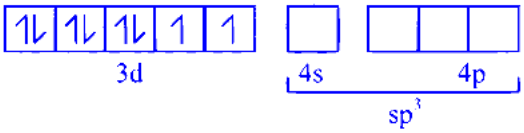
Valence orbitals of Ni^{+2} in pre-hybridisation state :



Square planar, Diamagnetic, $\mu = 0$ B.M.

(C) $[NiCl_4]^{2-}, Ni^{+2} : [Ar]3\ d^8 4\ s^0$

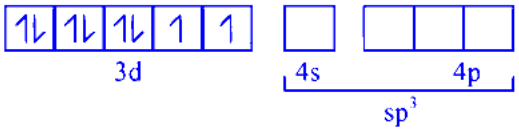
Valence orbitals of Ni^{+2} in ground state :



Tetrahedral, paramagnetic, $\mu = \sqrt{8} = 2.8$ B.M.

(D) $[MnBr_4]^{2-}, Mn^{+2} : [Ar]3\ d^5$

Valence orbitals of Mn^{+2} in ground state :



Tetrahedral, paramagnetic, $\mu = \sqrt{35} = 5.9$ B.M.

Question44

Yellow compound of lead chromate gets dissolved on treatment with hot NaOH solution. The product of lead formed is a : [27-Jan-2024 Shift 1]

Options:

A. Tetraanionic complex with coordination number six

B. Neutral complex with coordination number four

C. Dianionic complex with coordination number six

D. Dianionic complex with coordination number four

Answer: D

Solution:

Solution:

Question45

Identity the incorrect pair from the following:
[27-Jan-2024 Shift 2]

Options:

A. Photography - AgBr

B. Polythene preparation $-TiCl_4$, $Al(CH_3)_3$

C. Haber process - Iron

D. Wacker process $-Pt\,Cl_2$

Answer: D

Solution:

Solution:

Question46

Identify from the following species in which $d^2\,sp^3$ hybridization is shown by central atom:
[27-Jan-2024 Shift 2]

Options:

A. $[Co(NH_3)_6]^{3+}$

B. BrF_5

C. $[Pt(Cl)_4]^{2-}$

D. SF_6

Answer: A

Solution:

Solution:

Question47

The Spin only magnetic moment value of square planar complex $[Pt(NH_3)_2\,Cl(NH_2CH_3)]\,Cl$ is _____B.M. (Nearest integer)
(Given atomic number for Pt = 78)
[27-Jan-2024 Shift 2]

Answer: 0

Solution:

Solution:

Question48

In which one of the following metal carbonyls, CO forms a bridge between metal atoms? [29-Jan-2024 Shift 1]

Options:

A. $[Co_2(CO)_8]$

B. $[Mn_2(CO)_{10}]$

C. $[Os_3(CO)_{12}]$

D. $[Ru_3(CO)_{12}]$

Answer: A

Solution:

Solution:

Question49

The correct IUPAC name of K_2MnO_4 is [29-Jan-2024 Shift 2]

Options:

A. Potassium tetraoxopermanganate (VI)

B. Potassium tetraoxidomanganate (VI)

C. Dipotassium tetraoxidomanganate (VII)

D. Potassium tetraoxidomanganese (VI)

Answer: B

Solution:

Solution:

Question50

Aluminium chloride in acidified aqueous solution forms an ion having geometry [30-Jan-2024 Shift 1]

Options:

A. Octahedral

B. Square Planar

C. Tetrahedral

D. Trigonal bipyramidal

Answer: A

Solution:

Solution:

Question51

Choose the correct Statements from the following:
(A) Ethane-1 2-diamine is a chelating ligand.
(B) Metallic aluminium is produced by elecrtrolysis of aluminium oxide in presence of cryolite.
(C) Cyanide ion is used as ligand for leaching of silver.
(D) Phosphine act as a ligand in Wilkinson catalyst.
(E) The stability constants of Ca^{2+} and Mg^{2+} are similar with EDTA complexes.
Choose the correct answer from the options given below:
[30-Jan-2024 Shift 1]

Options:

A. (B), (C), (E) only

B. (C), (D), (E) only

C. (A), (B), (C) only

D. (A), (D), (E) only

Answer: C

Solution:

Solution:

Question52

The molecule/ion with square pyramidal shape is:
[30-Jan-2024 Shift 2]

Options:

A. $[Ni(CN)_4]^{2-}$

B. PCl_5

C. BrF_5

D. PF_5

Answer: C

Solution:

Solution:

Question53

The coordination geometry around the manganese in decacarbonyldimanganese(0)
[30-Jan-2024 Shift 2]

Options:

A. Octahedral

B. Trigonal bipyramidal

C. Square pyramidal

D. Square planar

Answer: A

Solution:

Solution:

Question54

Number of complexes which show optical isomerism among the following is

cis $-[Cr(ox)_2Cl_2]^{3-}$, $[Co(en)_3]^{3+}$

cis $-[Pt(en)_2Cl_2]^{2+}$, cis $-[Co(en)_2Cl_2]^{+}$

trans $-[Pt(en)_2Cl_2]^{2+}$, trans $-[Cr(ox)_2Cl_2]^{3-}$

[30-Jan-2024 Shift 2]

Answer: 4

Solution:

Solution:

Question55

The correct statements from following are:

A. The strength of anionic ligands can be explained by crystal field theory.

B. Valence bond theory does not give a quantitative interpretation of kinetic stability of coordination compounds.

C. The hybridization involved in formation of $[Ni(CN)_4]^{2-}$ complex is dsp^2.

D. The number of possible isomer(s) of cis- $[PtCl_2(en)_2]^{2+}$ is one

Choose the correct answer from the options given below:

[31-Jan-2024 Shift 1]

Options:

A. A, D only

B. A, C only

C. B, D only

D. B, C only

Answer: D

Solution:

Solution:

Question56

Select the option with correct property - [31-Jan-2024 Shift 2]

Options:

A. $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ both diamagnetic

B. $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ both paramagnetic

C. $[NiCl_4]^{2-}$ diamagnetic, $[Ni(CO)_4]$ paramagnetic

D. $[Ni(CO)_4]$ diamagnetic, $[NiCl_4]^{2-}$ paramagnetic

Answer: D

Solution:

Solution:

Question57

Choose the correct answer from the options given below :

	LIST - I (Complex ion)		LIST - II Electronic Configuration
A.	$[Cr(H_2O)_6]^{3+}$	I.	$t_{2g}^2 e_g^0$
B.	$[Fe(H_2O)_6]^{3+}$	II.	$t_{2g}^3 e_g^0$
C.	$[Ni(H_2O)_6]^{2+}$	III.	$t_{2g}^3 e_g^2$
D.	$[V(H_2O)_6]^{3+}$	IV.	$t_{2g}^6 e_g^2$

[31-Jan-2024 Shift 2]

Options:

A. A-III, B-II, C-IV, D-I

B. A-IV, B-I, C-II, D-III

C. A-IV, B-III, C-I, D-II

D. A-II, B-III, C-IV, D-I

Answer: D

Solution:

Solution:

Question58

Which of the following complex is homoleptic? [1-Feb-2024 Shift 1]

Options:

A. $[Ni(CN)_4]^{2-}$

B. $[Ni(NH_3)_2 Cl_2]$

C. $[Fe(NH_3)_4 Cl_2]^+$

D. $[Co(NH_3)_4 Cl_2]^+$

Answer: A

Solution:

Solution:

Question59

Given below are two statements:
Statement (I): A solution of $[Ni(H_2O)_6]^{2+}$ is green in colour.

Statement (II): A solution of $[Ni(CN)_4]^{2-}$ is colourless.

In the light of the above statements, choose the most appropriate answer from the options given below: [1-Feb-2024 Shift 1]

Options:

A. Both Statement I and Statement II are incorrect

B. Both Statement I and Statement II are correct

C. Statement I is incorrect but Statement II is correct

D. Statement I is correct but Statement II is incorrect

Answer: B

Solution:

Solution:

Question60

Given below are two statements :
Statement (I) : Dimethyl glyoxime forms a sixmembered covalent chelate when treated with $NiCl_2$ solution in presence of NH_4OH.

Statement (II) : Prussian blue precipitate contains iron both in (+2) and (+3) oxidation states. In the light of the above statements, choose the most appropriate answer from the options given below: [1-Feb-2024 Shift 2]

Options:

A. Statement I is false but Statement II is true

B. Both Statement I and Statement II are true

C. Both Statement I and Statement II are false

D. Statement I is true but Statement II is false

Answer: A

Solution:

Solution:

Question61

$[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are respectively known as:
[1-Feb-2024 Shift 2]

Options:

A. Spin free Complex, Spin paired Complex

B. Spin paired Complex, Spin free Complex

C. Outer orbital Complex, Inner orbital Complex

D. Inner orbital Complex, Spin paired Complex

Answer: B

Solution:

Solution:

Question62

1L, 0.02M solution of $[Co(NH_3)_5 SO_4]$ Br is mixed with 1L, 0.02M solution of $[Co(NH_3)_5 Br]SO_4$. The resulting solution is divided into two equal parts (X) and treated with excess $AgNO_3$ solution and $BaCl_2$ solution respectively as shown below:
1L Solution(X) + $AgNO_3$ solution (excess) $\rightarrow$Y
1L Solution (X) + $BaCl_2$ solution (excess) $\rightarrow$Z
The number of moles of Y and Z respectively are
[30-Jan-2023 Shift 2]

Options:

A. 0.02, 0.02

B. 0.01, 0.01

C. 0.02, 0.01

D. 0.01, 0.02

Answer: B

Solution:

Solution:

$$\underset{0.01\ mol}{[Co(NH_3)_5 SO_4]Br} + \underset{excess}{AgNO_3} \rightarrow \underset{0.01\ Mol}{AgBr\downarrow}$$

$$\underset{0.01\ mol}{[Co(NH_3)_5 Br]SO_4} + \underset{excess}{BaCl_2} \rightarrow \underset{0.01\ Mol}{BaSO_4\downarrow}$$

Question63

The $Cl-Co-Cl$ bond angle values in a fac$[Co(NH_3)_3 Cl_3]$ complex is/are:
[30-Jan-2023 Shift 2]

Options:

A. 90°&180°

B. 90°

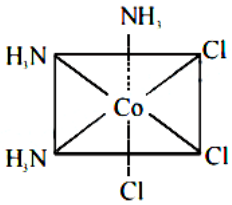
C. 180°

D. 90°&120°

Answer: B

Solution:

Solution:



The $Cl-Co-Cl$ bond angle in above octahedral complex is 90°

Question64

Cobalt chloride when dissolved in water forms pink colored complex X which has octahedral geometry. This solution on treating with cone HCl forms deep blue complex, Y which has a Z geometry. X, Y and Z, respectively, are
[31-Jan-2023 Shift 1]

Options:

A. X = $[Co(H_2O)_6]^{2+}$, Y = $[CoCl_4]^{2-}$, Z = Tetrahedral

B. X = $[Co(H_2O_6)]^{2+}$, Y = $[CoCl_6]^{3-}$, Z = Octahedral

C. X = $[Co(H_2O)_6]^{3+}$, Y = $[CoCl_6]^{3-}$, Z = Octahedral

D. X = $[Co(H_2O)_4\,Cl_2]^{+}$, Y = $[CoCl_4]^{2-}$, Z = Tetrahedral

Answer: A

Solution:

Solution:

$$CoCl_2 + 6H_2O \longrightarrow [Co(H_2O)_6]Cl_2$$

Pink(X)
octahedral
$\downarrow$ +HCl(conc.)
$[CoCl_4]^{2-}$
(Y)Blue solution
(Z)Tetrahedral

Question65

If the CFSE of $[Ti(H_2O)_6]^{3+}$ is $-96.0\ kJ/mol$, this complex will absorb maximum at wavelength ____ nm. (nearest integer)

Assume Planck's constant (h) = 6.4×10^{-34} Js Speed of light (c) = 3.0×10^{8}m / s and Avogadro's constant (N_A) = 6×10^{23} / mol
[31-Jan-2023 Shift 2]

Answer: 480

Solution:

Solution:
$(Ti^{+3}(H_2O)_6)^{3+}$
Ti^{+3} : $3d^1$
C.F.S.E. $= -0.4 \times \Delta_0$
$= -\frac{96 \times 10^3}{N_0}$ J
$\Delta_0 = \frac{96 \times 10^3}{0.4 \times 6 \times 10^{23}}$
$\Rightarrow \frac{hc}{\lambda} = \frac{96 \times 10^3}{0.4 \times 6 \times 10^{23}}$
$\lambda = \frac{0.4 \times 6 \times 10^{23} \times 6.4 \times 10^{-34} \times 3 \times 10^8}{96 \times 10^3}$
$= 0.48 \times 10^{-6}$m
$= 480 \times 10^{-9}$m
$= 480$ nm

Question66

Match List I with List II

List I Coordination entity	List II Wavelength of light absorbed in nm
A. $[CoCl(NH_3)_5]^{2+}$	I. 310
B. $[Co(NH_3)_6]^{3-}$	II. 475
C. $[Co(CN)_6]^{3-}$	III. 535
D. $[Cu(H_2O)_4]^{2-}$	IV. 600

Choose the correct answer from the options given below :-
[25-Jan-2023 Shift 2]

Options:

A. A-IV, B-I, C-III, D-II

B. A-III, B-II, C-I, D-IV

C. A-III, B-I, C-II, D-IV

D. A- II, B-III, C-IV, D-I

Answer: B

Solution:

Solution:

List I Coordination entity	List II Wavelength of light absorbed in nm
A. $[CoCl(NH_3)_5]^{2+}$	I. 535
B. $[Co(NH_3)_6]^{3+}$	II. 475
C. $[Co(CN)_6]^{3-}$	III. 310
D. $[Cu(H_2O)_4]^{2+}$	IV. 600

$E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$

$\Rightarrow \Delta(\text{ CFSE }) \propto \frac{1}{\lambda_{absorb}} \propto$ strength of ligand.

Question67

Total number of moles of AgCl precipitated on addition of excess of $AgNO_3$ to one mole each of the following complexes $[Co(NH_3)_4Cl_2]Cl$, $[Ni(H_2O)_6]Cl_2$, $[Pt(NH_3)_2Cl_2]$ and $[Pd(NH_3)_4]Cl_2$ is
[25-Jan-2023 Shift 2]

Answer: 5

Solution:

Solution:
$[Co(NH_3)_4Cl_2]Cl \Rightarrow$ Gives 1 mole AgCl
$[Ni(H_2O)_6]Cl_2 \Rightarrow$ Gives 2 moles AgCl
$[Pt(NH_3)_2Cl_2] \Rightarrow$ Gives No AgCl
$[Pd(NH_3)_4]Cl_2 \Rightarrow$ Gives 2 moles AgCl
Total number of moles of AgCl = 5 mole.

Question68

Chiral complex from the following is:
Here en = ethylene diamine
[29-Jan-2023 Shift 1]

Options:

A. cis $-[PtCl_2(en)_2]^{2+}$

B. trans - $[PtCl_2(en)_2]^{2+}$

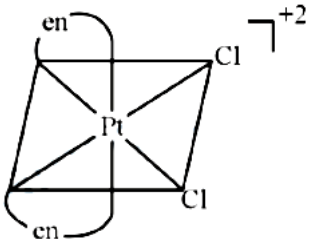
C. cis - $[PtCl_2(NH_3)_2]$

D. trans $-[Co(NH_3)_4Cl_2]^{+}$

Answer: D

Solution:

Solution:



this is chiral complex form

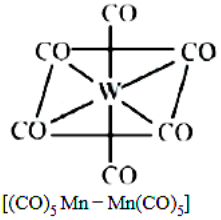
Question69

The sum of bridging carbonyls in $W(CO)_6$ and $Mn_2(CO)_{10}$ is _____.
[29-Jan-2023 Shift 1]

Answer: 0

Solution:

Solution:



$[(CO)_5Mn-Mn(CO)_5]$

Question70

Correct order of spin only magnetic moment of the following complex ions is:
(Given At. No. Fe: 26, Co:27)
[29-Jan-2023 Shift 2]

Correct order of spin only magnetic moment of the following complex ions is: (Given At. No. Fe: 26, Co:27) [29-Jan-2023 Shift 2]

Options:

A. $[FeF_6]^{3-} > [CoF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

B. $[Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-} > [FeF_6]^{3-}$

C. $[FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-}$

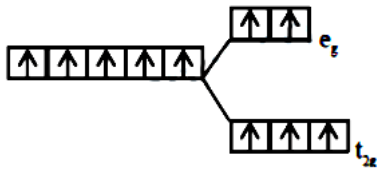
D. $[CoF_6]^{3-} > [FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

Answer: C

Solution:

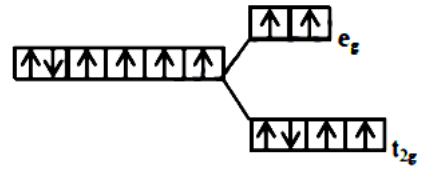
Solution:

$[FeF_6]^{3-}$: $Fe^{3+} = 3d^5 \Delta_0 < P$



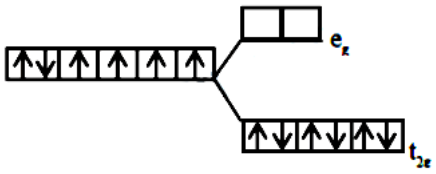
Number of unpaired $e^- = 5 \therefore \mu = \sqrt{35}$ BM

$[CoF_6]^{3-}$: $Co^{3+} = 3d^6 (\Delta_0 < P)$



Number of unpaired $e^- = 4 \therefore \mu = \sqrt{24}$ BM

$[Co(C_2O_4)_3]^{3-}$: $Co^{3+} = 3d^6 (\Delta_O > P)$



Number of unpaired $e^- = 0 \therefore \mu = 0$ BM

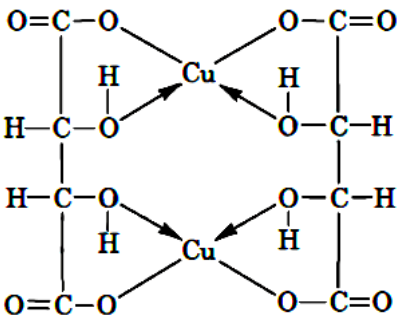
Question71

The denticity of the ligand present in the Fehling's reagent is _____.
Denticity = 2
[29-Jan-2023 Shift 2]

Answer: 4

Solution:

Solution:



Copper tartarate complex

Question72

Which of the following is correct order of ligand field strength?
[30-Jan-2023 Shift 1]

Options:

A. $CO < en < NH_3 < C_2O_4^{2-} < S^{2-}$

B. $S^{2-} < C_2O_4^{2-} < NH_3 < en < CO$

C. $NH_3 < en < CO < S^{2-} < C_2O_4^{2-}$

D. $S^{2-} < NH_3 < en < CO < C_2O_4^{2-}$

Answer: A

Solution:

Solution:
The increasing order of field strength of ligands (according to spectrochemical series)
$S^{2-} < C_2O_4^{2-} < NH_3 < en < CO$

Question73

To inhibit the growth of tumours, identify the compounds used from the following:
(A) EDTA
(B) Coordination Compounds of Pt
(C) D-Penicillamine
(D) Cis - Platin
Choose the correct answer from the option given below:
[30-Jan-2023 Shift 1]

Options:

A. B and D Only

B. C and D Only

C. A and B Only

D. A and C Only

Answer: B

Solution:

Solution:
Cis - Platin is used in chemotherapy to inhibits the growth of tumors. (cis $[Pt(NH_3)_2Cl_2]$)

Question74

The primary and secondary valencies of cobalt respectively in $[Co(NH_3)_5Cl]Cl_2$ are :
[24-Jan-2023 Shift 1]

Options:

A. 3 and 5

B. 2 and 6

C. 2 and 8

D. 3 and 6

Answer: D

Solution:

Solution:
$[Co(NH_3)_5Cl]Cl_2$
Oxidation number of Co is +3.

So primary valency is 3 .
It is an octahedral complex so secondary valency 6 or Co-ordination number 6 .

Question75

The d-electronic configuration of $[CoCl_4]^{2-}$ in tetrahedral crystal field is $e^m t_2^n$. Sum of ' m ' and 'number of unpaired electrons is_____
[24-Jan-2023 Shift 1]

Answer: 7

Solution:

Solution:
Co^{2+} : $3d^7 4s^0$, Cl^- :
Configuration $e^4 t_2^3$: m = 4
Number of unpaired electrons = 3
So, answer = 7

Question76

Which of the following cannot be explained by crystal field theory?
[24-Jan-2023 Shift 2]

Options:

A. The order of spectrochemical series

B. Magnetic properties of transition metal complexes

C. Colour of metal complexes

D. Stability of metal complexes

Answer: A

Solution:

Solution:
Crystal field theory introduce spectrochemical series based upon the experimental values of Δ but can't explain it's order. While other three points are explained by CFT. Specially when the CFSE increases thermodynamic stability of the complex increases.

Question77

Identify the correct statements about alkali metals.
A. The order of standard reduction potential (M^+ | M) for alkali metal ions is Na > Rb > Li.
B. CsI is highly soluble in water.
C. Lithium carbonate is highly stable to heat.
D. Potassium dissolved in concentrated liquid ammonia is blue in colour and paramagnetic.
E. All the alkali metal hydrides are ionic solids.
Choose the correct answer from the options given below
[24-Jan-2023 Shift 2]

Options:

A. A, B, D only

B. C and E only

C. A and E only

D. A, B and E only

Answer: C

Solution:

Solution:
(1) Na > Cs > Li− true { If considered with sign } The low solubility of CsI is due to smaller hydration enthalpy of it's two ions
Li_2CO_3 is highly stable to heat - false
In Conc. NH_3, K formed blue solution - true
All the alkali metal hydrides are ionic solid (True).

Question78

The number of paramagnetic species from the following is _____.

$[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[NiCl_4]^{2-}$

$[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$

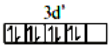
$[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{2+}$

[25-Jan-2023 Shift 1]

Answer: 4

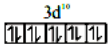
Solution:

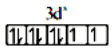
Solution:

$[Ni(CN)_4]^{2-}$: Ni^{+2} =  :

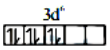
diamagnetic

^-CN : strong field ligand

$[Ni(CO)_4]$: Ni =  : diamagnetic

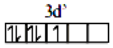
$[NiCl_4]^{2-}$: Ni^{2+} =  : paramagnetic

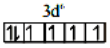
Cl^- : weak field ligand

$[Fe(CN)_6]^{4-}$: Fe^{2+}  : diamagnetic

^-CN : strong field ligand

$[Cu(NH_3)_4]^{+2}$: Cu^{+2} ⇒ one unpaired electron : paramagnetic

$[Fe(CN)_6]^{3-}$: Fe^{+3} :  :
paramagnetic, ^-CN : strong field ligand

$[Fe(H_2O)_6]^{2+}$: Fe^{2+} :  :
paramagnetic H_2O : Weak field ligand

Question79

Which of the following complex will show largest splitting of d-orbitals?
[1-Feb-2023 Shift 1]

Options:

A. $[Fe(C_2O_4)_3]^{3-}$

B. $[FeF_6]^{3-}$

C. $[Fe(CN)_6]^{3-}$

D. $[Fe(NH_3)_6]^{3+}$

Answer: C

Solution:

Solution:
CN is a strong field ligand so maximum splitting in d orbitals take place.

Question80

The complex cation which has two isomers is :
[1-Feb-2023 Shift 2]

Options:

A. $[Co(H_2O)_6]^{3+}$

B. $[Co(NH_3)_5\,Cl]^{2+}$

C. $[Co(NH_3)_5\,NO_2]^{2+}$

D. $[Co(NH_3)_5\,Cl]^{+}$

Answer: C

Solution:

Solution:
$([Co(NH_3)_5\,NO_2]^{2+}$.
Two linkage isomers possible
$NO_2 \rightarrow$ Ambidentate ligand

Question81

The spin only magnetic moment of $[Mn(H_2O)_6]^{2+}$ complexes is _____ B.M. (Nearest integer)
(Given : Atomic no. of Mn is 25)
[1-Feb-2023 Shift 2]

Answer: 6

Solution:

Solution:

($[Mn(H_2O)_6]^{2+}$.
$Mn^{2+} = 3d^5$
$\mu = \sqrt{5(5+2)} = 5.91$ BM

Question82

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The spin only magnetic moment value for $[Fe(CN)_6]^{3-}$ is 1.74 BM, whereas for $[Fe(H_2O)_6]^{3+}$ is 5.92 BM.
Reason R : In both complexes, Fe is present in +3 oxidation state.
In the light of the above statements, choose the correct answer from the options given below:
[6-Apr-2023 shift 1]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

B. A is false but R is true

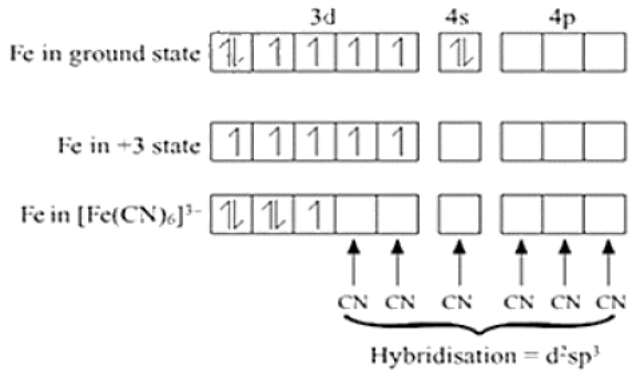
C. A is true but R is false

D. Both A and R are true and R is the correct explanation of A

Answer: A

Solution:

Solution:
$[Fe(CN)_6]^{3-}$



Unpaired electron = 1
$\mu = \sqrt{n(n+2)} = \sqrt{1 \times 3} = 1.74$ B.M.
$[Fe(H_2O)_6]^{3+}$ No pairing because H_2O is WFL
Number of unpaired electrons = 5, $\mu = 5.92$ BM
Assertion is true, Reason is true but not correct explanation.

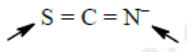
Question83

Number of ambidentate ligands in a representative metal complex $[M(en)(SCN)_4]$ is
[en = ethylenediamine]
[6-Apr-2023 shift 1]

Answer: 4

Solution:

Solution:
$[M(en)(SCN)_4]$



Ambidentate ligand means two ligand site, so ambidentate ligand is SCN^-.
Ans: 4

Question84

The IUPAC name of $K_3[Co(C_2O_4)_3]$ is :-
[6-Apr-2023 shift 2]

Options:

A. Potassium trioxalatocobaltate(III)

B. Potassium tris(oxalato)cobalt(III)

C. Potassium tris(oxalato)cobaltate(III)

D. Potassium trioxalatocobalt(III)

Answer: A

Solution:

Solution:
IUPAC name of $K_3[Co(C_2O_4)_3]$ is Potassium trioxalatocobaltate(III)

Question85

Given below are two statements : one is labelled as "Assertion A" and the other is labelled as "Reason R".
Assertion A : In the complex $Ni(CO)_4$ and $Fe(CO)_5$, the metals have zero oxidation state.
Reason R : Low oxidation states are found when a complex has ligands capable of π-donor character in addition to the σ-bonding.
In the light of the above statements, choose the most appropriate answer from the option given below.
[6-Apr-2023 shift 2]

Options:

A. A is correct but R is not correct

B. A is not correct but R is correct

C. Both A and R are correct but R is NOT the correct explanation of A

D. Both A and R are correct and R is the correct explanation of A.

Answer: A

Solution:

Solution:

Question86

Element not present in Nessler's reagent is:-
[6-Apr-2023 shift 2]

Options:

A. Hg

B. I

C. K

D. N

Answer: D

Solution:

Solution:
Nessler reagent is $-K_2[HgI_4]$

Question87

Which of the following complex is octahedral, diamagnetic and the most stable? [8-Apr-2023 shift 1]

Options:

A. $K_3[Co(CN)_6]$

B. $[Ni(NH_3)_6]Cl_2$

C. $[Co(H_2O)_6]Cl_2$

D. $Na_3[CoCl_6]$

Answer: A

Solution:

Solution:

$K_3[Co(CN)_6]$

$+3+x-6=0$

$x=+3$

$\Downarrow$

$Co^{-3} \rightarrow 3d^6$



$\because CN^-$ is SFL so pairing occur so



u-e $=0$

$\downarrow$

So diamagnetic

Question88

The correct order of spin only magnetic moments for the following complex ions is [8-Apr-2023 shift 1]

Options:

A. $[CoF_6]^{3-} < [MnBr_4]^{2-} < [Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-}$

B. $[Fe(CN)_6]^{3-} < [CoF_6]^{3-} < [MnBr_4]^{2-} < [Mn(CN)_6]^{3-}$

C. $[MnBr_4]^{2-} < [CoF_6]^{3-} < [Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-}$

D. $[Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-} < [CoF_6]^{2-} < [MnBr_4]^{2-}$

Answer: D

Solution:

Solution:

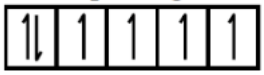
$[CoF_6]^{3-}$

↓

CO^{+3}

F^- WFL

So no pairing



u.e = 4

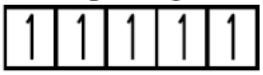
$[MnBr_4]^{2-}$

↓

Mn^{+2}

Br^- WFL

So no pairing



u.e = 5

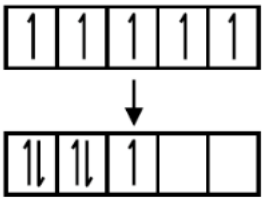
$[Fe(CN)_6]^{3-}$

↓

$3d^5$

CM^- SFL

Pairing occur



u.e = 1

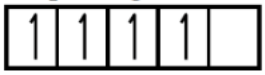
$[Mn(CN)_6]^{3-}$

↓

Mn^{+3}

CN^- is SFL

So pairing occur



u.e = 2

$[Fe(CN)_6]^{-3} < [Mn(CN)_6]^{-3} < [CoF_6]^{-3} < [MnBr_4]^{-2}$

Question89

Match list I with list II

LIST - I Coordination complex	LIST - II Number of unpaired electrons
A. $[Cr(CN)_6]^{3-}$	I. 0
B. $[Fe(H_2O)_6]^{2+}$	II. 3
C. $[Co(NH_3)_6]^{3+}$	III. 2
D. $[Ni(NH_3)_6]^{2+}$	IV. 4

Choose the correct answer from the options given below:
[8-Apr-2023 shift 2]

Options:

A. A-II, B-IV, C-I, D-III

B. A-IV, B-III, C-II, D-I

C. A-II, B-I, C-IV, D-III

D. A-III, B-IV, C-I, D-II

Answer: A

Solution:

Solution:
For option (A)
Cr^{+3} : $3d^3$
CN^- → SFL
⇒ No. of unpaired electrons = 3
For option (B)
Fe^{+2} : $3d^6$
H_2O : WFL
No. of unpaired electrons = 4
For option (C)
Co^{+3} : $3d^6$
NH_3 : SFL
No. of unpaired electrons = 0
For option (D)
Ni^{+2} : $3d^8$
NH_3 : SFL
No. of unpaired electrons = 2

Question90

The observed magnetic moment of the complex $[Mn(x\,NCS)_6]^{x-}$ is 6.06 BM. The numerical value of x is _____.
[8-Apr-2023 shift 2]

Answer: 4

Solution:

Solution:
$[Mn(NCS)_6]^{x-}$
Number of unpaired electron = 5
So, Mn must be in +2 oxidation state (Mn^{+2})
$\Rightarrow 2+(-6)=-x \quad \Rightarrow -4=-x \quad \Rightarrow x=4$

Question91

The sum of oxidation state of the metals in $Fe(CO)_5$, VO^{2+} and WO_3 is _____
[8-Apr-2023 shift 2]

Answer: 10

Solution:

Solution:
$\overset{(0)}{Fe}(CO)_5 \quad \overset{(+4)}{V}O^{2+} \quad \overset{(+6)}{W}O_3$
So, Sum of oxidation state $= 0 + 4 + 6 = 10$

Question92

The octahedral diamagnetic low spin complex among the following is
[10-Apr-2023 shift 1]

Options:

A. $[CoF_6]^{3-}$

B. $[CoCl_6]^{3-}$

C. $[Co(NH_3)_6]^{3+}$

D. $[NiCl_4]^{2-}$

Answer: C

Solution:

Solution:
(1) Paramagnetic, High Spin & Tetrahedral
(2) Paramagnetic, High Spin & Octahedral
(3) Paramagnetic, High Spin & Octahedral
(4) Diamagnetic, Low Spin & Octahedral
$[Co(NH_3)_6]^{3+}$, CN = 6 CN = 6 (Octahedral)
NH_3 = SFL
$Co^{+3} = [Ar]3d^6$



Diamagnetic & Low spin complex

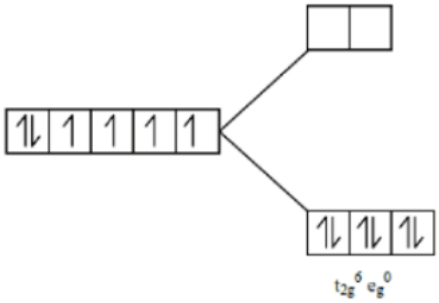
Question93

In potassium ferrocyanide, there are _____ pairs of electrons in the t_{2g} set of orbitals.
[10-Apr-2023 shift 1]

Answer: 3

Solution:

Solution:



$Fe^{+2} = [Ar]3d^6$

$CN^- = SFL$

t_{2g} contain 6 electron so it become 3 pairs

Question94

Match List I with List II

LIST I Complex	LIST II Crystal Field splitting energy (Δ_0)
A $[Ti(H_2O)_6]^{2+}$	I. -1.2
B $[V(H_2O)_6]^{2+}$	II. -0.8
C $[Mn(H_2O)_6]^{3-}$	III. 0
D $[Fe(H_2O)_6]^{3+}$	IV. -0.8

Choose the correct answer from the options given below: [10-Apr-2023 shift 2]

Options:

A. A-IV, B-I, C-II, D-III

B. A-IV, B-I, C-III, D-II

C. A-II, B-IV, C-III, D-I

D. A-II, B-IV, C-I, D-III

Answer: A

Solution:

Solution:

$$[Ti(H_2O)_6]^{+2}\ [V(H_2O)_6]^{+2}\ [Mn(H_2O)_6]^{+3}\ [Fe(H_2O)_6]^{3+}$$

$\downarrow$

Ti^{+2} V^{+2} Mn^{+3} Fe^{+3}

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

$3d^2$ $3d^3$ $3d^4$ $3d^5$

$CFSE = -0.4 \times t_{2g} + 0.6 \times eg + xp$

$= -0.4 \times 2 + 0.6 \times 0 + xp$

(A) $= -0.8 \rightarrow Ti^{+2}$

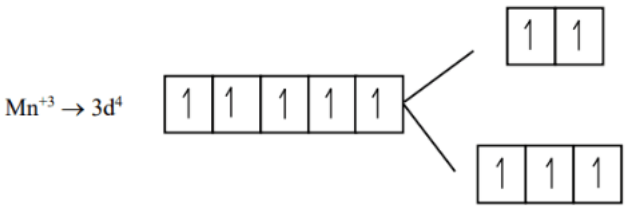
(B) $V^{+2} \rightarrow 3d^3$

$CFSE = -0.4 \times t_{2g} + 0.6 \times eg + xp$

$= -0.4 \times 3 + 0.6 \times 0 + xp$

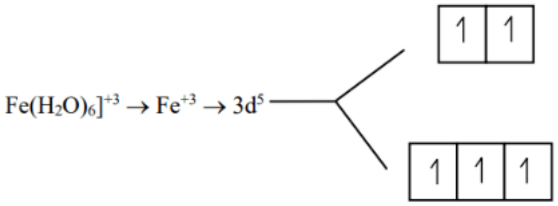
$= -1.2$

(C)



$CFSE = -0.4 \times t_{2g} + 0.6 \times eg + xp$
$-0.4 \times 3 + 0.6 \times 1 + xp$
$= -1.2 + 0.6 = 0.6$

(D)



$CFSE = -0.4 \times t_{2g} + 0.6 \times eg + xp$
$= -0.4 \times 3 + 0.6 \times 2$
$= -1.2 + 1.2$
$= 0$

Question95

The correct order of the number of unpaired electrons in the given complexes is

A. $[Fe(CN)_6]^{3-}$

B. $[FeF_6]^{3-}$

C. $[CoF_6]^{3-}$

D. $[Cr(\ oxalate\ )_3]^{3-}$

E. $[Ni(CO)_4]$

Choose the correct answer from the options given below:
[10-Apr-2023 shift 2]

Options:

A. E < A < D < C < B

B. A < E < C < B < D

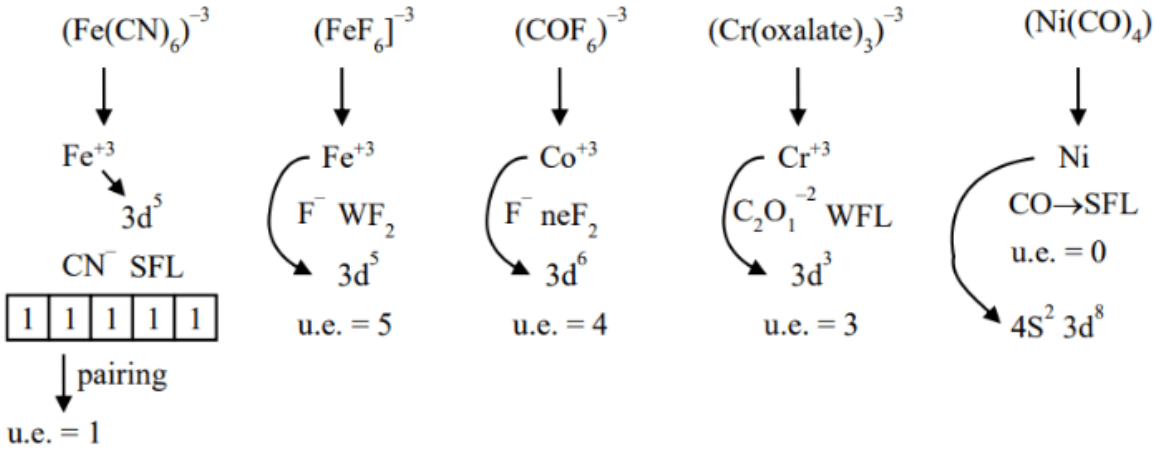
C. A < E < D < C < B

D. E < A < B < D < C

Answer: A

Solution:

Solution:



Question96

The complex that dissolves in water is
[11-Apr-2023 shift 1]

Options:

A. $[Fe_3(OH)_2(OAc)_6]\,Cl$

B. $Fe_4[Fe(CN)_6]_3$

C. $K_3[Co(NO_2)_6]$

D. $(NH_4)_3[As(Mo_3O_{10})_4]$

Answer: A

Solution:

Solution:
$Fe_4[Fe(CN)_6]_3$ Prussian Blue-water insoluble
$K_3[Co(NO_2)_6]$ very poorly water soluble
$(NH_4)_3[As(MO_3O_{10})_4]$ water insoluble
ammonium arseno molybdate
$[Fe_3(OH)_2(OAc)_6]\,Cl$ is water soluble.

Question97

The set which does not have ambidentate ligand (s) is
[11-Apr-2023 shift 1]

Options:

A. $C_2O_4^{2-}$, NO_2^-, NCS^-

B. $EDTA^{4-}$, NCS^-, $C_2O_4^{2-}$

C. NO_2^-, $C_2O_4^{2-}$, $EDTA^{4-}$

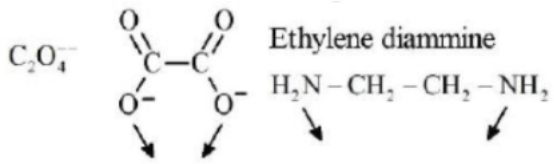
D. $C_2O_4^{2-}$, ethylene diamine, H_2O

Answer: D

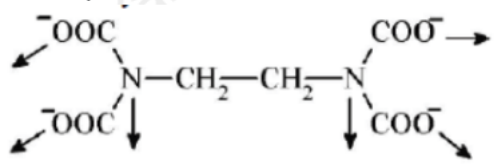
Solution:

Solution:

NO_2^- NCS^- are ambidentate ligand



EDTA Ethylene diamine tetra acetate



Question98

Which of the following complex has a possibility to exist as meridional isomer? [11-Apr-2023 shift 1]

Options:

A. $[Co(en)_2 Cl_2]$

B. $[Pt(NH_3)_2 Cl_2]$

C. $[Co(en)_3]$

D. $[Co(NH_3)_3(NO_2)_3]$

Answer: D

Solution:

Solution:

$[MA_3B_3]$ type of compound exists as facial and meridional isomer.



Question99

The ratio of spin-only magnetic moment values μ_{eff} $[Cr(CN)_6]^{3-}$ / μ_{eff} $[Cr(H_2O)_6]^{3+}$ is _____. [11-Apr-2023 shift 1]

Answer: 1

Solution:

Solution:

Spin magnetic moment of $[Cr(CN)_6]^{3-}(t_{2g}^3 eg_g^0)$

$\mu_1 = \sqrt{3(3+2)} = \sqrt{15}$ BM

Spin magnetic moment of $[Cr(H_2O)_6]^{3+}(t_{2g}^3 eg_g^0)$

$\mu_2 = \sqrt{3(3+2)} = \sqrt{15}$ BM

$\frac{\mu_1}{\mu_2} = \frac{\sqrt{51}}{\sqrt{51}} = 1$

Question100

If Ni^{2+} is replaced by Pt^{2+} in the complex $[NiCl_2Br_2]2^-$, which of the following properties are expected to get changed?
A. Geometry
B. Geometrical isomerism
C. Optical isomerism
D. Magnetic properties
[11-Apr-2023 shift 2]

Options:

A. A, B and C

B. A and D

C. B and C

D. A, B and D

Answer: D

Solution:

Solution:
$[NiBr_2Cl_2]^{2-}$ → This complex species is tetrahedral as $Br^{\ominus}$&$Cl^{\ominus}$ are weak field ligands.
$[PtBr_2Cl_2]^{2-}$ → As Pt belongs to 5d series. This complex species is square planar.
Both the complex species are optically inactive.
$[NiBr_2Cl_2]^{2-}$, being tetrahedral does not show Geometrical Isomerism.
$[PtBr_2Cl_2]^{2-}$ shows two Geometrical Isomers.

Question101

Match List I with List II

List I Complex	List I Colour
A. $Mg(NH_4)PO_4$	I. Brown
B. $K_3[Co(NO_2)_6]$	II. White
C. $MnO(OH)_2$	III. Yellow
D. $Fe_4[Fe(CN)_6]_3$	IV. blue

Choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-II, B-IV, C-I, D-III

C. A-III, B-IV, C-II, D-I

D. A-II, B-III, C-I, D-IV

Answer: D

Solution:

Solution:
$Mg(NH_4)PO_4 \Rightarrow$ White
$K_3[Co(NO_2)_6] \Rightarrow$ Yellow
$MnO(OH)_2 \Rightarrow$ Brown
$Fe_4[Fe(CN)_6]_3 \Rightarrow$ Blue

Question102

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : $[CoCl(NH_3)_5]^{2+}$ absorbs at lower wavelength of light with respect to $[CoCl(NH_3)_5(H_2O)]^{3+}$
Reason R : It is because the wavelength of the light absorbed depends on the oxidation state of the metal ion.
In the light of the above statements, choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

B. A is true but R is false

C. Both A and R are true and R is the correct explanation of A

D. A is false but R is true

Answer: D

Solution:

Solution:
Since H_2O is strong field ligand compared to chloride and Co^{3+} ion is present.
$\therefore$ CFSE is higher for $[Co(NH_3)_5H_2O]^{+3}$, hence it will absorb at lower wavelength.

Question103

The magnetic moment is measured in Bohr Magneton (BM). Spin only magnetic moment of Fe in $[Fe(H_2O)_6]^{3+}$ and $[Fe(CN)_6]^{3-}$ complexes respectively is:
[11-Apr-2023 shift 2]

Options:

A. 3.87 B. M. and 1.732 B.M.

B. 6.92 B.M. in both

C. 5.92 B.M. and 1.732 B.M.

D. 4.89 B.M. and 6.92 B.M.

Answer: C

Solution:

Solution:

$[Fe(H_2O)_6]^{3+}$

$Fe^{3+} \Rightarrow [Ar]3d^5 4s^0$

No pairing

1	1	1	1	1

$\therefore$ Unpaired $e^- = 5$

$\mu = \sqrt{n(n+2)}$

$= \sqrt{5(5+2)}$

$\mu = \sqrt{35} = 5.92$ B.M.

$[Fe(CN)_6]^{-3}$

1↓	1↓	1	1	1

$\therefore$ Unpaired $e^- = 1$

$\mu = \sqrt{n(n+2)}$

$= \sqrt{1(1+2)} = \sqrt{3} = 1.732$ B.M.

Question104

Match List I with List II

LIST I Complex		LIST II CFSE (Delta(Δ_o)	
A.	$[Cu(NH_3)_6]^{2+}$	I.	−0.6
B.	$[Ti(H_2O)_6]^{3+}$	II.	-2.0
C.	$[Fe(CN)_6]^{3-}$	III.	-1.2
D.	$[NiF_6]^{4-}$	IV.	-0.4

Choose the correct answer from the options given below: [12-Apr-2023 shift 1]

Options:

A. A-III, B-IV, C-I, D-II

B. A-II, B-III, C-I, D-IV

C. A-I, B-IV, C-II, D-III

D. A-I, B-II, C-IV, D-III

Answer: C

Solution:

Solution:

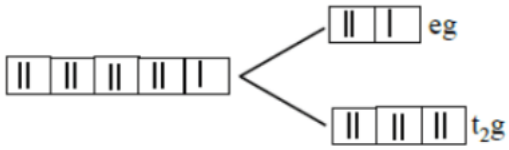
$(Cu(NH_3)_6)^{+2}$

↑

Cu+2

↑

3d9



$CSf: -0.4 \times 6 + 3 \times 0.6 + xp$

$-2.4 + 1.8$

$= -0.6$

$Ti(H_2O)_6^{+3}$

↓ , $CFse = -0.4$

$Ti^{+3} \rightarrow 3d^1$

$Ti(H_2O)_6^{+3}$

↓

$Ti^{+3} \rightarrow 3d^1$

Question105

The mismatched combinations are
A. Chlorophyll - Co
B. Water hardness - EDTA
C. Photography $-[Ag(CN)_2]^-$
D. Wilkinson catalyst - $[(Ph_3P)_3$ RhCl]
E. Chelating ligand - D-Penicillamine
Choose the correct answer from the options given below :
[13-Apr-2023 shift 1]

Options:

A. A and C Only

B. D and E Only

C. A and E Only

D. A, C, and E Only

Answer: A

Solution:

Solution:
Mg is present in chlorophyll and in black and white photography the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion $[Ag(S_2O_3)_2]^{3-}$

Question106

The covalency and oxidation state respectively of boron in $[BF_4]^-$, are :
[13-Apr-2023 shift 2]

Options:

A. 3 and 5

B. 4 and 3

C. 4 and 4

D. 3 and 4

Answer: B

Solution:

Solution:



Number of covalent bond formed by Boron is 4
Oxidation number of fluorine is -1 ,
Oxidation number of $B + 4 \times (-1) = -1$,
Thus, Oxidation number of $B = +3$

Question107

Which of the following complexes will exhibit maximum attraction to an applied magnetic field? [13-Apr-2023 shift 2]

Options:

A. $[Zn(H_2O)_6]^{2+}$

B. $[Ni(H_2O)_6]^{2+}$

C. $[Co(en)_3]^{3+}$

D. $[Co(H_2O)_6]^{2+}$

Answer: D

Solution:

Solution:
Complex with maximum number of unpaired electron will exhibit maximum attraction to an applied magnetic field
$[Zn(H_2O)_6]^{2+} \rightarrow d^{10}$ system $\rightarrow t_{2g}^{6}$ eg ^{4}, 0 unpaired e^-
$[Co(H_2O)_6]^{2+} \rightarrow d^{7}$ system $\rightarrow t_{2g}^{5}$ eg ^{2}, 3 unpaired e^-
$[Co(en)_3]^{3+} \rightarrow d^{6}$ system $\rightarrow t_{2g}^{6} eg^{0}$, 0 unpaired e^-
$[Ni(H_2O)_6]^{2+} \rightarrow d^{8}$ system $\rightarrow t_{2g}^{6}$ eg ^{2}, 2 unpaired e^-

---

Question108

The total number of stereoisomers for the complex $[Cr(ox)_2 ClBr]^{3-}$(where ox = oxalate) is : [13-Apr-2023 shift 2]

Options:

A. 3

B. 1

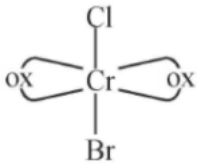
C. 4

D. 2

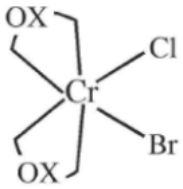
Answer: A

Solution:

Solution:
$[Cr(Ox)_2 ClBr]^{-3}$
- No. of isomers-



- This structure has plane of symmetry, So no optical isomerism will be shown.



- This structure does not contain plane of symmetry, So two forms d as well as 1 will be shown.

Question109

The complex with highest magnitude of crystal field splitting energy (Δ_0) is [15-Apr-2023 shift 1]

Options:

A. $[Mn(OH_2)_6]^{3+}$

B. $[Fe(OH_2)_6]^{3+}$

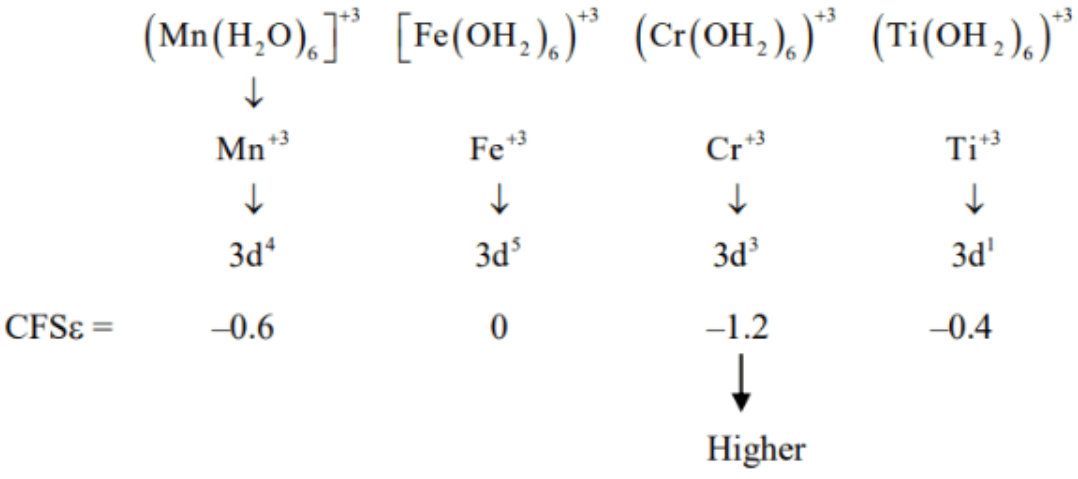
C. $[Cr(OH_2)_6]^{3+}$

D. $[Ti(OH_2)_6]^{3+}$

Answer: C

Solution:

Solution:



Question110

The volume (in mL) of 0.1M$AgNO_3$ required for complete precipitation of chloride ions present in 20 mL of 0.01M solution of $[Cr(H_2O)_5Cl]Cl_2$ as silver chloride is _____.
[15-Apr-2023 shift 1]

Answer: 4

Solution:

Solution:
$[Cr(H_2O)_5Cl]Cl_2 + 2AgNO_3 \rightarrow$
0.01M, 20 mL 0.1M
For 0.2 milimole $AgNO_3$ required
= 0.4 milimole
0.4 = 0.1 × V(ml)
V = 4 mL

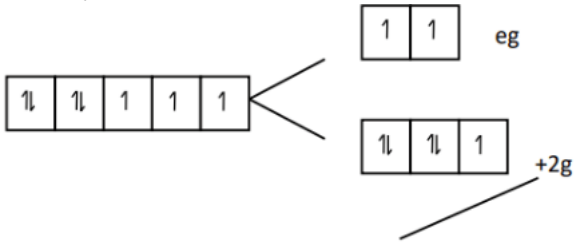
Question111

The homoleptic and octahedral complex of Co^{2+} and H_2O has _____ unparied electrons(s) in the t_{2g} set of orbitals.
[15-Apr-2023 shift 1]

Answer: 1

Solution:

Solution:
$(Co(H_2O)_6)^{+2} \rightarrow CO^{+2} \rightarrow 3d^7$



u.e. in t_2g is = (1)

Question112

List-1		List -II	
(A)	$[PtCl_4]^{2-}$	I	sp^3d
(B)	BrF_5	II	d^2sp^3
(C)	PCl_5	III	dsp^2
(D)	$[CoNH_{36}]^{3+}$	IV	sp^3d^2

Choose the most appropriate answer from the options given below :

[24-Jun-2022-Shift-1]

Options:

A. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

B. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

C. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

D. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Answer: B

Solution:

Solution:

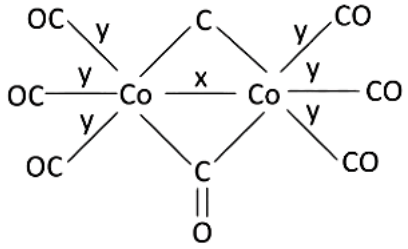
Question113

In the cobalt-carbonyl complex : $[CO_2(CO)_8]$, number of Co − Co bonds is "X" and terminal CO ligands is "Y". X + Y =
[24-Jun-2022-Shift-1]

Answer: 7

Solution:

Solution:



x = 1
y = 6
∴x + y = 7

Question114

Transition metal complex with highest value of crystal field splitting (Δ_0) will be
[24-Jun-2022-Shift-2]

Options:

A. $[Cr(H_2O)_6]^{3+}$

B. $[Mo(H_2O)_6]^{3+}$

C. $[Fe(H_2O)_6]^{3+}$

D. $[Os(H_2O)_6]^{3+}$

Answer: D

Solution:

Solution:
Crystal field splitting (Δ_0) for octahedral complexes depends on oxidation state of the metal as well as to which transition series the metal belongs. For the same oxidation state, the crystal field splitting (Δ_0) increases as we move from $3d \rightarrow 4d \rightarrow 5d$

Cr^{3+} and Fe^{3+} belong to $3d$ series, Mo^{3+} belongs to $4d$ series and Os^{3+} belongs to $5d$ series.
Therefore crystal field splitting (Δ_0) is highest for $[Os(H_2O)_6]^{3+}$.

Question115

White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of : [25-Jun-2022-Shift-1]

Options:

A. $[Ag(NH_3)_4]Cl_2$

B. $[Ag(Cl)_2(NH_3)_2]$

C. $[Ag(NH_3)_2]\,Cl$

D. $[Ag(NH_3)\,Cl]\,Cl$

Answer: C

Solution:

Solution:
$AgCl + 2NH_3 \rightarrow \underset{\text{Soluble}}{[Ag(NH_3)_2]^+Cl^-}$

Question116

If $[Cu(H_2O)_4]^{2+}$ absorbs a light of wavelength 600 nm for d − d transition, then the value of octahedral crystal field splitting energy for $[Cu(H_2O)_6]^{2+}$ will be ____$\times 10^{-21}$J. [Nearest Integer] (Given : h = 6.63×10^{-34} Js and c = $3.08 \times 10^8 ms^{-1}$) [25-Jun-2022-Shift-1]

Answer: 745

Solution:

Solution:
$[Cu(H_2O)_4]^{2+}$ is tetrahedral
$[Cu(H_2O)_6]^{2+}$ is octahedral
$\because \Delta_t = \frac{4}{9} \times \Delta_0$
$\Delta_t = \frac{6.63 \times 10^{-34} \times 3.08 \times 10^8}{600 \times 10^{-9}}$
$\Delta_0 = \frac{9}{4} \times \frac{6.63 \times 10^{-34} \times 3.08 \times 10^8}{600 \times 10^{-9}} \approx 765 \times 10^{-21} J$

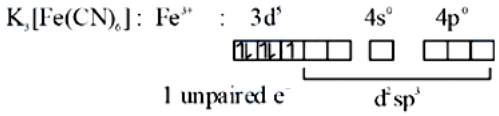
Question117

Amongst $FeCl_3 \cdot 3H_2O$, $K_3[Fe(CN)_6]$ and $[Co(NH_3)_6]Cl_3$, the spin-only magnetic moment value of the innerorbital complex that absorbs light at shortest wavelength is_____ B.M. [nearest integer] [25-Jun-2022-Shift-2]

Answer: 2

Solution:

Solution:
$[Fe(H_2O)_3Cl_3]$, $\underset{\text{inner orbital complexes}}{K_3[Fe(CN)_6]}$, $[Co(NH_3)_6]Cl_3$
$K_3[Fe(CN)_6]$ has more value of Δ_0 than that of $[Co(NH_3)_6]Cl_3$; as $\overline{CN}$ is stronger ligand.
More $\Delta_0 \Rightarrow$ smaller value of absorbed λ



Spin only magnetic moment $(\mu) = \sqrt{3}$ BM $= 1.732$ BM
Rounding off $\Rightarrow 2$

Question118

Which statement is not true with respect to nitrate ion test? [26-Jun-2022-Shift-1]

Options:

A. A dark brown ring is formed at the junction of two solutions.

B. Ring is formed due to nitroferrous sulphate complex.

C. The brown complex is $[Fe(H_2O)_5(NO)]SO_4$.

D. Heating the nitrate salt with conc. H_2SO_4, light brown fumes are evolved.

Answer: B

Solution:

Solution:
Brown ring test
$NO_3^- + 3Fe^{+2} + 4H^+ \rightarrow NO + 3Fe^{+3} + 2H_2O$
$[Fe(H_2O)_6]^{2+} NO \rightarrow \underset{\text{Brown ring}}{[Fe(H_2O)_5 NO]^{2+}} H_2O$

Question119

The spin-only magnetic moment value of an octahedral complex among $CoCl_3 \cdot 4NH_3$, $NiCl_2 \cdot 6H_2O$ and $PtCl_4 \cdot 2HCl$, which upon reaction with excess of $AgNO_3$ gives 2 moles of AgCl is ____B.M. (Nearest integer) [26-Jun-2022-Shift-1]

Answer: 3

Solution:

Solution:

$CoCl_3 \cdot 4NH_3 \xrightarrow[\text{excess}]{AgNO_3} [Co(NH_3)_4 \cdot Cl_2] + AgCl$

$NiCl_2 \cdot 6H_2O \xrightarrow[\text{excess}]{AgNO_3} [Ni(H_2O)_6]^{2+} + 2\,AgCl$

$PtCl_4 \cdot 2\,HCl \longrightarrow [PtCl_6]^{4-} + \text{No AgCl ppt}$

$[Ni^{+2}(H_2O)_6] = d^8 = t2_g e_g^{\,2} = 2$ unpaired electrons

Magnetic moment $= \sqrt{2(2+2)}$

$2\sqrt{8} \approx 3$

Question120

Reaction of $[Co(H_2O)_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g}-orbitals of the product is__
[26-Jun-2022-Shift-2]

Answer: 6

Solution:

Solution:

$[Co(H_2O)_6]^{2+} + NH_3\ (\text{excess}) \rightarrow [Co(NH_3)_6]^{3+} + 6H_2O$

Diamagnetic

↓

Low spin complex

$Co^{3+} \Rightarrow 3d^6 4s^0$

$\Rightarrow t_{2g}^{\,6} e_g^{\,0}$

Total number electrons $= 6$

Question121

Which of the following will have maximum stabilization due to crystal field?
[27-Jun-2022-Shift-1]

Options:

A. $[Ti(H_2O)_6]^{3+}$

B. $[Co(H_2O)_6]^{2+}$

C. $[Co(CN)_6]^{3-}$

D. $[Cu(NH_3)_4]^{2+}$

Answer: C

Solution:

Solution:

The given complexes are:

$[Ti(H_2O)_6]^{3+}$, $[Co(H_2O)_6]^{2+}$, $[Co(CN)_6]^{-3}$, $[Cu(NH_3)_4]^{2+}$

CN^- is the strongest ligand among the given complexes CFSE value for the $[Co(CN)_6]^{-3}$ complex will be highest as it has d^6 configuration with a CFSE value of $-2.40\ \Delta_0 + 2P$, where P represents pairing energy and Δ_0 represents splitting energy in octahedral field.

The value of Δ_0 is high for cyanide complexes.

Question122

Arrange the following coordination compounds in the increasing order of magnetic moments. (Atomic numbers : Mn = 25; Fe = 26)

A. $[FeF_6]^{3-}$

B. $[Fe(CN)_6]^{3-}$

C. $[MnCl_6]^{3-}$ (high spin)

D. $[Mn(CN)_6]^{3-}$

Choose the correct answer from the options given below :
[27-Jun-2022-Shift-2]

Options:

A. A < B < D < C

B. B < D < C < A

C. A < C < D < B

D. B < D < A < C

Answer: B

Solution:

Solution:

Coordination Compound	Number of unpaired e^-(n)	Magnetic moment (μ)(B . M)
A. $[FeF_6]^{3-} - d^5$	5	5.91
B. $[Fe(CN)_6]^{3-} - d^5$	1	1.73
C. $[MnCl]_6]^{3-} - d^4$	4	4.89
D. $[Mn(CN)_6]^{3-} - d^4$	2	2.82

Hence, correct order of magnetic moment is $2 < 4 < 3 < 1$

Question123

Given below are two statements :
Statement I : $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[Ni(CO)_4]$ is tetrahedral, paramagnetic and with sp^3-hybridication for Ni.

Statement II : $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ both have same d-electron configuration have same geometry and are paramagnetic.
In light the above statements, choose the correct answer from the options given below :
[28-Jun-2022-Shift-1]

Options:

A. Both Statement I and Statement II are true.

B. Both Statement I and Statement II are false.

C. Statement I is correct but Statement II is false.

D. Statement I is false but Statement II is correct.

Answer: B

Solution:

Solution:
$[Ni(CN)_4]^{2-}$: d^8 configuration, SFL, sq. planar splitting (dsp^2), diamagnetic.
$[Ni(CO)_4]$: d^{10} config (after excitation), SFL, tetrahedral splitting (sp^3), diamagnetic.
$[NiCl_4]^{2-}$: d^8 config, WFL, tetrahedral splitting (sp^3), paramagnetic (2. unpaired e^-).

Question124

Number of complexes which will exhibit synergic bonding amongst, $[Cr(CO)_6]$, $[M n(CO)_5]$ and $[M n_2(CO)_{10}]$ is___
[28-Jun-2022-Shift-1]

Answer: 3

Solution:

Solution:
Carbonyl complex compounds have tendency to show synergic bonding.

Question125

$[Fe(CN)_6]^{4-}$
$[Fe(CN)_6]^{3-}$
$[Ti(CN)_6]^{3-}$
$[Ni(CN)_4]^{2-}$
$[Co(CN)_6]^{3-}$
Among the given complexes, number of paramagnetic complexes is____
[28-Jun-2022-Shift-2]

Answer: 2

Solution:

Solution:
$[Fe(CN)_6]^{4-}$ Diamagnetic
$[Fe(CN)_6]^{3-}$ Paramagnetic (1 unpaired electron)
$[Ti(CN)_6]^{3-}$ Paramagnetic (1 unpaired electron)
$[Ni(CN)_4]^{2-}$ Diamagnetic
$[Co(CN)_6]^{3-}$ Diamagnetic

Question126

(a) $CoCl_3 \cdot 4NH_3$,

(b) $CoCl_3.5NH_3$,

(c) $CoCl_3 \cdot 6NH_3$ and

(d) $CoCl(NO_3)_2 \cdot 5NH_3$.

Number of complex(es) which will exist in cis-trans form is/are_____
[28-Jun-2022-Shift-2]

Answer: 1

Solution:

Solution:
$CoCl_3 \cdot 4NH_3 \Rightarrow [Co(NH_3)_4\,Cl_2]\,Cl$
$CoCl_3 \cdot 5NH_3 \Rightarrow [Co(NH_3)_5\,Cl]Cl_2$
$CoCl_3 \cdot 6NH_3 \Rightarrow [Co(NH_3)_6]Cl_3$
Only $[Co(NH_3)_4\,Cl_2]$ can show geometrical isomerism. Hence can exist in cis-trans form.

Question127

Given below are two statements.
- Statement I : In $CuSO_4 \cdot 5H_2O$, Cu − O bonds are present.

- Statement II : In $CuSO_4 \cdot 5H_2O$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the correct answer from the options given below.
[29-Jun-2022-Shift-2]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

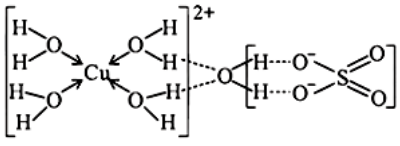
C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: C

Solution:

Solution:
Statement I is true but statement II is false. Only oxygen atom forms a Co-ordinate bond with Cu^{+2} in $CuSO_4 \cdot 5H_2O$



Question128

Spin only magnetic moment of $[MnBr_6]^{4-}$ is _____ B.M. (round off to the closest integer)
[29-Jun-2022-Shift-2]

Answer: 6

Solution:

Solution:
$Mn^{2+} \rightarrow t_{2g^{111}}e_{g^{11}}$
$\mu_s = \sqrt{35}$
$= 5.91$
$= 6$

Question129

Consider the following metal complexes:
$[Co(NH_3)_6]^{3+}$
$[CoCl(NH_3)_5]^{2+}$
$[Co(CN)_6]^{3-}$
$[Co(NH_3)_5(H_2O)]^{3+}$
The spin-only magnetic moment value of the complexthat absorbes light with shortest wavelength is B. M. (Nearest integer)
[25-Jul-2022-Shift-1]

Answer: 0

Solution:

Solution:
$\Delta_0 \propto \frac{1}{\lambda}$
Here, CN^- being SFL will have maximum CFSE
So, $[Co(CN)_6]^{3-}$ will be $d^2 sp^3$, $\mu = 0$

Question130

The correct order of energy of absorption for the following metal complexes is
A : $[Ni(en)_3]^{2+}$, B : $[Ni(NH_3)_6]^{2+}$, C : $[Ni(H_2O)_6]^{2+}$
[25-Jul-2022-Shift-2]

Options:

A. C < B < A

B. B < C < A

C. C < A < B

D. A < C < B

Answer: A

Solution:

Solution:
Stronger is ligand attached to metal ion, greater will be the splitting between t_2g and eg (hence greater will be ΔU)
$\therefore$ greater will be absorption of energy.
Hence correct order
$[Ni(en)_3]^{2+} > [Ni(NH_3)_6]^{2+} > [Ni(H_2O)_6]^{2+}$

Question131

The difference between spin only magnetic moment values of $[Co(H_2O)_6]Cl_2$ and $[Cr(H_2O)_6]Cl_3$ is _____.
[26-Jul-2022-Shift-1]

Answer: 0

Solution:

Solution:
$Co \rightarrow 4s^2 3d^7$
H_2O is weak field ligand.
$Co^{+2} \rightarrow 3d^7$
n = 3 μ_1 \;\; =√{n({\n}+2)} $
= $\sqrt{15}$ B.M. $
$Cr \rightarrow 4s^1 3d^5$
$Co^{+3} \rightarrow 3d^3$
n = 3 $\mu_2 = \sqrt{15}$ B.M.
$\mu_1 - \mu_2 = 0$

Question132

The metal complex that is diamagnetic is (Atomic number: Fe, 26; Cu, 29)
[26-Jul-2022-Shift-2]

Options:

A. $K_3[Cu(CN)_4]$

B. $K_2[Cu(CN)_4]$

C. $K_3[Fe(CN)_4]$

D. $K_4[FeCl_6]$

Answer: A

Solution:

Solution:
⇒$K_3[Cu(CN)_4]$ is diamagnetic
Cu(I) ⇒ d^{10} configuration ⇒ No unpaired electrons.
⇒$K_2[Cu(CN)_4]$, $K_3[Fe(CN)_4]$ and $K_4[FeCl_6]$ are paramagnetic in nature

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Question133

The conductivity of a solution of complex with formula $CoCl_3(NH_3)_4$ corresponds to 1 : 1 electrolyte, then the primary valency of central metal ion is _____.
[27-Jul-2022-Shift-1]

Answer: 3

Solution:

Solution:
In 1: 1 type of electrolyte the ions have $+1$ and -1 charge on them
$\therefore$ Possible compound is $\rightarrow[Co(NH_3)_4Cl_2]^+Cl^-$
Oxidation state of central atom represents the total number of primary valency
$\therefore$ Primary valency will be 3 .

Question134

Low oxidation state of metals in their complexes are common when ligands: [27-Jul-2022-Shift-2]

Options:

A. have good π-accepting character

B. have good σ-donor character

C. are having good π-donating ability

D. are having poor σ-donating ability

Answer: A

Solution:

Solution:
Ligands like :CO, are sigma donor and π-acceptor and they make stronger bond with lower oxidation state metal ion, in this case back bonding is more effective

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Question135

Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of : [27-Jul-2022-Shift-2]

Options:

A. $[Fe(H_2O)_6]_2[Fe(CN)_6]$

B. $Fe_2[Fe(CN)_6]_2$

C. $Fe_3[Fe(OH)_2(CN)_4]_2$

D. $Fe_4[Fe(CN)_6]_3$

Answer: D

Solution:

Solution:
$4Fe^{3+} + 3[Fe(CN)_6]^{-4} \rightarrow Fe_4[Fe(CN)_6]_3$

Question136

The spin only magnetic moment of the complex present in Fehling's reagent is _____ B.M. (Nearest integer). [27-Jul-2022-Shift-2]

Answer: 2

Solution:

Solution:
Fehling solution is a complex of Cu^{++}
$Cu^{++} = 3d^9$
No. of unpaired $e^- = 1$
M.M $= \sqrt{1(1+2)} = \sqrt{3} = 1.73$ BM

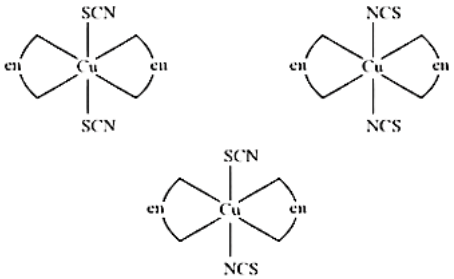
Question137

Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be _____.
[28-Jul-2022-Shift-1]

Answer: 3

Solution:

Solution:
$[Cu(en)_2(SCN)_2]$



Question138

Match List I with List II

List - I (Complex)	List - II (Hybridization)
(A) $Ni(CO)_4$	(I) sp^3
(B) $[Ni(CN)_4]^{2-}$	(II) sp^3d^2
(C) $[Co(CN)_6]^{3-}$	(III) d^2sp^3
(D) $[CoF_6]^{3-}$	(IV) dsp^2

Choose the correct answer from the options given below:
[28-Jul-2022-Shift-2]

Options:

A. A-IV, B-I, C-III, D-II

B. A-I, B-IV, C-III, D-II

C. A-I, B-IV, C-II, D-III

D. A-IV, B-I, C-II, D-III

Answer: B

Solution:

Solution:
$Ni(CO)_4$ Hybridisation sp^3
$[Ni(CN)_4]^{2-}$ Hybridisation dspe2
$[Co(CN)_6]^{3-}$ Hybridisation $d^2 sp^3$
$[Co(F)_6]^{3-}$ Hybridisation sp^3d^2

Question139

$[Fe(CN)_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is (−) _____ Δ_0. (Nearest integer)
[29-Jul-2022-Shift-1]

Answer: 2

Solution:

Solution:
In $[Fe(CN)^6]^{3-}$, Fe is present in (+3) oxidation state $Fe(III) \Rightarrow$ inner orbital complex $\Rightarrow d^5$ (with pairing)
Configuration $\Rightarrow t_{2g}^5$
$CFSE = 5\times \frac{-2}{5}\Delta_0 = -2\Delta_0$

Question140

Octahedral complexes of copper(II) undergo structural distortion (Jahn-Teller). Which one of the given copper (II) complexes will show the maximum structural distortion? (en - ethylenediamine; $.H_2N_CH_2 - CH_2 - NH_2$)
[29-Jul-2022-Shift-2]

Options:

A. $[Cu(H_2O)_6]SO_4$

B. $[Cu(en)(H_2O)_4]SO_4$

C. cis- $[Cu(en)_2 Cl_2]$

D. trans$-[Cu(en)_2 Cl_2]$

Answer: A

Solution:

Solution:
According to John teller any nonlinear molecular system in a degenerate electronic state will be unstable and will undergo some kind of distortion which will lower its symmetry and energy and split the degenerate state.
In case of octahedral d^9 configuration, the last electron may occupy either d^2 or $d_{x^2y^2}$ orbitals of e_g set.

If it occupies dz^2 orbital most of the electron density will be concentrated between the metal and the two ligands on the z axis. Thus there will be greater electrostatic repulsion associated with these ligands than with the other four on xy plane.
The Jahn Teller effect is mostly observed in octahedral environments. The considerable distortions are usually observed in high spin d^4, low spin d^7 and d^9 configuration.

Question141

Sum of oxidation state (magnitude) and coordination number of cobalt in $Na \cdot [Co(bpy)Cl_4]$ is _____.
[29-Jul-2022-Shift-2]

Answer: 9

Solution:

Solution:
$Na[Co(bpy)Cl_4]$
Oxidation state of cobalt $= +3$
Coordination number of cobalt $= 6$
[As bpy is bidentate]
So, sum $= 9$

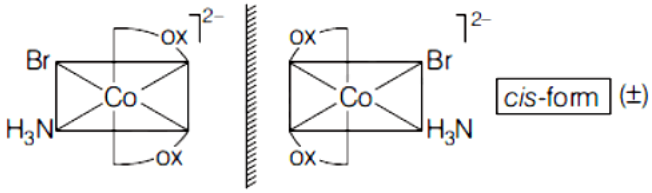
Question142

The number of stereoisomers possible for $[Co(OX)_2(Br)(NH_3)]^{2-}$ is ..... .[ox = Oxalate].
[26 Feb 2021 Shift 2]

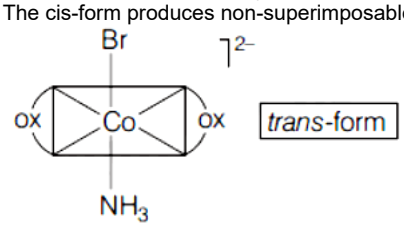
Answer: 3

Solution:

Solution:
The coordination compound $[\overset{III}{Co}(ox)_2(Br)(NH_3)]^{2-}$ of general formula $[M(A-A)_2BC]$ can show both geometrical and optical isomerism (stereoisomerism).



The cis-form produces non-superimposable mirror images, i.e. enantiomeric pairs (optically active)



The trans-form is optically inactive.
So, total number of stereoisomers possible
$= cis(\pm) + trans = 3$

Question143

Given below are two statements.
Statement I The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH.
Statement II The dimethyl glyoxime is a bidentate neutral ligand.
In the light of the above statements, choose the correct answer from the options given below.
[25 Feb 2021 Shift 2]

Options:

A. Both statements I and II are true.

B. Both statements I and II are false.

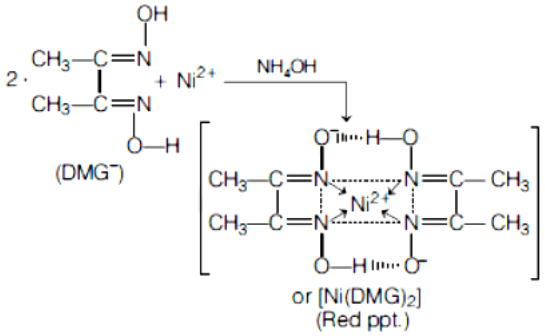
C. Statement I is true but statement II is false.

D. Statement I is false but statement II is true.

Answer: A

Solution:

Solution:
Both statements are true. Dimethyl glyoxime (DMG) is a neutral bidentate ligand (DMG^-).
Ni^{2+} ion is identified with DMG in presence of NH_4OH to give a red ppt. Its reaction is as follows.



Question144

The stepwise formation of $[Cu(NH_3)_4]^{2+}$ is given below

$$Cu^{2+} + NH_3 \overset{K_1}{\rightleftharpoons} [Cu(NH_3)_2]^{2+}$$

$$[Cu(NH_3)]^{2+} + NH_3 \overset{K_2}{\rightleftharpoons} [Cu(NH_3)_2]^{2+}$$

$$[Cu(NH_3)_2]^{2+} + NH_3 \overset{K_3}{\rightleftharpoons} [Cu(NH_3)_3]^{2+}$$

$$[Cu(NH_3)_3]^{2+} + NH_3 \overset{K_4}{\rightleftharpoons} [Cu(NH_3)_4]^{2+}$$

The value of stability constants K_1, K_2, K_3 and K_4 are 10^4, 1.58×10^3, 5×10^2 and 10^2 respectively. The overall equilibrium constants for dissociation of $[Cu(NH_3)_4]^{2+}$ is $x \times 10^{-12}$. The value of x is___
(Rounded off to the nearest integer)
[24 Feb 2021 Shift 1]

Answer: 1

Solution:

Solution:
$Cu^{2+} + NH_3 \overset{K_1}{\rightleftharpoons} [Cu(NH_3)]^{2+}$
$[Cu(NH_3)]^{2+} + NH_3 \overset{K_2}{\rightleftharpoons} [Cu(NH_3)_2]^{2+}$

$[Cu(NH_3)_2]^{2+} + NH_3 \overset{K_3}{\rightleftharpoons} [Cu(NH_3)_3]^{2+}$

$[Cu(NH_3)_3]^{2+} + NH_3 \overset{K_4}{\rightleftharpoons} [Cu(NH_3)_4]^{2+}$

$Cu^{2+} + 4NH_3 \overset{K}{\rightleftharpoons} [Cu(NH_3)_4]^{2+}$

So, $K = K_1 \times K_2 \times K_3 \times K_4$

$= 10^4 \times 1.58 \times 10^3 \times 5 \times 10^2 \times 10^2$

$\therefore K = 7.9 \times 10^{11}$

Where K = Equilibrium constant for formation of $[Cu(NH_3)_4]^{2+}$

So, equilibrium constant (K') for dissociation

of $[Cu(NH_3)_4]^{2+}$ is $\frac{1}{K}$

$K' = \frac{1}{K}$

$K' = \frac{1}{7.9 \times 10^{11}} = 1.26 \times 10^{-12} = (x \times 10^{-12})$

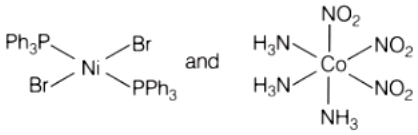
So, $x = 1.26 \approx 1.0$.

Question145

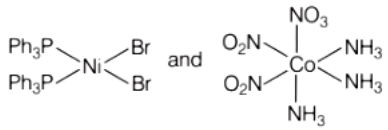
The correct structures of trans-$[NiBr(PPh_3)_2]$ and meridonial-$[Co(NH_3)_3(NO_2)_3]$, respectively are [18 Mar 2021 Shift 1]

Options:

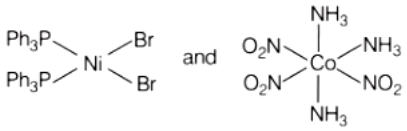
A.



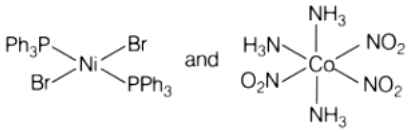
B.



C.



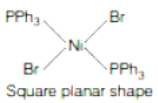
D.



Answer: D

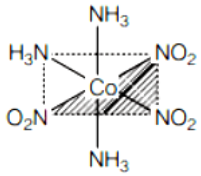
Solution:

Solution:
Trans $[NiBr_2(PPh_3)_2]$



Square planar shape

and meridional $[Co(NH_3)_3(NO_2)_3]$



Meridional (*mer*)
[*trans*]

Question146

Match List-I and List-II.

List I	List II
A. $[Co(NH_3)_6][Cr(CN)_6]$	1. Linkage isomerism
B. $[Co(NH_3)_3(NO_2)_3]$	2. Solvate isomerism
C. $[Cr(H_2O)_6]Cl_3$	3. Co-ordination isomerism
D. $cis-[CrCl_2(ox)_2]^{3-}$	4. Optical isomerism

[17 Mar 2021 Shift 2]

Options:

A. A-3 ,B-1 ,C-2 ,D-4

B. A-4 ,B-2 ,C-3 ,D-1

C. A-2 ,B-1 ,C-3 ,D-4

D. A-1 ,B-2 ,C-3 ,D-4

Answer: A

Solution:

Solution:
A. Coordination isomerisms occurs in coordination salts in which there is an interchange of ligands between the metal in the cationic coordination spheres and the metal in the anionic coordination sphere.
$[Co(NH_3)_6][Cr(CN)_6] \Leftrightarrow [Co(CN)_6][Cr(NH_3)_6]$ (Coordinate isomerism)
B. Linkage isomerism occurs in coordination compounds that have the same composition but differ in the connectivity of the ligand to the metal.
$[Co(NH_3)_3(NO_2)_3]$ shows linkage isomerism. NO_2 can connect through N and O donor sites.
C. Solvate isomerism is also known as 'hydrate isomerism' where water behaves as a ligand and water of crystallisation. e.g., The complex $[Cr(H_2O)_6]Cl_3$ (violet) contains water as a ligand and its solvate isomer $[Cr(H_2O)_5Cl]Cl_2H_2O$ (grey- green) contain water as a water of crystallisation.
D. Optical isomerism occurs mainly in compounds that are mirror images of each other. $cis[CrCl_2(ox)_2]^{3+}$ is chiral compound, hence shows optical isomerism.

Question147

The correct pair(s) of the ambident nucleophiles is(are)
A. AgCN /K CN
B. RCOOAg/RCOOK
C. $AgNO_2/KNO_2$
D. AgI /K I
[17 Mar 2021 Shift 2]

Options:

A. B and C

B. Only A

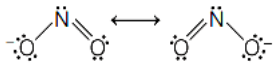
C. A and C

D. Only B

Answer: C

Solution:

Solution:
The correct pair(s) of ambident nucleophile are (A) KCN and $AgCN$ and (C) $AgNO_2$ and KNO_2.
Structure are as follows:
$\bar{C}N$ can bind through C and N atoms.
$:\overset{\ominus}{C} \equiv N: \leftrightarrow :C = \ddot{N}:^{\ominus}$
NO_2^- can bind through N and O.



Question148

The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $COCl_3 \cdot 4NH_3$ is (Round off to the nearest integer).
[16 Mar 2021 Shift 1]

Answer: 2

Solution:

Solution:
$[Co(NH_3)_4Cl_2]Cl + 2en \longrightarrow [Co(en)_2Cl_2] + 4NH_3$
NH_3 is the neutral monodentate ligand. Ethylene diamine is a neutral didentate ligand.

$H_2\ddot{N} - CH_2 - CH_2 - \ddot{N}H_2 (\ en\ )$
So, two ethylene diamine are equivalent to four NH_3 ligand.

Question149

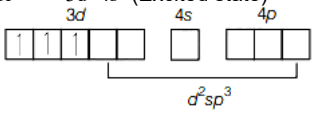
The total number of unpaired electrons present in the complex $K_3[Cr\ (oxalate\ )_3]$ is_____.
[18 Mar 2021 Shift 1]

Answer: 3

Solution:

Solution:
$K_3[Cr(ox)_3]$
Let, x be the oxidation state of Cr in $K_3[Cr(C_2O_4)_3]$. Since, the overall charge on the complex is 0, the sum of oxidation states of all elements in it should be equal to 0. Therefore,
$+3 + x + 3(-2) = 0$ (oxalate is bidentate ligand and having -2 charge)
or, $x = +3$.
Hence, chromium is in $+3$ oxidation state.
$Cr \longrightarrow 3d^5 4s^1$ (ground state electronic configuration)

$Cr^{3+} \longrightarrow 3d^3 4s^0$ (Excited state)



CN = 6
∴ oxalate is weak field ligand
∴3 unpaired electron in d-orbital. The hybridisation of chromium in the complex is d^2sp^3.

Question150

On complete reaction of $FeCl_3$ with oxalic acid in aqueous solution containing KOH , resulted in the formation of product A. The secondary valency of Fe in the product A is (Round off to the nearest integer).
[17 Mar 2021 Shift 2]

Answer: 6

Solution:

Solution:
Secondary valency is equal to coordination number of central metal ion. In this reaction, central metal atom iron is attached with 3-oxalato ligand, and oxlaic acid it is a bidentate ligand i.e. have two donor sites. So, coordination number or secondary valency $= 3 \times 2 = 6$

$$FeCl_3 + 3H_2C_2O_4 + 6KOH \longrightarrow \underset{(A)}{K_3[Fe(C_2O_4)_3]} + 3KCl + 6H_2O$$

Hence, product A have iron with secondary valency = 6

Question151

What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25 , in it's aqueous solution?
[17 Mar 2021 Shift 1]

Options:

A. 5.92

B. 5.0

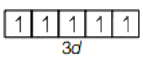
C. zero

D. 5.26

Answer: A

Solution:

Solution:
Atomic number 25 is for manganese (Mn).
Configuration of $Mn = [Ar]4s^2 3d^5$
In aqueous medium, $Mn^{2+} = [Ar]3d^5$



Total number of unpaired electrons = 5
μ (magnetic moment) $= \sqrt{n(n+2)}$ BM
where, n = number of unpaired electron
$\therefore \mu = \sqrt{5(5+2)} = \sqrt{35}BM = 5.92BM$

Question152

3 moles of metal complex with formula $Co(en)_2Cl_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.
(Round off to the nearest integer)
[27 Jul 2021 Shift 2]

Answer: 6

Solution:

Solution:
$3[Co(en)_2Cl_2]Cl + \underset{(excess)}{AgNO_3} \rightarrow \underset{(white\ ppt.)}{3AgCl}$
Secondary valency of Co = 6
(C. N.)

Question153

Given below are two statements:
Statement I : $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ are d^2sp^3 hybridised.
Statement II : $[MnCl_6]^{3-}$ and $[FeF_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons, respectively.
In the light of the above statements, choose the correct answer from the options given below :
[27 Jul 2021 Shift 2]

Options:

A. Statement I is correct but statement II is false

B. Both statement I and statement II are false

C. Statement I is incorrect but statement II is true

D. Both statement I and statement II are are true

Answer: D

Solution:

Solution:
$[Mn(CN)_6]^{3-}$ $[Fe(CN)_6]^{3-}$ $[Co(C_2O_4)_3]^{3-}$
⇓
$Mn^{3+}CN^-$ Fe^{3+},CN^- $Co^{3+},C_2O_4^{2-}$

d^4 configuration, SFL d^5 configuration, SFL d^6 configuration, Chelating ligand
⇒ All will have larger splitting hence d^2sp^3 hybridisation

$[MnCl_6]^{3-}$ and $[FeF_6]^{3-}$
d^4 configuration, Cl^- d^5 configuration, F^-
WFL WFL
↑ _ ↑ ↑
↑ ↑ ↑ ↑ ↑ ↑
⇩ ⇩
4 unpaired electrons 5 unpaired electrons

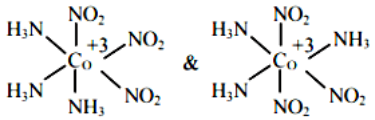
Question154

The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____.
[27 Jul 2021 Shift 1]

Answer: 2

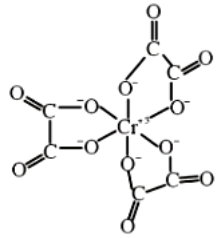
Solution:

Solution:
Triamminetrinitrocobalt(III) $\rightarrow [Co(NO_2)_3(NH_3)_3]$
trioxalatochromate(III) ion $\rightarrow [Cr(C_2O_4)_3]^{3-}$
$[Co(NO_2)_3(NH_3)_3]$



Two geometrical isomers (X)

$[Cr(C_2O_4)_3]^{3-}$



Zero geometrical isomer (Y)

$X + Y = 2 + 0 = 2.0$

Question155

An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with $AgNO_3$ [Carius Method], the percentage of chlorine in compound A is _____ when it forms 0.3849 g of AgCl. (Round off to the Nearest Integer)
(Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)
[27 Jul 2021 Shift 1]

Answer: 19

Solution:

Solution:
n_{cl} in compound $= n_{AgCl} = \frac{0.3849g}{(107.87+35.5)} g/mol$
$\Rightarrow$ mass of chlorine $= n_{Cl} \times 35.5 = 0.0953gm$
$\Rightarrow$ % wt of chlorine $= \frac{0.0953}{0.5} \times 100$
$= 19.06\%$

Question156

The number of geometrical isomers found in the metal complexes $[PtCl_2(NH_3)_2]$, $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are :

[27 Jul 2021 Shift 1]

Options:

A. 1, 1, 1, 1

B. 2, 1, 2, 2

C. 2, 0, 2, 2

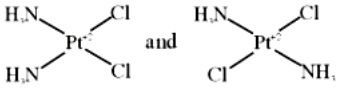
D. 2, 1, 2, 1

Answer: C

Solution:

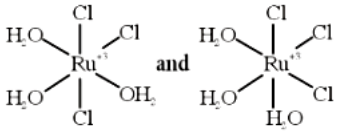
Solution:

$[PtCl_2(NH_3)_2]$

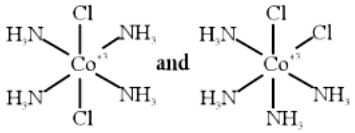


$[Ni(CO)_4]$ → All ligands are same — Zero Geometrical isomers

$[Ru(H_2O)_3Cl_3]$



$[CoCl_2(NH_3)_4]^+$



Question157

The type of hybridisation and magnetic property of the complex $[M\,nCl_6]^{3-}$, respectively, are:

[27 Jul 2021 Shift 1]

Options:

A. sp^3d^2 and diamagnetic

B. d^2sp^3 and diamagnetic

C. d^2sp^3 and paramagnetic

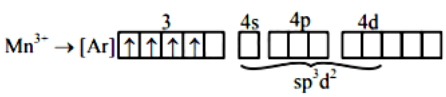
D. sp^3d^2 and paramagnetic

Answer: D

Solution:

Solution:

$[M\,nCl_6]^{3-}$



Paramagnetic and having 4 unpaired electrons.

Question158

Which one of the following metal complexes is most stable?
[25 Jul 2021 Shift 2]

Options:

A. $[Co(en)(NH_3)_4]Cl_2$

B. $[Co(en)_3]Cl_2$

C. $[Co(en)_2(NH_3)_2]Cl_2$

D. $[Co(NH_3)_6]Cl_2$

Answer: B

Solution:

Solution:
Complex $[Co(en)_3]Cl_2$ is most stable complex among the given complex compounds because more number of chelate rings are present in this complex as compare to others.

(1) $[Co(en)(NH_3)_4]Cl_2$	1 chelate ring
(2) $[Co(en)_3]Cl_2$	3 chelate ring
(3) $[Co(en)_2(NH_3)_2]Cl_2$	2 chelate ring
(4) $[Co(NH_3)_6]Cl_2$	0 chelate ring

Question159

Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $CrCl_3.3NH_3.3H_2O$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is _____.
[25 Jul 2021 Shift 1]

Answer: 0

Solution:

Solution:
Mole of AgCl precipitated is equal the mole of Cl^- present in ionization sphere.

$$[Cr(H_2O)_3(NH_3)_3]Cl_3 \rightarrow [Cr(H_2O)_3(NH_3)_3]^{3+} + 3Cl^-$$

1 mole 1 mole 3 mole

$AgNO_3$ (Excess) ↓ AgCl (3mole)

Since none of Cl^- is present in the co-ordination sphere. Therefore answer is zero.

Question160

Which one of the following species responds to an external magnetic field?
[25 Jul 2021 Shift 1]

Options:

A. $[Fe(H_2O)_6]^{3+}$

B. $[Ni(CN)_4]^{2-}$

C. $[Co(CN)_6]^{3-}$

D. $[Ni(CO)_4]$

Answer: A

Solution:

Solution:
1. $[Fe(H_2O)_6]^{3+}$
Fe^{3+} : $[Ar]3d^5$
Hybridisation : sp^3d^2
Magnetic nature : Paramagnetic (so this complex response to external magnetic field)
2. $[Ni(CN)_4]^{2-}$
Ni^{2+} : $[Ar]3d^8$
Hybridisation: dsp^2
Magnetic nature : diamagnetic
3. $[Co(CN)_6]^{3-}$
Co^{3+} : $[Ar]3d^6$
Hybridisation: d^2sp^3
Magnetic nature : diamagnetic
4. $[Ni(CO)_4]$
Ni : $[Ar]3d^8 4s^2$
Hybridisation : sp^3
Magnetic nature : diamagnetic

Question161

The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is
[22 Jul 2021 Shift 2]

Answer: 1

Solution:

Solution:
$[Co(NH_3)_6]Cl_2$
Co^{2+} : $[Ar]3d^7 4s^0 4p^0$
For this complex Δ_0< P.E., so pairing of electron does not take place.
sp^3d^2 hybridisation
Total 3 unpaired electrons are present.
$[Co(NH_3)_6]Cl_3$
Co^{3+} : $[Ar]3d^6 4s^0 4p^0$
d^2sp^3 hybridisation
NH_3 acts as SFL because $\Delta_0 > P.E$
So here all electrons becomes paired.

Question162

The correct order of intensity of colors of the compounds is:
[20 Jul 2021 Shift 1]

Options:

A. $[Ni(CN)_4]^{2-} > [NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+}$

B. $[Ni(H_2O)_6]^{2+} > [NiCl_4]^{2-} > [Ni(CN)_4]^{2-}$

C. $[NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+} > [Ni(CN)_4]^{2-}$

D. $[NiCl_4]^{2-} > [Ni(CN)_4]^{2-} > [Ni(H_2O)_6]^{2+}$

Answer: C

Solution:

Solution:

	$[NiCl_4]^{2-}$	>	$[Ni(H_2O)_6]^{2+}$	>	$[Ni(CN)_4]^{2-}$
Splitting energy order	Δ_t	<	Δ_0	<	Δ_{sq}
absorbed energy order	$[NiCl_4]^{2-}$	<	$[Ni(H_2O)_6]^{2+}$	<	$[Ni(CN)_4]^{2-}$
intensity of colour of compound	$[NiCl_4]^{2-}$	>	$[Ni(H_2O)_6]^{2+}$	>	$[Ni(CN)_4]^{2-}$

Question163

According to the valence bond theory the hybridization of central metal atom is $d sp^2$ for which one of the following compounds?
[20 Jul 2021 Shift 1]

Options:

A. $NiCl_2 . 6H_2O$

B. $K_2[Ni(CN)_4]$

C. $[Ni(CO)_4]$

D. $Na_2[NiCl_4]$

Answer: B

Solution:

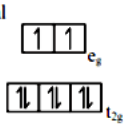
Solution:

1) $NiCl_2.6H_2O$

$Ni^{+2} \rightarrow [Ar]_{18}3d^8 4s^0$

C.N. = 6 octahedral

splitting



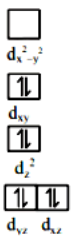
Hybridisation sp^3d^2

2) $K_2[Ni(CN)_4]$

C.N. 4 $Ni^{-2} \rightarrow [Ar]_{18}3d^8 4s^0$

$CN^- \rightarrow$ Strong field ligand

Square planar splitting

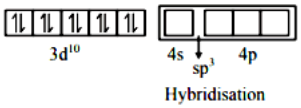


Hybridisation dsp^2

3) $Ni(CO)_4$

CO - Strong field ligand

$Ni \rightarrow [Ar]_{18}$



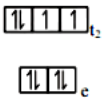
Hybridisation

4) $Na_2[NiCl_4]$ $Ni \rightarrow [Ar]_{18}3d^8 4s^0$

$Cl^{\ominus} \rightarrow$ weak field ligand

C.N. 4

tetrahedral splitting



Hybridisation sp^3

Question164

An aqueous solution of $NiCl_2$ was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[Ni(CN)_6]^{2-}$. The total change in number of unpaired electrons on metal centre is _____.

[20 Jul 2021 Shift 2]

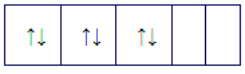
Answer: 2

Solution:

Solution:

$[Ni(CN)_6]^{2-}$

$Ni^{+4} \rightarrow d^6$ strong field ligand



Pairing will be there zero unpaired electron

$NiCl_2 \rightarrow Ni^{2+} \rightarrow d^8$



$\rightarrow$ two unpaired e^-

Change = 2

Question165

Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$is
[31 Aug 2021 Shift 2]

Options:

A. 5.92

B. 0

C. 1

D. 1.73

Answer: D

Solution:

Solution:
In $[Fe(CO)_4(C_2O_4)]^+$, oxidation number of $Fe = +3$
$\{x + 4 \times 0 + (-2) = 1\}$
$\therefore x = 3$
$Fe^{3+} = [Ar]3d^5 4s^0$
CO is a strong field ligand, so pairing is favoured
$Fe^{3+} = t_{2g}^{2, 2, 1}, e_g^{0, 0}$
Number of unpaired electron (n) = 1
$\mu = \sqrt{n(n+2)}$ BM
$= \sqrt{1(1+2)} = \sqrt{3} = 1.73$ BM

Question166

The denticity of an organic ligand, biuret is
[31 Aug 2021 Shift 1]

Options:

A. 2

B. 4

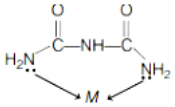
C. 3

D. 6

Answer: A

Solution:

Solution:
Biuret is bidentate ligand and denticity of this organic ligand is 2.



Hence, correct option is (a).

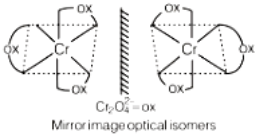
Question167

The number of optical isomers possible for $[Cr(C_2O_4)_3]^{3-}$ is ...
[27 Aug 2021 Shift 2]

Answer: 2

Solution:

Solution:
The structure of $[Cr(C_2O_4)_3]^{3-}$ is



Mirror image optical isomers

Oxalate is bidentate symmetrical ligand. The complex does not possess any element of symmetry and are optically active. The two optical isomers are shown above.
∴ Answer is two (2).

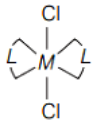
Question168

1 mol of an octahedral metal complex with formula $MCl_3.2L$ on reaction with excess of $AgNO_3$ gives 1 mol of AgCl. The denticity of ligand L is ..... .(Integer answer)
[27 Aug 2021 Shift 1]

Answer: 2

Solution:

Solution:
Since, the compound is octahedral, the central metal M will form 6 bonds. On reaction with excess of $AgNO_3$, only 1 mol of AgCl is precipitated that means only $1Cl^-$ is present as counter ion and $2Cl^-$ ions are present in coordinate sphere.
So 'L' should by bidentate to form an octahedral complex along with $2Cl^-$ ions.



$[MCl_2L_2]Cl$

Question169

Indicate the complex/complex ion which did not show any geometrical isomerism.
[26 Aug 2021 Shift 2]

Options:

A. $[CoCl_2(en)_2]$

B. $[Co(CN)_5(NC)]^{3-}$

C. $[Co(NH_3)_3(NO_2)_3]$

D. $[Co(NH_3)_4Cl_2]^+$

Answer: B

Solution:

Solution:
(a) $[CoCl_2(en)_2]$ shows cis-trans isomerism.
(b) $[Co(CN)_5(NC)]^{3-}$ is of MA_5B type complex, which will not show any isomerism.
(c) $[Co(NH_3)_3(NO_2)_3]$ shows facial and meridonial isomerism.
(d) $[Co(NH_3)_4Cl_2]^+$ shows geometrical (cis-trans) isomerism. In cis isomer, two Cl ligands are adjacent to each other. In trans

isomer two Cl ligands are opposite to each other.
∴ Option (b) is correct.

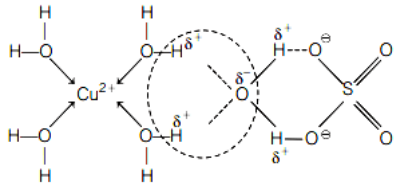
Question170

The number of hydrogen bonded water molecule(s) associated with stoichiometry $CuSO_4.5H_2O$ is/are. [31 Aug 2021 Shift 1]

Answer: 1

Solution:

Solution:
The number of hydrogen bonded water molecule associated with stoichiometry $CuSO_4.5H_2O$ is 1.



Question171

In the structure of the dichromate ion, there is a [31 Aug 2021 Shift 1]

Options:

A. linear symmetrical Cr − O − Cr bond.

B. non-linear symmetrical Cr − O − Cr bond.

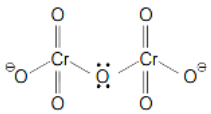
C. linear unsymmetrical Cr − O − Cr bond.

D. non-linear unsymmetrical Cr − O − Cr bond.

Answer: B

Solution:

Solution:
The structure of dichromate ion $(Cr_2O_7^{2-})$ contains a non-linear symmetrical Cr − O − Cr bond.



Hence, correct option is (b).

Question172

Acidic ferric chloride solution on treatment with excess of potassium ferrocyanide gives a prussian blue coloured colloidal species. It is [27 Aug 2021 Shift 1]

Options:

A. $Fe_4[Fe(CN)_6]_3$

B. $K_5 Fe[Fe(CN)_6]_2$

C. $HFe[Fe(CN)_6]$

D. $KFe[Fe(CN)_6]$

Answer: A

Solution:

Solution:
When a solution of ferric chloride is treated with potassium ferrocyanide solution, a prussian blue precipitate of ferric ferrocyanide is obtained.

$$\underset{\text{Ferric chloride}}{4\,FeCl_3} + \underset{\text{Potassium ferrocyanide}}{3K_4[Fe(CN)_6]} \rightarrow \underset{\text{Ferric ferrocyanide}}{Fe_4[Fe(CN)_6]_3} + 12\,KCl$$

Question173

The overall stability constant of the complex ion $[Cu(NH_3)_4]^{2+}$ is 2.1×10^{13}. The overall dissociations constant is $y \times 10^{-14}$. Then, y is ..... . (Nearest integer)
[26 Aug 2021 Shift 2]

Answer: 5

Solution:

Solution:
Given, $K_f = 2.1 \times 10^{13}$

Dissociation constant $(K_d) = \frac{1}{K_f} = 4.7 \times 10^{-14}$

$\therefore y = 4.7 \simeq 5.$

Question174

Arrange the following cobalt complexes in the order of increasing crystal field stabilisation energy (CFSE) value.
Complexes $\underset{(A)}{[CoF_6]^{3-}}$, $\underset{(B)}{[CO(H_2O)_6]^{3+}}$, $\left[\underset{(C)}{Co_{(NH_3)_6}}\right]^{3+}$ **and** $\left[\underset{(D)}{Co_{(en)_3}}\right]^{3+}$
Choose the correct option.
[26 Aug 2021 Shift 2]

Options:

A. A < B < C < D

B. B < A < C < D

C. B < C < D < A

D. C < D < B < A

Answer: A

Solution:

Solution:
For the same central metal ion, the CFSE increases with increase in strength of the ligand. The order of strength of ligand is given by the spectrochemical series:
$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^-$
$< C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < CN^- < CO$

Thus, the order of CFSE of given complexes will be :
$\underset{(A)}{[CoF_6]^{3-}} < \underset{(B)}{[CO(H_2O)_6]^{3+}} < \underset{(C)}{\left[Co(NH_3)_6\right]^{3+}} < \underset{(D)}{\left[Co(en)_3\right]^{3+}}$.

Question175

Which one of the following complexes is violet in colour?
[26 Aug 2021 Shift 1]

Options:

A. $[Fe(CN)_6]^{4-}$

B. $[Fe(SCN)_6]^{4-}$

C. $Fe_4[Fe(CN_6)]_3 \cdot H_2O$

D. $[Fe(CN)_5 NOS]^{4-}$

Answer: D

Solution:

Solution:
The complex $[Fe(CN)_5 NOS]^{4-}$ is violet in colour. It is formed when sodium nitroprusside reacts with sulphide ion to give violet colour as follows
$S^{2-} + [Fe(CN)_5 NO]^{2-} \rightarrow \underset{\text{(Violet colour)}}{[Fe(CN)_5 NOS]^{4-}}$

Question176

The potassium ferrocyanide solution gives a prussian blue colour, when added to
[1 Sep 2021 Shift 2]

Options:

A. $CoCl_3$

B. $FeCl_2$

C. $CoCl_2$

D. $FeCl_3$

Answer: D

Solution:

Solution:
Potassium ferrocyanide solution $\{K_4[Fe(CN)_6]\}$ will give prussian blue colour when added to ferric chloride $(FeCl_3)$ due to formation of iron (III) hexacyanoferrate (II).
$FeCl_3 + K_4[Fe(CN)_6] \longrightarrow Fe_4[Fe(CN)_6]_3$ Iron (III) hexacyanido - ferrate (II) (Prussian blue colour)

Question177

The crystal field stabilisation energy (CFSE) and magnetic moment (spin - only) of an octahedral aqua complex of a metal ion (M^+) are $-0.8\Delta_0$ and 3.87 BM, respectively.

Identify (M^{2+}).
[1 Sep 2021 Shift 2]

Options:

A. V^{3+}

B. Cr^{3+}

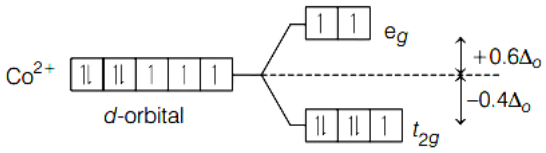
C. Mn^{4+}

D. Co^{2+}

Answer: D

Solution:

Solution:
The crystal field stabilisation energy (CFSE) and magnetic moment (spin - only) of CO^{2+} octahedral aqua complex are $-0.8\Delta_0$ and $3.87\,BM$ respectively.



CFSE $= [(e^- \text{in } t_{2g}) \times (-0.4) + (e^- \text{in } e_g) \times 0.6]\,\Delta_0$
$= [5 \times -0.4 + 2 \times 0.6]\Delta_0$
$= -0.8\Delta_0$
Magnetic moment (spin only) can be calculated as:
$\mu = \sqrt{n(n+2)}$
(where, μ = spin only magnetic moment and
n = number of unpaired electrons = 3
$= \sqrt{3(3+2)} = \sqrt{15} = 3.87\,BM$

Question178

The volume (in mL) of 0.125M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3g of $[Co(NH_3)_6]Cl_3$ is _____.
$M_{[Co(NH_3)_6]Cl_3} = 267.46 g/mol$
$M_{AgNO_3} = 169.87 g/mol$
[NV, Jan. 08 ,2020(I)]

Answer: 26.92

Solution:

Solution:
$$\underset{0.3g}{[ML_6]Cl_3} + \underset{V\,ml\,0.125\,M}{3AgNO_3} \longrightarrow 3AgCl$$
Number of moles of the complex $= \dfrac{0.3}{267.46}$
Number of moles of $AgNO_3 = 0.125 \times V \times 10^{-3}$
$\dfrac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3}$
Or, $V = \dfrac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 mL$

Question179

The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is:
(A) $Ni(CO)_4$

(B) $[Ni(H_2O)_6]Cl_2$
(C) $Na_2[Ni(CN)_4]$
(D) $PdCl_2(PPh_3)_2$
[Jan. 08, 2020(II)]

Options:

A. (A) ≈ (C) < (B) ≈ (D)

B. (C) < (D) < (B) < (A)

C. (C) ≈ (D) < (B) < (A)

D. (A) ≈ (C) ≈ (D) < (B)

Answer: D

Solution:

Solution:
(A) $Ni(CO)_4$
$Ni = 3d^8 4s^2$
CO is strong field ligand. So pairing of elections happens.
$\therefore$ Number of unpaired electrons $= 0$
$\therefore \mu_{spin} = 0$

(B) $[Ni(H_2O)_6]Cl_2$
$Ni^{+2} = 3d^8 4s^0$
H_2O is weak field ligand. So no pairing of electrons happens.
Number of unpaired electron $= 2$
$\therefore \mu_{spin} = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8}$ B.M

(C) $Na_2[Ni(CN)_4]$
$Ni^{+2} = 3d^8 4s^0$
CN^- is strong field ligand. So pairing of electrons happens.
Number of unpaired electron $= 0$
$\therefore \mu_{spin} = 0$

(D) $PdCl_2(PPh_3)_2$
$Pd^{2+} = 4d^8$
This is dsp^2 complex. And shape is square planar.
$\therefore \mu_{spin} = 0$

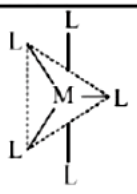
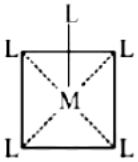
Question180

Complexes (ML_5) of metals Ni and Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively. The sum of the 90°, 120° and 180° L − M − L angles in the two complexes is _____.
[NV, Jan. 08, 2020(II)]

Answer: 20

Solution:

Solution:

 (Trigonal bipyramidal)	 (Square pyramidal)
$\angle 120° = 3; \angle 90° = 6;$ $\angle 180° = 1$ $\Rightarrow$ Total = 10	$\angle 90° = 8; \angle 180° = 2$ $\Rightarrow$ Total = 10

Total number of 180°, 90° and $120^{\circ} L - M - L$ bond angles $= 10 + 10 = 20$

Question181

The theory that can completely\/properly explain the nature of bonding in $[Ni(CO)_4]$ is: [Jan. 07,2020(I)]

Options:

A. Werner's theory

B. Molecular orbital theory

C. Crystal field theory

D. Valence bond theory

Answer: B

Solution:

Solution:
The covalant character of the bonding ($M - C\sigma$ and $M - C\pi$ bonding) which exists between the metal and the carbon atom of the CO can only be explained by the molecular orbital theory.

Question182

Complex X of composition $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83BM . It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is: [Jan. 09,2020 (I)]

Options:

A. Hexaaqua chromium (III) chloride

B. Tetraaquadichlorido chromium (IV) chloride dihydrate

C. Dichloridotetraaqua chromium (IV) chloride dihydrate

D. Tetraaquadichlorido chromium(III) chloride dihydrate

Answer: D

Solution:

Solution:
$\mu = \sqrt{n(n+2)}$ B.M. $= 3.83$ B.M.
$n = 3$($n =$ No. of unpaired e^-)
Therefore, oxidation number of Cr should be +3 .
Hence complex is $Cr(H_2O)_6Cl_3$
Complex shows geometrical isomerism therefore formula of complex is $[Cr(H_2O)_4Cl_2]Cl.2H_2O$

IUPAC Name: Tetraaquadichlorido chromium(III)
Chloride dihydrate

Question183

The isomer(s) of $[Co(NH_3)_4Cl_2]$ that has/have a $Cl-Co-Cl$ angle of 90°, is/are: [Jan. 09,2020 (II)]

Options:

A. meridional and trans

B. cis and trans

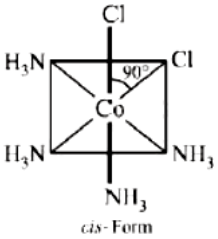
C. trans only

D. cis only

Answer: D

Solution:

Solution:



cis-Form

Question184

The complex that can show f ac - and mer-isomers is: [Jan. 08,2020 (I)]

Options:

A. $[Co(NH_3)_4Cl_2]^+$

B. $[Pt(NH_3)_2Cl_2]$

C. $[CoCl_2(en)_2]$

D. $[Co(NH_3)_3(NO_2)_3]$

Answer: D

Solution:

Solution:
$[Ma_3b_3]$ type complex shows facial and meridional isomerism.
So, the complex $[CO(NH_3)_3(NO_2)_3]$ will show fac-and merisomers.

Question185

Among (a) - (d) , the complexes that can display geometrical isomerism are:
(a) $[Pt(NH_3)_3Cl]^+$

(b) $[Pt(NH_3)_3Cl]^+$
(c) $[Pt(NH_3)_2Cl(NO_2)]$
(d)$[Pt(NH_3)_4ClBr]^{2+}$
[Jan. 08,2020 (II)]

Options:

A. (b) and (c)

B. (d) and (a)

C. (c) and (d)

D. (a) and (b)

Answer: C

Solution:

Solution:
$[Pt(NH_3)_2Cl(NO_2)]$ and $[Pt(NH_3)_4ClBr]^{2+}$ are the Ma_4bc and Ma_2bc type of complexes.
Each shows two geometrical isomers i.e., cis and trans isomers.

Question186

The IUPAC name of the complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is:
[Jan. 07,2020 (I)]

Options:

A. Diamminechlorido(methanamine) platinum (II)chloride

B. Diammine (methanamine) chloridoplatinum(II)chloride

C. Diamminechlorido(aminomethane) platinum (II)chloride

D. Bisammine(methanamine) chlorido platinum(II)chloride

Answer: A

Solution:

Solution:
$[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$
Diamine chlorido (methanamine) platinum (II) chloride

Question187

The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is:
Note: A and B are unidentate neutral and unidentate monoanionic ligands, respectively.
[Jan. 07,2020 (II)]

Options:

A. 0 and 2

B. 2 and 2

C. 0 and 0

D. 0 and 1

Answer: C

Solution:

Solution:
$[M A_2B_2]$ will not exhibit optical isomerism in both conditions.

Question188

Among the statements (A) − (D), the incorrect ones are:
(A) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments.
(B) When $\Delta_0 < P$, the d -electron configuration of Co(I I I) in an octahedral complex is $t_{2g}^4 e_g^2$
(C) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^3$
(D) If the Δ_0 for an octahedral complex of Co(I I I) is 18,000 cm^{-1}, the Δ_i for its tetrahedral complex with the same ligand will be 16, 000cm^{-1}.
[Jan. 07,2020 (II)]

Options:

A. (A) and (D) only

B. (C) and (D) only

C. (A) and (B) only

D. (B) and (C) only

Answer: A

Solution:

Solution:
Co^{3+} with strong field ligand forms complex of low magnetic moment.
$\Delta_t = \left(\frac{4}{9}\right)\Delta_0$
$\Rightarrow \Delta_1 = \frac{4}{9} \times 18000 = 8000 cm^{-1}$

Question189

$[Pd(F)(Cl)(Br)(I)]^2$ has n number of geometrical isomers.
Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[Fe(CN)_6]^{n-6}$, respectively, are:
[Note : Ignore the pairing energy]
[Jan. 09,2020(I)]

Options:

A. 2.84BM and $-1.6\Delta_0$

B. 5.92BM and 0

C. 1.73BM and $-2.0\Delta_0$

D. 0BM and $-2.4\Delta_0$

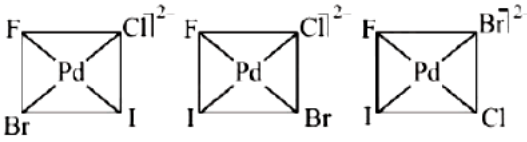
Answer: C

Solution:

Solution:

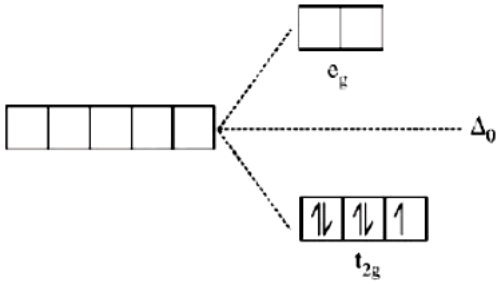
Number of Geometrical isomers (n) in square planar

$[Pd(F)(Cl)(Br)(I)]^{2-} = 3$



$[Fe(CN)_6]^{3-6} = [Fe(CN)_6]^{3-}$

$Fe^{3+} = 3d^5$



$\mu = \sqrt{n(n+2)} = 1.73B.M$

$CFSE = -0.4\,\Delta_0 \times nt_{2g} + 0.6\,\Delta_0 \times n_{2g}$

$= -0.4\,\Delta_0 \times 5 = -2.0\Delta_0$

Question190

The correct order of the spin-only magnetic moments of the following complexes is:
(A) $[Cr(H_2O)_6]Br_2$
(B) $Na_4[Fe(CN)_6]$
(C) $Na_3[Fe(C_2O_4)_3](\Delta_0 > P)$
(D) $(Et_4N)_2[CoCl_4]$
[Jan. 09,2020(II)]

Options:

A. (C) > (A) > (D) > (B)

B. (C) > (A) > (B) > (D)

C. (A) > (D) > (C) > (B)

D. (B) ≈ (A) > (D) > (C)

Answer: C

Solution:

Solution:

As, $\mu_s = \sqrt{n(n+2)}$

$Na_4[Fe(CN)_6] \rightarrow Fe^{2+}$

$\Rightarrow$ S.F.L. $\Rightarrow t_{2g}^6 e_g^0 \Rightarrow \mu_s = 0$

$[Cr(H_2O)_6]Br_2 \rightarrow Cr^{2+}$

$\Rightarrow$ W.F.L. $\Rightarrow t_{2g}^3 e_g^1 \Rightarrow \mu_s = \sqrt{24}$ B.M.

$(Et_4N)_2[CoCl_4] \rightarrow Co^{2+}$

$\Rightarrow$ W.F.L. $\Rightarrow e^4 t_2^3 \Rightarrow \mu_s = \sqrt{15}$ B.M

$Na_3[Fe(C_2O_4)_3] \rightarrow Fe^{3+}$

$\Rightarrow$ S.F.L. $\Rightarrow t_{2g}^5 e_g^0 \Rightarrow \mu_s = \sqrt{3}$ B.M

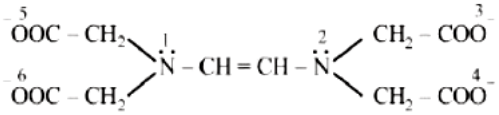
Question191

The total number of coordination sites in ethylenediaminetetraacetate ($EDTA^+$) is_____.
[NV, Sep. 05,2020 (I)]

Answer: 6

Solution:

Solution:



EDTA ^{4} is a hexadentate ligand, since hexa means six and the ligand attaches six times.

Question192

Consider the complex ions, trans-$[Co(en)_2Cl_2]^+$(A) and cis $-[Co(en)_2Cl_2]^+$(B). The correct statement regarding them is :
[Sep. 05,2020 (II)]

Options:

A. both (A) and (B) cannot be optically active.

B. (A) can be optically active, but (B) cannot be optically active.

C. both (A) and (B) can be optically active.

D. (A) cannot be optically active, but (B) can be optically active.

Answer: D

Solution:

Solution:
trans-$[Co(en)_2Cl_2]^+$ (A) contains a plane of symmetry so (A) cannot be optically active, whereas, (B) cis $[Co(en)_2Cl_2]^-$ does not contain any plane of symmetry, so (B) will be optically active.

Question193

The number of isomers possible for $[Pt(en)(NO_2)_2]$ is
[Sep. 04, 2020 (I)]

Options:

A. 2

B. 4

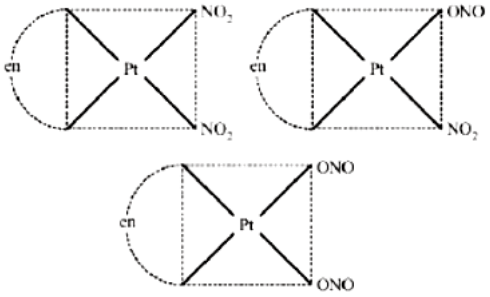
C. 1

D. 3

Answer: D

Solution:

Solution:
$[Pt(en)(NO_2)_2]$



So, total possible isomers are 3.

Question194

The complex that can show optical activity is: [Sep. 03,2020 (I)

Options:

A. $\text{trans} - [Cr(Cl_2)(ox)_2]^3$

B. $\text{trans-}[Fe(NH_3)_2(CN)_4]$

C. $\text{cis} - [Fe(NH_3)_2(CN)_4]$

D. $\text{cis} - [CrCl_2(ox)_2]^{3-}(\text{ox} = \text{ oxalate})$

Answer: D

Solution:

Solution:
Only cis- $[CrCl_2(ox)_2]^{3-}$ shows optical isomerism while its trans form do not show optical isomerism due to presence of plane of symmetry.

Question195

ComplexA has a composition of $H_{12}O_6Cl_3Cr$. If the complex on treatment with conc. H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is: [Given : atomic mass of Cr = 52 amu and Cl = 35 amu] [Sep. 03,2020 (II)

Options:

A. $[Cr(H_2O)_6]Cl_3$

B. $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$

C. $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$

D. $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$

Answer: D

Solution:

Solution:
Conc. H_2SO_4 acts as dehydrating agent. Molar mass of given complex $= 266.5 g/mol$ On treating with conc. H_2SO_4 the mass lost by the complex $= \frac{13.5}{100}(266.5) \approx 36g = 2mol\ es\ of\ H_2O$
Formula of the complex $= [Cr(H_2O)_4Cl_2]Cl.2H_2O$

Question196

The one that is not expected to show isomerism is:
[Sep. 02,2020 (II)]

Options:

A. $[Ni(NH_3)_4(H_2O)_2]^{2+}$

B. $[Ni(en)_3]^{2+}$

C. $[Ni(NH_3)_2Cl_2]$

D. $[Pt(NH_3)_2Cl_2]$

Answer: C

Solution:

Solution:
(a) $[Ni(NH_3)_4(H_2O)_2]^{2+}$ shows geometrical isomerism.
(b) $[Ni(en)_3]^{2+}$ shows optical isomerism.
(c) $[Ni^{2+}(NH_3)_2Cl_2] \Rightarrow Ni^{2+} \Rightarrow 3d^8 4s^0$
$\Rightarrow sp^3$ hybridisation
$\Rightarrow$ tetrahedral
So, $[Ni(NH_3)_2Cl_2]$ does not show isomerism.
(d) $[Pt(NH_3)_2Cl_2]$ shows geometrical isomerism.

Question197

The species that has a spin-only magnetic moment of 5.9 BM , is : T_d = tetrahedral)
[Sep. 06, 2020 (I)]

Options:

A. $[Ni(CN)_4]^{2-}$ (square planar)

B. $[NiCl_4]^{2-}(T_d)$

C. $Ni(CO)_4(T_d)$

D. $[MnBr_4]^{2-}(T_d)$

Answer: D

Solution:

Solution:
$[MnBr_4]^{2-} \Rightarrow Mn^{2+} \Rightarrow d^5(Td)$
[$\because$ Br is weak field ligand]
$d^5(Td)$ is high spin complex.
So, $\mu = \sqrt{5(5+2)} = 5.91B \cdot M$

Question198

For a d^4 metal ion in an octahedral field, the correct electronic configuration is :
[Sep. 06, 2020 (II)]

Options:

A. $t_2^3 e_g^1$ when $\Delta_0 < P$

B. $t_{2g}^3 e_g^1$ when $\Delta_0 > P$

C. $t_{2g}^4 e_g^0$ when $\Delta_0 < P$

D. $e_g^2 t_{2g}^2$ when $\Delta_0 < P$

Answer: A

Solution:

Solution:
If $\Delta_0 > P$ then fot $d^4 \Rightarrow t_{2g}^4 e_g^0$
If $\Delta_0 < P$ then for $d^4 \Rightarrow t_{2g}^3 e_g^1$

Question199

The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are:
[Sep. 05, 2020 (I)]

Options:

A. $-0.4\Delta_0$ and $-0.6\Delta_t$

B. $-2.4\Delta_o$ and $-0.6\Delta_t$

C. $-1.6\Delta_0$ and $-0.4\Delta_t$

D. $-0.4\Delta_0$ and $-0.27\Delta_t$

Answer: A

Solution:

Solution:
For d^6 configuration, high spin complex.
(i) In case of octahedral field, $t_{2g}^4 e_g^2$
CF SE $= [-0.4p + 0.6q]\ \Delta_0 + n(P)$
$= [-0.4 \times 4 + 0.6 \times 2]\ \Delta_0 + 0 = -0.4\Delta_0$
(ii) In case of tetrahedral field, e^3, t_2^3
CF SE $= [-0.6p + 0.4q]\ \Delta t$
$= [-0.6 \times 3 + 0.4 \times 3]\ \Delta t = -0.6\ \Delta t.$

Question200

Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[Ru(H_2O)_6]^{2+}$ would be _____.
[NV Sep. 05, 2020 (II)]

Answer: 0

Solution:

Solution:
$Ru^{2+} = 4d^6 = t_{2g}^6 e_g^0$ since $\Delta_0 > P$
No. of unpaired electrons $= 0$
$\therefore$ Magnetic moment $= 0$B.M.

Question201

The pair in which both the species have the same magnetic moment (spin only) is: [Sep. 04, 2020 (I)]

Options:

A. $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

B. $[Co(OH)_4]^{2-}$ and $[Fe(NH_3)_6]^{2+}$

C. $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)]^{2+}$

D. $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

Answer: A

Solution:

Solution:
$[Cr(H_2O)_6]^{2+} \Rightarrow Cr^{2+} \Rightarrow 3d^4 \Rightarrow t_{2g}^{1,1,1}, e_g^{1,0}$
No. of unpaired e : 4
$[Fe(H_2O)_6]^{2+} \Rightarrow Fe^{2+} \Rightarrow 3d^6 \Rightarrow t_{2g}^{2,1,1}, e_g^{1,1}$
No. of unpaired e : 4
$[Fe(NH_3)_6]^{2+} \Rightarrow Fe^{2+} \Rightarrow 3d^6 \Rightarrow t_{2g}^{2,1,1}, e_g^{1,1}$
No. of unpaired e : 4
$[Co(OH)_4]^{2-} \Rightarrow Co^{2+} \Rightarrow 3d^T \Rightarrow e^{2,2}, t_2^{1,1,1}$
No. of unpaired e : 3
$[CoCl_4]^{2-} \Rightarrow Co^{2+} \Rightarrow 3d^7 \Rightarrow e^{2,2}, t_2^{1,1,1}$
No. of unpaired e : 3
$[Mn(H_2O)_6]^{2+} \Rightarrow Mn^{2+} \Rightarrow 3d^5 \Rightarrow t_{2g}^{1,1,1}, e_g^{1,1}$
No. of unpaired e : 5
So $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$ have same magnetic moment (spin only).

Question202

The Crystal Field Stabilization Energy (CFSE) of $[CoF_3(H_2O)_3]$($\Delta_0 < P$) is : [Sep. 04, 2020 (II)]

Options:

A. $-0.8\ \Delta_0 + 2P$

B. $-0.4\Delta_0$

C. $-0.8\Delta_0$

D. $-0.4\ \Delta_0 + P$

Answer: B

Solution:

Solution:
$[CoF_3(H_2O)_3] \Rightarrow Co^{3+} \Rightarrow d^6$ or $t_{2g}^4 e_g^2$
$CFSE = -[0.4p + 0.6q]\ \Delta_0 + n(P)$

$= [-4 \times 0.4 + 2 \times 0.6]\ \Delta_0 + 0 = -0.4\Delta_0$

Question203

The one that can exhibit highest paramagnetic behaviour among the following is :
gly = glycinato; bpy = 2, 2′-bipyridine
[Sep. 04, 2020 (II)]

Options:

A. $[Pd(gly)_2]$

B. $[Fe(en)(bpy)(NH_2)_2]^{2+}$

C. $[Co(OX)_2(OH)_2](\Delta_0 > P)$

D. $[Ti(NH_3)_6]^{3+}$

Answer: C

Solution:

Solution:

	No, of unpaired electrons
$[Fe(en)(bpy)(NH_3)_2]^{2+}$	$Fe^{2+} - 3d^6 0$
$[Pd(gly)_2]$	$Pd^{2+} - 3d^8 0$
$[Co(OX)_2(OH)_2]$	$Cos^{5+} - 3d^4 2$
$[Ti(NH_3)_6]^3$	$Ti^{3+} - 3d^1 1$

Thus, $[Co(OX)_2(OH)_2]$ exhibits highest paramagnetic behaviour due to highest number of unpaired electrons.

Question204

The electronic spectrum of $[Ti(H_2O)_6]^{3+}$ shows a single broad peak with a maximum at 20, 300cm^{-1}.
The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1}, is:
$\left(1\ kJ\ mol^{-1} = 83.7 cm^{-1} \right)$
[Sep. 03, 2020 (I)]

Options:

A. 145.5

B. 242.5

C. 83.7

D. 97

Answer: D

Solution:

Solution:
$[Ti(H_2O)_6]^{3+} \Rightarrow Ti^{3+} = 3d^1 4s^\circ$
$\therefore$ Electronic configuration is $t_{2g}^1 e_g^0$

$$\text{CPSE} = \left[-0.4n_{t_{2g}} + 0.6n_{eg}\right]\Delta_0 + n(p)$$
$$= [(-0.4)\times 1 + 0]20300$$
$$= -8120\text{cm}^{-1}$$
$$= \frac{-8120}{83.7}\text{kJ/mol} = -97\text{kJ/mol}$$

Question205

The d -electron configuration of $[Ru(en)_3]Cl_2$ and $[Fe(H_2O)_6]Cl_2$, respectively are: [Sep. 03, 2020 (II)]

Options:

A. $t_{2g}^6 e_g^0$ and $t_{2g}^6 e_g^0$

B. $t_{2g}^4 e_g^2$ and $t_{2g}^6 e_g^0$

C. $t_{2g}^6 e_g^0$ and $t_{2g}^4 e_g^2$

D. $t_{2g}^4 e_g^2$ and $t_{2g}^4 e_g^2$

Answer: C

Solution:

Solution:
$[Ru(en)_3]Cl_2 \Rightarrow Ru^{2+} = 4d^6 = t_{2g}^6, e_g^0$
$[Fe(H_2O)_6]^{2+} \Rightarrow Fe^{2+} = 3d^6 = t_{2g}^4, e_g^2$
So, correct answer is (c).

Question206

Consider that a d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90BM . The geometry and the crystal field stabilization energy of the complex is: [Sep. 02, 2020 (I)]

Options:

A. octahedral and $-2.4\,\Delta_0 + 2P$

B. tetrahedral and $-0.6\Delta_t$

C. octahedral and $-1.6\Delta_0$

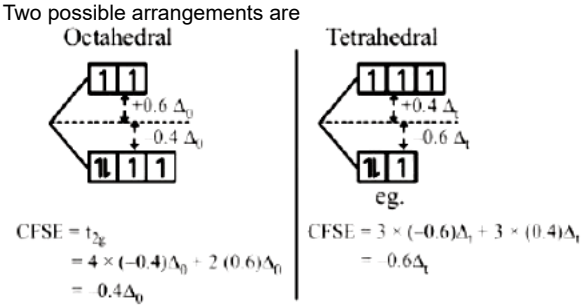
D. tetrahedral and $-1.6\,\Delta_t + 1P$

Answer: B

Solution:

Solution:
Spin only magnetic moment $= 4 \cdot 9 = \sqrt{n(n+2)}$
$\therefore$ number of unpaired electrons $= 4$

Two possible arrangements are



So, option (b) is correct.

Question207

For octahedral M n(II) and tetrahedral Ni(II) complexes, consider the following statements:
(I) both the complexes can be high spin.
(II) N i(II) complex can veryrarely below spin.
(III) with strong field ligands, Mn(II) complexes can be low spin.
(IV) aqueous solution of M n(II) ions is yellow in color.
The correct statements are:
[Sep. 02, 2020 (I)]

Options:

A. (I) and (II) only

B. (I), (III) and (IV) only

C. (I), (II) and (III) only

D. (II), (III) and (IV) only

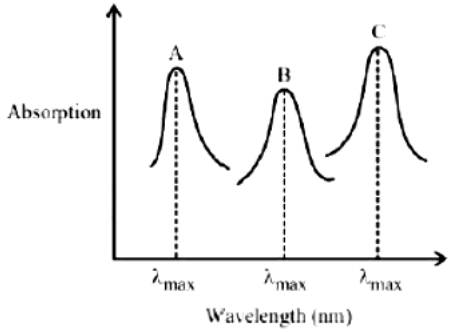
Answer: C

Solution:

Solution:
With weak field ligands Mn(II) will be of high spin and with strong field ligands it will be of low spin. Ni(II) tetrahedral complexes will be generally of high spin due to sp^3 hybridisation. $M n(I I)$ is of light pink colour in aqueous solution.

Question208

Simplified absorption spectra of three complexes ((i), (ii) and (iii)) of M^{n+} ion are provided below; their λ_{max} values are marked as A, B and C respectively. The correct match between the complexes and their λ_{max} values is :



(i) $[M (N CS)_6]^{(-6+a)}$

(ii) $[M F_6]^{(-6+n)}$

(iii) $[M(NH_3)_6]^{n+}$
[Sep. 02, 2020 (II)]

Options:

A. A − (iii), B − (i), C − (ii)

B. A − (ii), B − (i), C -(iii)

C. A − (ii), B − (iii), C − (i)

D. A − (i), B − (ii), C − (iii)

Answer: A

Solution:

Solution:
Stronger the ligand greater is splitting of d orbitals and smaller will be wavelength of light absorbed. According to spectrochemical series, the splitting power of ligands is $NH_3 > NC\bar{S} > F$.
Thus, order of wavelength of light absorbed is
$\lambda_{NH_3} < \lambda_{NC\bar{S}} < \lambda_{F^-}$

Question209

The coordination number of Th in $K_4[Th(C_2O_4)_4(H_2O)_2]$ is: ($C_2O_4^{2-}$ = oxalato)
[Jan. 11,2019(II)]

Options:

A. 14

B. 6

C. 8

D. 10

Answer: D

Solution:

Solution:
$K_4[Th(C_2O_4)_4(H_2O)_2]$
$C_2O_4^{2-}$ (oxalato): bidentate ligand
H_2O (aqua): monodentate
∴ Co-ordination no. of $Th = 2 \times 4 + 2 = 10$

Question210

The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is:
[Jan. 10, 2019(I)]

Options:

A. 16

B. 8

C. 4

D. 12

Answer: D

Solution:

Solution:
Possible complexes No. of isomers
$[M(F)(Cl)(SCN)(NO_2)] \rightarrow 3$
$[M(F)(Cl)(SCN)(ONO)] \rightarrow 3$
$[M(F)(Cl)(NCS)(NO_2)] \rightarrow 3$
$[M(F)(Cl)(NCS)(ONO)] \rightarrow 3$
Total = 12

Question211

A reaction of cobalt(III) chloride and ethylenediamine in a 1: 2 mole ratio generates two isomeric products A (violet coloured) and B (green coloured). A can show optical activity, but, B is optically inactive. What type of isomers does A and B represent? [Jan. 10,2019(II)]

Options:

A. Geometrical isomers

B. Coordination isomers

C. Linkage isomers

D. Ionisation isomers

Answer: A

Solution:

Solution:
Reaction for the given condition can be written as:
$[CoCl_6]^3 + 2(en) \longrightarrow [CoCl_2(en)_2]^+$
(1 : 2 mole ratio) (cis-trans-isomer)
A = optically active (cis-isomer), violet
B = optically inactive (trans-isomer), green

Question212

The metal d -orbitals that are directly facing the ligands in $K_3[Co(CN)_6]$ are : [Jan.12, 2019(I)]

Options:

A. d_{xy} and $d_{x^2-y^2}$

B. $d_{x^2-y^2}$ and d_{z^2}

C. d_{xz}, d_{yz} and d_{z^2}

D. d_{xy}, d_x and d_{yz}

Answer: B

Solution:

Solution:
$K_3[Co(CN)_6]$ is an octahedral complex.
During splitting of d orbitals in octahederal complexes, $d_{x^2-y^2}$ and d_{z^2} orbitals point towards the direction of ligands (i.e. they experience more repulsion and will be raised in energy by $\frac{3}{5}\Delta_0$).

Question213

The magnetic moment of an octahedral homoleptic Mn(II) complex is 5.9BM . The suitable ligand for this complex is:
[Jan.12, 2019(I)]

Options:

A. Ethylenediamine

B. CN

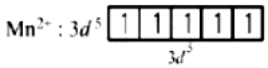
C. NCS

D. CO

Answer: C

Solution:

Solution:
Electronic configuration of Mn^{2+} is,



Presence of 5 unpaired e^- shows that the complex of Mn^{2+} has only weak field ligand (NCS^-).

Question214

Match the metals (column I) with the coordination compound(s)/enzyme(s) (column II) :

(column I)	(column II)
Metals	**Coordination compound(s)/enzyme(s)**
(A) Co	(i) Wilkinson catalyst
(B) Zn	(ii) Chlorophyll
(C) Rh	(iii) Vitamin B_{12}
(D) Mg	(iv) Carbonic anhydrase

[Jan.11, 2019(I)]

Options:

A. (A) – (iii); (B) – (iv); (C) – (i); (D) – (ii)

B. (A) – (i); (B) – (ii); (C) – (iii); (D) – (iv)

C. (A) – (ii); (B) – (i); (C) – (iv); (D) – (iii)

D. (A) – (iv); (B) – (iii); (C) – (i); (D) – (ii)

Answer: A

Solution:

Solution:
Wilkinson catalyst: $[Rh(PPh)_3Cl]$
Chlorophyll: $C_{55}H_{72}O_5N_4Mg$
Vitamin B_{12} contains Co.
Carbonic anhydrase contains a Zn ion.

Question215

The difference in the number of unpaired electrons of a metal ion in its high-spin and low-spin octahedral complexes is two. The metal ion is:
[Jan.10, 2019(II)]

Options:

A. Ni^{2+}

B. Fe^{2+}

C. Co^{2+}

D. Mn^{2+}

Answer: C

Solution:

Solution:

Metal ion	High spin	Low spin	Difference in the unpaired electrons
$Ni^{2+}(3d^8)$	2	2	0
$Mn^{2+}(3d^5)$	5	1	4
$Fe^{2+}(3d^6)$	4	0	4
$Co^{2+}(3d^7)$	3	1	2

Question216

Two complexes $[Cr(H_2O)_6]Cl_3$ (A) and $[Cr(NH_3)_6]Cl_3$(B) are violet and yellow coloured, respectively. The incorrect statement regarding them is:
[Jan.9, 2019(I)]

Options:

A. Δ_0 values of (A) and (B) are calculated from the energies of violet and yellow light, respectively.

B. both are paramagnetic with three unpaired electrons.

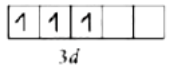
C. both absorb energies corresponding to their complementary colors.

D. Δ_0 value for (A) is less than that of (B).

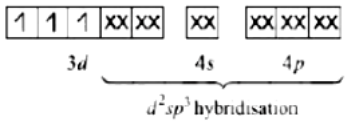
Answer: A

Solution:

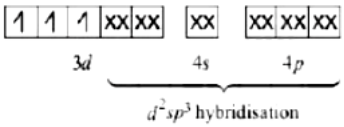
Solution:
E.C. of $Cr^{3+}(3d^3)$:



For complex $A[Cr(H_2O)_2]^{3+}$



For complex B $[Cr(NH3)_6]^{3+}$:



Here, both the complexes (A) and (B) are paramagnetic with 3 unpaired electrons each. H_2O is also a weak field ligand, which causes lesser splitting than NH_3, which is comparatively stronger field ligand. Hence, the (Δ_0) value of (A) and (B) are calculated from the wavelengths of light absorbed and not from the wavelengths of light emitted.

Question217

The complex that has highest crystal field splitting energy (Δ) is: [Jan.9, 2019(II)]

Options:

A. $[Co(NH_3)_5(H_2O)]Cl_3$

B. $K_2[CoCl_4]$

C. $[Co(NH_3)_5Cl]Cl_2$

D. $K_3[Co(CN)_6]$

Answer: D

Solution:

Solution:
In case of similar metal atom or ion, the value of coordination number and the strength of the ligands determine the value of crystal field splitting energy. Greater the co-ordination number and strength of value of the ligand, greater will be the value of CFSE. Strength of ligands:
$CN^- > NH_3 > H_2O > Cl$
$\therefore K_3[Co(CN)_6]$ has the highest crystal field splitting energy.

Question218

Homoleptic octahedral complexes of a metal ion M^{3+1} with three monodentate ligands L_1, L_2 and L_3 absorb wavelengths in the region of green, blue and red respectively. The increasing order of the ligand strength is: [Jan.9, 2019(II)]

Options:

A. $L_3 < L_1 < L_2$

B. $L_3 < L_2 < L_1$

C. $L_1 < L_2 < L_3$

D. $L_2 < L_1 < L_3$

Answer: A

Solution:

Solution:
Lesser the wavelength of light absorbed (more ener- gy) greater will be ligand strength.
Energy : Blue > Green > Red
Ligand strength: L_2 > L_1 > L_3

Question219

$Mn_2(CO)_{10}$ is an organometallic compound due to the presence of : [Jan. 12,2019(I)]

Options:

A. $Mn-C$ bond

B. Mn-Mn bond

C. $Mn-O$ bond

D. C-O bond

Answer: A

Solution:

Solution:
Compounds having atleast one carbon metal $(M-C)$ bond are known as organometallic compounds. It contains $Mn-C$ bond.

Question220

The number of bridging CO ligand(s) and Co-Co bond(s) in $Co_2(CO)_8$, respectively are: [Jan. 11,2019(I)]

Options:

A. 2 and 1

B. 2 and 0

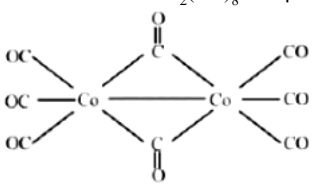
C. 0 and 2

D. 4 and 0

Answer: A

Solution:

Solution:
The structure of $Co_2(CO)_8$ is represented as



It contains two bridging CO ligands and one metal-metal $(Co-Co)$ bond.

Question221

Wilkinson catalyst is: [Jan. 10,2019(I)]

Options:

A. $[(Ph_3P)_3IrCl]$

B. $[(Et_3P)_3RhCl]$

C. $[(Ph_3P)_3RhCl]$

D. $[(Et_3P)_3IrCl]$

Answer: C

Solution:

Solution:
Wilkinson's catalyst is $[Rh(PPh_3)_3Cl]$

Question222

The coordination numbers of Co and Al in $[Co(Cl)(en)_2]Cl$ and $K_3[Al(C_2O_4)_3]$, respectively, are: (en = ethane- 1,2 -diamine) [April 12,2019 (II)]

Options:

A. 5 and 3

B. 3 and 3

C. 6 and 6

D. 5 and 6

Answer: D

Solution:

Solution:
$[Co(en)_2Cl]Cl$
Cl^- - monodentate ligand
en - bidentate ligand
∴ Co-ordination Number of Co = $(2 \times 2) + 1 = 5$
$K_3[Al(C_2O_4)_3]$
$C_2O_4^{2-}$ - bidentate ligand
∴ Co-ordination Number of Al = $2 \times 3 = 6$

Question223

The species that ean have a trans-isomer is: (en = ethane −1, 2− diamine, Ox = oxalate) [April 10,2019 (I)]

Options:

A. $[Zn(en)Cl_2]$

B. $[Pt(en)Cl_2]$

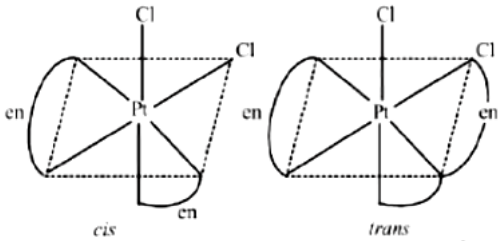
C. $[Cr(en)_2(Ox)]^+$

D. $[Pt(en)_2Cl_2]^{2+}$

Answer: D

Solution:

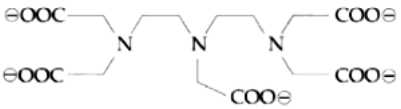
Solution:



cis-trans Isomerism is possible with $[Pt(en)_2Cl_2]^{2+}$. $[Cr(en)_2Ox]^{+}$ shows only optical isomerism. The other two complexes, i.e., $[Pt(en)Cl_2]$ and $[Zn(en)Cl_2]$ do not show stereoisomerism.

Question224

The maximum possible denticities of a ligand given below towards a common transition and inner-transition metal ion, respectively, are:



[April 9,2019 (II)]

Options:

A. 8 and 6

B. 6 and 8

C. 6 and 6

D. 8 and 8

Answer: B

Solution:

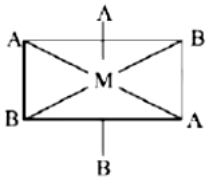
Solution:
The maximum possible denticites of the given ligand towards transition metal ion is 6 and towards inner transition metal ion (due to greater ionic radii and more atomic orbitals) is 8 .

Question225

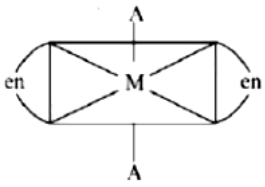
The one that will show optical activity is:
(en = ethane 1,2 -diamine)
[April 9,2019 (I)]

Options:

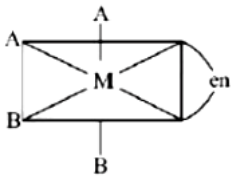
A.



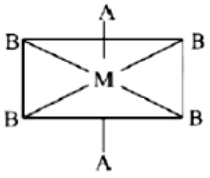
B.



C.



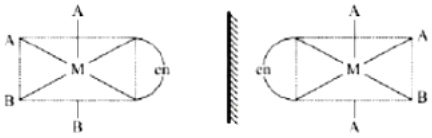
D.



Answer: C

Solution:

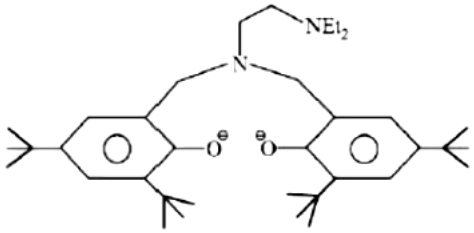
Solution:



No plane of symmetry or centre of symmetry is present. Hence, it will be optically active.

Question226

The following ligand is:



[April 8,2019 (I)]

Options:

A. hexadentate

B. tetradentate

C. bidentate

D. tridentate

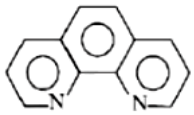
Answer: B

Solution:

Solution:
It has four atoms containing lone pair of e^-, therefore, it will be able to donate these lone pairs and acts as a tetradentate ligand.

Question227

The complex ion that will lose its crystal field stabilization energy upon oxidation of its metal to +3 state is:

(Phen = 

and
ignore pairing energy)

[April 12, 2019 (I)]

Options:

A. $[Co(phen)_3]^{2^2}$

B. $[Ni(phen)_3]^2$

C. $[Zn(phen)_3]^{2+}$

D. $[Fe(phen)_3]^2$

Answer: D

Solution:

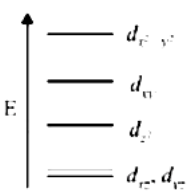
Solution:
$CFSE \Rightarrow [-0.4p + 0.6q]\Delta_0$
$Ni^{2+}(d^8) \longrightarrow Ni^{3+}(d^7)$
$t_{2g}^6 e_B^2 \qquad t_{2g}^6 e_B^1$
$CFSE \Rightarrow -1.2 \quad -1.8$
$Co^{2+}(d^7) \longrightarrow Co^{3+}(d^6)$
$t_{2g}^6 e_g^1 \qquad t_2^6 e^0 g$
$CFSE \Rightarrow -1.8 \quad -2.4$
$Zn^{2+}(d^{10}) \longrightarrow Zn^{3+}(d^9)$
$t_{2g}^6 e_g^4 \qquad t_{2g}^6 e^3$
$CFSE \Rightarrow 0.0 \quad -0.6$
$Fe^{2+}(d^6) \rightarrow Fe^{3+}(d^5)$
$t_{2g}^6 e_g^0 \qquad t_{2g} e_g^0$
$CFSE \Rightarrow -2.4 \quad -2.0$
So, only Fe^{2+} will lose crystal field stabilisation energy upon oxidation to +3 , others will gain crystal field stabilisation energy.

Question228

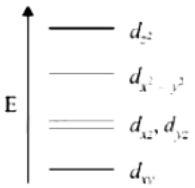
Complete removal of both the axial ligands (along the zaxis) from an octahedral complex leads to which of the following splitting patterns? (relative orbital energies not on scale)
[April 12, 2019 (I)]

Options:

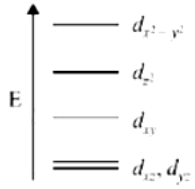
A.



B.



C.



D.



Answer: A

Solution:

Solution:
(a) After removal of both axial ligands from octahedral complex the field becomes square planar and leads to following energy patterns.
$d_{x^2-y^2} > d_{xy} > d_{z^2} > d_{xz} = d_{yz}$

Question229

Three complexes, $[CoCl(NH_3)_5]^{2+}$(I), $[Co(NH_3)_5H_2O]^{3+}$(II) and $[Co(NH_3)_6]^{3+}$ (III) absorb light in the visible region. The correct order of the wavelength of light absorbed by them is :
[April 10, 2019 (I)]

Options:

A. (III) > (I) > (II)

B. (III) > (II) > (I)

C. (II) > (D) > (III)

D. (I) > (II) > (III)

Answer: D

Solution:

Solution:
Wavelength of the energy absorbed by the coordination compound is inversely proportional to ligand field strength of the given co-ordination compound. The strong field ligand causes higher splitting of the d -orbitals. The decreasing order of ligand field strength is $NH_3 > H_2O > Cl$.
Therefore decreasing order of absorbed wavelength is $(I) > (II) > (III)$

Question230

The crystal field stabilization energy (CFSE) of $[Fe(H_2O)_6]Cl_2$ and $K_2[NiCl_4]$, respectively are:
[April 10, 2019 (II)]

Options:

A. $-0.6\Delta_0$ and $-0.8\Delta_z$

B. $-0.4\Delta_o$ and $-0.8\Delta_\tau$

C. $-2.4\Delta_o$ and $-1.2\Delta_e$

D. $-0.4\Delta_o$ and $-1.2\Delta_r$

Answer: B

Solution:

Solution:
$[Fe(H_2O)_6]^{2+} t_2^4 ge_g^2$
$CFSE = (-4 \times 0.4 + 2 \times 0.6)A_0 = -0.4\Delta_0$
$[NiCl_4]^{2-} e^4 t_2^4$ $CFSE = (-4 \times 0.6 + 4 \times 0.4)\Delta_1 = -0.8\Delta_1$

Question231

The degenerate orbitals of $[Cr(H_2O)_6]^{3+}$ are:
[April 9, 2019 (I)]

Options:

A. d_{xz} and $d_{y=}$

B. d_{yz} and d_{z^2}

C. d_{z}^2 and d_{xz}

D. $dx^2 - y^2$ and d_{xy}

Answer: A

Solution:

Solution:
Cr^{3+} has d^3 configuration and forms an octahedral inner orbitals complex, therefore the set of degenerate orbitals are (d_{xy}, d_{yz} and d_{xz}) and ($d_x^2 - y^2$ and d_{z2}).

Question232

The correct statements among I to III are:
(I) Valence bond theory cannot explain the color exhibited by transition metal complexes.
(II) Valence bond theory can predict quantitatively the magnetic properties of transition metal complexes.
(III) Valence bond theory cannot distinguish ligands as week and strong field ones.
[April 9, 2019 (I)]

Options:

A. (II) and (III) only

B. (I), (II) and (III)

C. (I) and (III) only

D. (I) and (II) only

Answer: C

Solution:

Solution:
Valence bond theory cannot distinguish between weak field ligands and strong field ligands. Therefore, it cannot predict quantitatively the magnetic properties of transition metal complexes.

Question233

The correct order of the spin-only magnetic moment of metal ions in the following low-spin complexes, $[V(CN)_6]^4$, $[Fe(CN)_6]^4$, $[Ru(NH_3)_6]^{3+}$, and $[Cr(NH_3)_6]^{2+}$, is
[April 8, 2019 (I)]

Options:

A. $Cr^{2+} > Ru^{3+} > Fe^{2+} > V^2$

B. $V^{2+} > Cr^{2+} > Ru^{3+} > Fe^2$

C. $V^{2+} > Ru^{3+} > Cr^{2+} > Fe^{2+}$

D. $Cr^{2+} > V^{2+} > Ru^{3+} > Fe^{2+}$

Answer: B

Solution:

Solution:

Complex	Metal ion electrons	No. of unpaired
$[V(CN)_6]^4$	V^{2+}	3
$[Ru(NH_3)_6]^{3+}$	Ru^{3+}	1
$[Fe(CN)_6]^+$	Fe^{2+}	0
$[Cr(NH_3)_6]^{2+}$	Cr^{2+}	2

∵ Spin magnetic moment ∝ no. of unpaired electrons. So the order of spin magnetic moment is:
$V^{2+} > Cr^{2+} > Ru^{3+} > Fe^{2+}$

Question234

The calculated spin-only magnetic moments (BM) of the anionic and cationic species of $[Fe(H_2O)_6]_2$ and $[Fe(CN)_6]$, respectively, are:
[April 8, 2019 (II)]

Options:

A. 0 and 4.9

B. 2.84 and 5.92

C. 4.9 and 0

D. 0 and 5.92

Answer: C

Solution:

Solution:
$[Fe(H_2O)_6]^{2+}$: cationic species of $[Fe(H_2O)_6]_2$
$[Fe(CN)_6]^{4-}$: anionic species of $[Fe(CN)_6]$
Magnetic Moment $(\mu) = \sqrt{n(n+2)}$
Where n = no. of unpaired electrons
Now, $[Fe(H_2O)_6]^{2+}[Fe(CN)_6]^4$
$n = 4 \quad n = 0$
$\therefore \mu = \sqrt{4(4+2)} \quad \mu = \sqrt{0(0+2)}$
$\mu = 4.9 \quad \mu = 0$

Question235

The compound used in the treatment of lead poisoning is: [April 12, 2019 (II)]

Options:

A. D-penicillamine

B. desferrioxime B

C. Cis-platin

D. EDTA

Answer: D

Solution:

Solution:
EDTA is used in the treatment of lead poisoning.Deferrioxime B is used in treatment of iron poisoning and D-penicillamine is used in treatment of heavymetal poisoning, while cis -platin is used for treating cancer.

Question236

The compound that inhibits the growth of tumors is : [April 8, 2019 (II)]

Options:

A. trans-$[Pt(Cl)_2(NH_3)_2]$

B. cis $-[Pd(Cl)_2(NH_3)_2]$

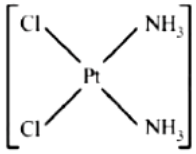
C. cis $-[Pt(Cl)_2(NH_3)_2]$

D. trans-$[Pd(Cl)_2(NH_3)_2]$

Answer: C

Solution:

Solution:
cis-Platin is used as an anti-cancer drug.



cis-$[Pt(Cl)_2(NH_3)_2]$

Question237

Consider the following reaction and statements: $[Co(NH_3)_4Br_2]^+ + Br^- \rightarrow [Co(NH_3)_3Br_3] + NH_3$
(I) Two isomers are produced if the reactant complex ion is a cis-isomer.
(II) Two isomers are produced if the reactant complex ion is a trans -isomer
(III) Only one isomer is produced if the reactant complex ion is a trans-isomer
(IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.
The correct statements are:
[2018]

Options:

A. (I) and (II)

B. (I) and (III)

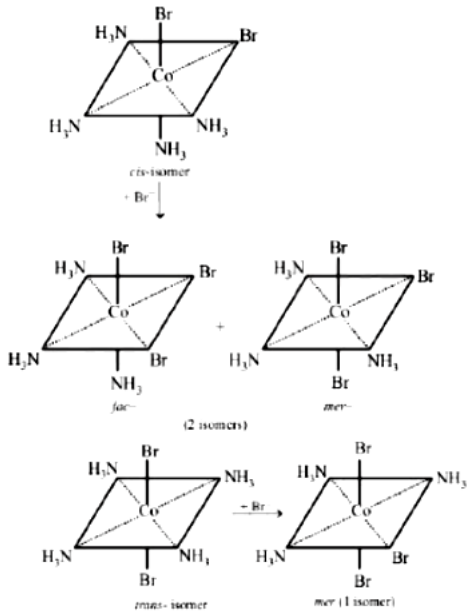
C. (III) and(IV)

D. (II) and (IV)

Answer: B

Solution:

Solution:



Question238

Which of the following complexes will show geometrical isomerism?
[Online April 16, 2018]

Options:

A. Potassium tris(oxalato) chromate (III)

B. Pentaaquachlorochromium (III) chloride

C. Aquachlorobis (ethylenediamine) cobalt(II) chloride

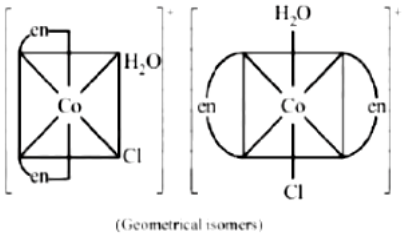
D. Potassium amminetrichloroplatinate (II)

Answer: C

Solution:

Solution:

$[Co(H_2O)Cl(cn)_2]Cl$



(Geometrical isomers)

Question239

The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$ is: [Online April 15, 2018(II)]

Options:

A. 16

B. 12

C. 8

D. 24

Answer: B

Solution:

Solution:
The square planar complex of the type [Mabed] $^{n+}$, where all four ligands are different, has 3 geometrical isomers. But if one of the ligands is ambidentate, then $2 \times 3 = 6$ geometrical isomers are possible. But if two ligands are ambidentate, then $4 \times 3 = 12$ geometrical isomers are possible.
In the given example, NO_2^- and SCN are ambidentate ligands

Question240

In Wilkinson's catalyst, the hybridisation of central metal ion and its shape are respectively: [Online April 16, 2018]

Options:

A. sp^3d, trigonal bipyramidal

B. d^2sp^3, octahedral

C. dsp^2, squarc planar

D. sp^3, tetrahedral

Answer: C

Solution:

Solution:
Wilkinson catalyst is $[RhCl(PPh_3)_3]$.
Rh belongs to $4d$ series, so it terms square planar complexes and contains dsp^2 hybridisation.

Question241

The correct combination is:
[Online April 15, 2018(I)]

Options:

A. $[NiCl_4]^{2-}$ – square-planar; $[Ni(CN)_4]^2$ -paramagnetic

B. $[Ni(CN)_4]^{2-}$ -tetrahedral; $[Ni(CO)_4]$ -paramagnetic

C. $[NiCl_4]^{2-}$ -paramagnetic; $[Ni(CO)_4]$ -tetrahedral

D. $[NiCl_4]^{2-}$ – dimagnetic; $[Ni(CO)_4]$ -square-planar

Answer: C

Solution:

Solution:
$[Ni(CN)_4]^{2-}$ is square planar, diamagnetic (0 unpaired electrons) with dsp^2 hybridisation.
$[Ni(CO)_4]$ - is tetrahedral, diamagnetic (0 unpaired electrons) with sp^3 hybridisation.
$[NiCl_4]^2$ is tetrahedral, paramagnetic (2 unpaired electrons) with sp^3 hybridisation.
Hence, the option (c) is the correct answer.

Question242

The correct order of spin-only magnetic moments among the following is:
(Atomic number: Mn = 25, Co = 27, Ni = 28, Zn = 30)
[Online April 15, 2018(II)]

Options:

A. $[ZnCl_4]^{2-} > [NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

B. $[CoCl_4]^2 > [MnCl_4]^2 > [NiCl_4]^{2-} > [ZnCl_4]^2$

C. $[NiCl_4]^2 > [CoCl_4]^{2-} > [MnCl_4]^{2-} > [ZnCl_4]^2$

D. $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Answer: D

Solution:

Solution:
The complex having highernumber of unpaired electrons will have higher value of spin only magnetic moment.
In all these complexes, the central metal ion is in +2 oxidation state.
Zn^{2+} has $3d^{10}$ outer electronic configuration with 0 unpaired electron.
Ni^{2+} has $3d^{t}$ outer electronic configuration with 2 unpaired electrons.
Co^{2+} has $3d^7$ outer electronic configuration with 3 unpaired electrons.
Mn^{2+} has $3d^6$ outer electronic configuration with 5 unpaired electrons
Hence the correct order of spin-only magnetic moments is $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Question243

On treatment of 100mL of 0.1M solution of $CoCl_3.6H_2O$ with excess $AgNO_3$; 1.2×10^{22} ions are precipitated. The complex is:
[2017]

Options:

A. $[Co(H_2O)_4Cl_2]Cl.2H_2O$

B. $[Co(H_2O)_3Cl_3] \cdot 3H_2O$

C. $[Co(H_2O)_6]Cl_3$

D. $[Co(H_2O)_5Cl]Cl_2, H_2O$

Answer: D

Solution:

Solution:

Moles of complex $= \frac{\text{Molarity} \times \text{Volume(mL)}}{1000}$

$= \frac{100 \times 0.1}{1000} = 0.01 \text{mole}$

Moles of ions precipitated with excess of

$AgNO_3 = \frac{1.2 \times 10^{22}}{6.02 \times 10^{23}} = 0.02 \text{moles}$

$\therefore$ Number of Cl^- present in ionisation sphere

$= \frac{\text{Number of moles of ions precipitated}}{\text{Number of moles of complex}}$

$= \frac{0.02}{0.01} = 2$

It means $2Cl^-$ ions present in ionization sphere.

Thus formula of the complex is

$[Co(H_2O)_5Cl]Cl_2H_2O$

Question244

The pair of compounds having metals in their highest oxidation state is : [Online April 8, 2017]

Options:

A. MnO_2 and CrO_2Cl_2

B. $[NiCl_4]^{2-}$ and $[CoCl_4]^{2-}$

C. $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_2]^{2-}$

D. $[FeCl_4]^-$ and Co_2O_3

Answer: A

Solution:

Solution:

$\overset{2-}{CrO_2}\overset{-1}{Cl_2}$ $Cr = +6$; $\underset{(\text{Highest oxidation state})}{MnO_2^{2-}}$ $Mn = +4$

$\overset{-1}{[NiCl_4]^{3-}}$ $Ni = +2$; $\overset{-1}{[CoCl_4]^{2-}}$ $Co = +2$

$\overset{-1}{[Fe(CN)_6]^{-3}}$ $Fe = +3$; $\overset{-1}{[Cu(CN)_4]^{2-}}$ $Cu = +2$

(4) $\overset{-1}{[FeCl_4]^-}$ $Fe = +3$; $\overset{-2}{Co_2O_3}$ $Co = +3$

Question245

$[Co_2(CO)_8]$ displays: [Online April 9,2017]

Options:

A. oneCo-Cobond, six terminal CO and two bridging CO

B. oneCo-Cobond, four terminal CO and four bridging CO

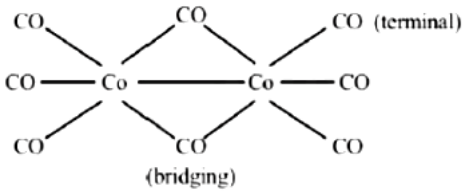
C. no Co-Co bond, six terminal CO and two bridging CO

D. noCo-Cobond, four terminal CO and four bridging CO

Answer: A

Solution:

Solution:



Question246

Which one of the following complexes shows optical isomerism? [2016]

Options:

A. trans$[Co(en)_2Cl_2]Cl$

B. $[Co(NH_3)_4Cl_2]Cl$

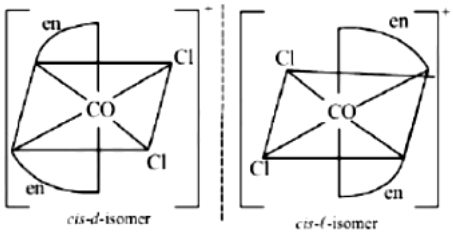
C. $[Co(NH_3)_3Cl_3]$

D. cis$[Co(en)_2Cl_2]Cl$ (en = ethylenediamine)

Answer: D

Solution:

Solution:
Optical isomerism occurs when a molecule is nonsuper imposable with its mirror image hence the complex cis-$[Co(en)_2Cl_2]Cl$ is optically active.



Question247

Which one of the following complexes will consume more equivalents of aqueous solution of $AgNO_3$? [Online April 9,2016]

Options:

A. $Na_2[CrCl_5(H_2O)]$

B. $Na_3[CrCl_6]$

C. $[Cr(H_2O)_5Cl]Cl_2$

D. $[Cr(H_2O)_6]Cl_3$

Answer: D

Solution:

Solution:
More equivalents of $AgNO_3$ aqueous solution will be consumed if complex will furnish more Cl^- ions in solution. Hence complex $[Cr(H_2O)_6]Cl_3$ will consume more equivalents of aqueous solution of $AgNO_3$.

Question248

Which of the following is an example of homoleptic complex? [Online April 10, 2016]

Options:

A. $[Co(NH_3)_6]Cl_3$

B. $[Pt(NH_3)_2Cl_2]$

C. $[Co(NH_3)_4Cl_2]$

D. $[Co(NH_3)_5Cl]Cl_2$

Answer: A

Solution:

Solution:
Complexes having only one type of ligands are examples of homoleptic complex.

Question249

The pair having the same magnetic moment is: [At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27] [2016]

Options:

A. $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$

B. $[CoCl_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$

C. $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2}$

D. $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

Answer: D

Solution:

Solution:

	Metal ion electron	Unpaired moment	Magnetic
(i)	Cr^{2+}	4	$\sqrt{24}$ B.M.
(ii)	Fe^{2}	4	$\sqrt{24}$ B.M.
(iii)	Co^{2+}	3	$\sqrt{15}$ B.M.
(iv)	Mn^{2+}	5	$\sqrt{35}$ B.M.

Since (i) and (ii), cach has 4 unpaired electron, so they will exhibit same magnetic moment. Thus option (d) is correct.

Question250

Identify the correct trend given below: (Atomic No. = T i : 22, Cr : 24 and M o : 42) [Online April 9, 2016]

Options:

A. Δ_o of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ Δ_o of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$

B. Δ_o of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$

C. Δ_0 of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$

D. Δ_0 of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$

Answer: D

Solution:

Solution:
The splitting is affected by the oxidation state of the central metal ion. A higher oxidation state leads to larger splitting, hence
Δ_o of $[Ti(H_2O)_6]^{3+} < \Delta_0$ of $[Ti(H_2O)_6]^{2+}$
Further Δ_o also depends of Z_{eff} and Z_{eff}. of 4d series is more than 3d series. Hence
Δ_0 of $[Cr(H_2O)_6]^{2+} < \Delta_o$ of $[Mo(H_2O)_6]^{2+}$

Question251

The number of geometric isomers that can exist for square planar complex $[Pt(Cl)(py)(NH_3)(NH_2OH)]^{+}$ is (py = pyridine): [2015]

Options:

A. 4

B. 6

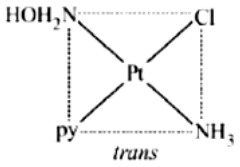
C. 2

D. 3

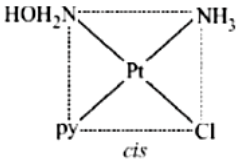
Answer: D

Solution:

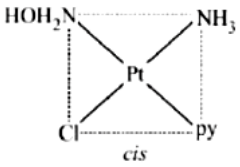
Solution:
Square planar complexes of type M[ABCD] form three isomers. Their position may be obtained by fixing the position of one ligand and placing at the trans position any one of the remaining three ligands one by one.



trans



cis



cis

Question252

The correct statement on the isomerism associated with the following complex ions,
(A) $[Ni(H_2O)_5NH_3]^{2+}$,
(B) $[Ni(H_2O)_4(NH_3)_2]^{2+}$
(C) $[Ni(H_2O)_3(NH_3)_3]^{2+}$ is :
[Online April 10,2015]

Options:

A. (A) and (B) show only geometrical isomerism

B. (A) and(B) show geometrical and optical isomerism

C. (B) and (C) show geometrical and optical isomerism

D. (B) and (C) show only geometrical isomerism

Answer: D

Solution:

Solution:
It does not show optical and geometrical isomerism (B) and (C) shows only geometrical isomerism.

Question253

Which of the following compounds is not colored yellow?
[2015]

Options:

A. $(NH_4)_3[As(Mo_3O_{10})_4]$

B. $BaCrO_4$ [2015]

C. $Zn_2[Fe(CN)_6]$

D. $K_3[Co(NO_2)_6]$

Answer: C

Solution:

Solution:

Question254

The color of $KMnO_4$ is due to :
[2015]

Options:

A. $L \rightarrow M$ charge transfer transition

B. $\sigma - \sigma^*$ transition

C. $M \rightarrow L$ charge transfer transition

D. $d - d$ transition

Answer: A

Solution:

Solution:
$L \rightarrow M$ charge transfer spectra. $KMnO_4$ is colored because it absorbs light in the visible range of electromagnetic radiation. The permanganate ion is the source of color, as a ligand to metal $(L \rightarrow M)$ charge transfer takes place between oxygen's p orbitals and the empty d orbitals on the metal. This charge transfer takes place when a photon of light is absorbed which leads to the purple color of the compound.

Question255

Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals?
[Online April 11, 2015]

Options:

A. $[FeF_6]^3$

B. $[Mn(CN)_6]^4$

C. $[CoF_6]^3$

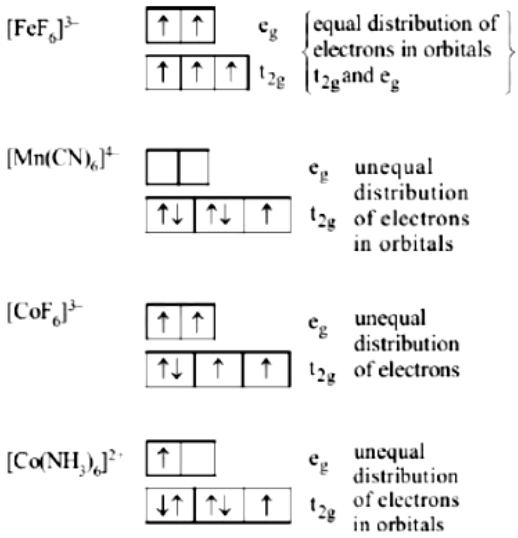
D. $[Co(NH_3)_6]^{2+}$

Answer: A

Solution:

Solution:

Symmetrically filled t_2 and e_g are those, which contain equal distribution of electrons.



Question256

Which molecule/ion among the following cannot act as a ligand in complex compounds? [Online April 10,2015]

Options:

A. CH_4

B. CO

C. CN

D. Br

Answer: A

Solution:

Solution:
The donor atoms, molecules or anions which donate a pair of electrons to the metal atom or ion and form a coordinate bond with it are called ligands.
In methane there is no electrons for donation to central metal atom/ ion, it is stable with complete octet configuration.

Question257

Which of the following name formula combinations is not correct?

	Formula	
(1)	$K_2[Pt(CN)_4]$	Potassium tetracyanoplatinate (II)
(2)	$[Mn(CN)_5]^{2-}$	Pentacyanomagnate (II) ion
(3)	$K[Cr(NH_3)_2Cl_4]$	Potassium diammine tetrachlorochromate (III)
(4)	$[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammine aquaioda cobalt (III) sulphate.

[Online April 11,2014]

Answer: 2

Solution:

Solution:
Pentacyanomangnate (III) is the correct IUPAC name of $[Mn(CN)_5]^{2}$.

Question258

An octahedral complex with molecular composition $M.5\,NH_3 \cdot Cl\,.SO_4$ has two isomers, A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is:
[Online April 19,2014]

Options:

A. Linkage isomerism

B. Ionisation isomerism

C. Coordinate isomerism

D. Geometrical isomerism

Answer: B

Solution:

Solution:
The two possible isomers for the given octahedral complex are $[M(NH_3)_5SO_4]Cl$ and $[M(NH_3)_5Cl]SO_4$. They respectively give chloride ion (indicated by precipitation with $BaCl_2$) and SO_4 ion (indicated by precipitation with $AgNO_3$). Hence the type of isomerism exhibited by the complex is ionisation isomerism.

Question259

The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:
[2014]

Options:

A. $L_4 < L_3 < L_2 < L_1$

B. $L_1 < L_3 < L_2 < L_4$

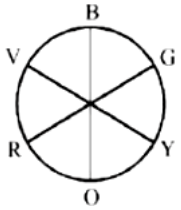
C. $L_3 < L_2 < L_4 < L_1$

D. $L_1 < L_2 < L_4 < L_3$

Answer: B

Solution:

Solution:



For a given metal ion, weak field ligands create a complex with smaller Δ, which will absorbs light of longer λ and thus lower frequency. Conservely, stronger field ligands create a larger Δ, absorb light of shorter λ and thus higher vi.e. higher energy.
So order of ligand strength is
$L_1 < L_3 < L_2 < L_4$

Question260

An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is:
[Online April 9, 2014]

Options:

A. sp^3d^2

B. dsp^2

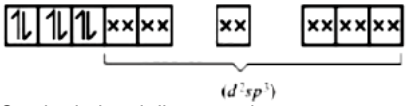
C. d^2sp^3

D. sp^3d

Answer: C

Solution:

Solution:
$[Co(NH_3)_6]^{3+}$



Octahedral and diamagnetic.

Question261

The correct statement about the magnetic properties of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ is ($Z = 26$):
[Online April 9, 2014]

Options:

A. both are paramagnetic.

B. both are diamagnetic.

C. $[Fe(CN)_6]^{3-}$ is diamagnetic, $[FeF_6]^{3-}$ is paramagnetic.

D. $[Fe(CN)_6]^{3-}$ is paramagnetic, $[FeF_6]^{3-}$ is diamagnetic.

Answer: A

Solution:

Solution:
Both are paramagnetic, the only difference is that CN is a strong field ligand whereas F^- is a weak field ligand.

Question262

Consider the coordination compound, $[Co(NH_3)_6]Cl_3$. In the formation of this complex, the species which acts as the Lewis acid is:
[Online April 11, 2014]

Options:

A. $[Co(NH_3)_6]^{3+}$

B. Cl

C. Co^{3+}

D. NH_3

Answer: C

Solution:

Solution:
$\underset{\text{Lewis acid}}{Co^{3+}} + 6\underset{\text{Lewis base}}{NH_3} \rightarrow \underset{\text{abduct}}{[Co(NH_3)_6]^{3+}}$

Question263

Among the following species the one which causes the highest CFSE, Δ_o as a ligand is:
[Online April 12, 2014]

Options:

A. CN^-

B. NH_3

C. F^-

D. CO

Answer: D

Solution:

Solution:
Following is the order of increasing value of Δ_0 for the ligands $I^- < Br^- < Cl^- < F^- < H_2O < Py = NH_3 < en < NO_2^- < < CN^{-<} < CO$

Question264

Which one of the following complexes will most likely absorb visible light?
(At nos. Sc = 21, Ti = 22, V = 23, Zn = 30)
[Online April 12, 2014]

Options:

A. $[Sc(H_2O)_6]^{3+}$

B. $[Ti(NH_3)_6]^{4+}$

C. $[V(NH_3)_6]^{3+}$

D. $[Zn(NH_3)_6]^{2+}$

Answer: C

Solution:

Solution:
The absorption of visible light is responsible for coloured nature of the transition metal cation due to the promotion of one or more unpaired $-d$ -electron from a lower to higher level within same d -subshell. Hence higher the number of unpaired electron higher will be the absorpion in visible light.
The electronic configuration of the given elements is $Sc^{3+}(18) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^0 4s^0$ – no unpaired e^-
$Ti^{4+}(18) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^0 4s^0$ – no unpaired e^-
$V^{3+}(20) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^0$ – Two unpaired e^-
$Zn^{2+}(28) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^0$ – no unpaired e.
hence $[V(NH_3)_6]^{3+}$ will most likely absorb visible light.

Question265

Nickel (Z = 28) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^{2}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively:
[Online April 19, 2014]

Options:

A. sp^3, two

B. dsp^2, zero

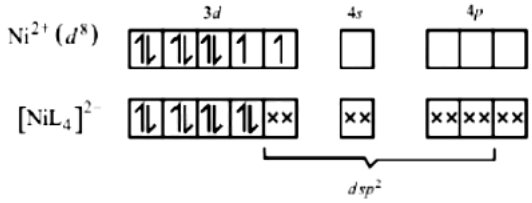
C. dsp^2, one

D. sp^3, zero

Answer: B

Solution:

Solution:
$[NiL_4]^{2-} \Rightarrow$ Diamagnetic in nature (given)



So, no. of unpaired electron $= 0$
hybridisation $-dsp^2$

Question266

Which of the following complex species is not expected to exhibit optical isomerism?
[2013]

Options:

A. $[Co(en)_3]^{3+}$

B. $[Co(en)_2Cl_2]^{+}$

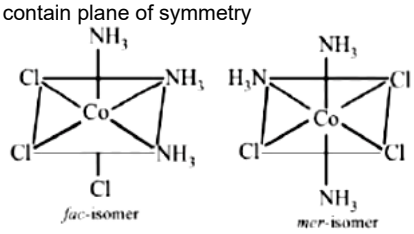
C. $[Co(NH_3)_3Cl_3]$

D. $[Co(en)(NH_3)_2Cl_2]^+$

Answer: C

Solution:

Solution:
Octahedral coordination entities of the type Ma_3b_3 exhibit geometrical isomerism. The compound exists both as facial and meridional isomers, both contain plane of symmetry



Question267

Type of isomerism which exists between $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ is: [Online April 9,2013]

Options:

A. Linkage isomerism

B. Coordination isomerism

C. Ionisation isomerism

D. Solvate isomerism

Answer: A

Solution:

Solution:
The compound shows linkage isomerism because the ligand in the compound is an ambidenate ligand that can bond at more than one atomic site. i.e., $:NCS^-$ and $:SCN^-$

Question268

Which of the following is diamagnetic? [Online April 22, 2013]

Options:

A. $[Fe(CN)_6]^{3-}$

B. $[Co(ox)]^{3-}$

C. $[FeF_6]^{3-}$

D. $[Co(F_6)]^{3-}$

Answer: B

Solution:

Solution:
Among four ligands CN^- and OX^{2-} are strong field ligands. So pairing takes places in $[(FeCN)_6]^3$ and $[Co(OX)_3]^{3-}$ complexes. Fe^{3+} having one unpaired electron in its complex white Co^{3+} does not have unpaired electron in its complex. So option b is correct.

Question269

The magnetic moment of the complex anion $[Cr(NO)(NH_3)(CN)_4]^{2-}$ is:
[Online April 23, 2013]

Options:

A. 5.91BM

B. 3.87BM

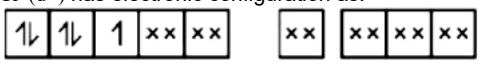
C. 1.73BM

D. 2.82BM

Answer: C

Solution:

Solution:
In $[Cr(NO)(NH_3)(CN)_4]^2$,
$Cr^+(d^5)$ has electronic configuration as:



So, 1 unpaired electron is present.
Thus, $\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.73BM$

Question270

In which of the following octahedral complex species the magnitude of Δ_0 will be maximum?
[Online April 25, 2013]

Options:

A. $[Co(H_2O)_6]^{2+}$

B. $[Co(CN)_6]^{3-}$

C. $[Co(C_2O_4)_3]^{3-}$

D. $[Co(NH_3)_6]^{3+}$

Answer: B

Solution:

Solution:
Crystal field splitting depends upon the nature of ligand. The nature of ligand decreases as shown below
$C_2O_4^{2-} < H_2O < NH_3 < CN^-$
Hence the crystal field splitting will be maximum for $[Co(CN)_6]^{3-}$

Question271

The structure of which of the following chloro species can be explained on the basis of dsp^2 hybridisation?
[Online April 25, 2013]

Options:

A. $PdCl_4^{2-}$

B. $FeCl_4^{2-}$

C. $CoCl_4^{2-}$

D. $NiCl_4^{2-}$

Answer: A

Solution:

Solution:
$[PdCl_4]^{2-}$ is dsp^2 hybridised and square planar in shape.

Question272

Which among the following will be named as dibromido bis (ethylenediamine) chromium (III) bromide?
[2012]

Options:

A. $[Cr(en)_3]Br_3$

B. $[Cr(en)_2Br_2]Br$

C. $[Cr(en)Br_4]$

D. $[Cr(en)Br_2]Br$

Answer: B

Solution:

Solution:
$[Cr(en)_2Br_2]Br$
dibromido bis(ethylenediamine) chromium (III) Bromide.

Question273

Which of the following complex ions will exhibit optical isomerism? (en = 1, 2 -diamine ethane).
[Online May 12,2012]

Options:

A. $[Cr(NH_3)_2Cl_2]^+$

B. $[Co(en)_2Cl_2]^+$

C. $[Co(NH_3)_4Cl_2]^+$

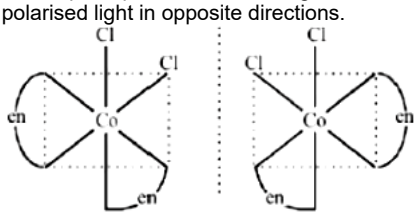
D. $[Zn(en)_2]^{2+}$

Answer: B

Solution:

Solution:

Non superimposable mirror images are called optical isomers and may be described as "chiral'. They are also called enantiomers and rotate plane polarised light in opposite directions.



Question274

The complex ion $[Pt(NO_2)(Py)(NH_3)(NH_2OH)]^+$ will give [Online May 26,2012]

Options:

A. 2 isomers (Geometrical)

B. 3 isomers (Geometrical)

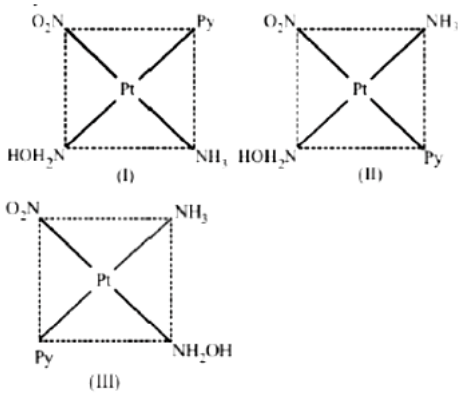
C. 6 isomers (Geometrical)

D. 4 isomers (Geometrical)

Answer: B

Solution:

Solution:
Complexes of the type M_{ABCD} may exist in three isomeric form. Similarly $[Pt(NO_2)(Py)(NH_3)(NH_2OH)^+]$ may exist in three isomeric form.



Question275

The d -electron configurations of Cr^{2+}, Mn^{2+}, Fe^{2+} and Co^{2+} are d^4, d^5, d^6 and d^7 respectively. Which one of the following will exhibit the lowest paramagnetic behaviour? (Atomic no. Cr = 24, Mn = 25, Fe = 26, Co = 27) [Online May 7,2012]

Options:

A. $[Co(H_2O)_6]^{2+}$

B. $[Cr(H_2O)_6]^{2+}$

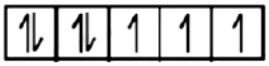
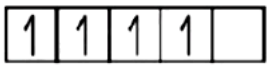
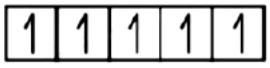
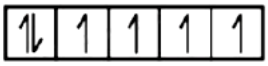
C. $[Mn(H_2O)_6]^{2+}$

D. $[Fe(H_2O)_6]^{2+}$

Answer: A

Solution:

Solution:

	Electronic Configuration	No. of unpairedelectrons
Co^{2+}	d^7 	3
Cr^{2+}	d^4 	4
Mn^{2+}	d^5 	5
Fe^{2+}	d^6 	4

∴ Since Co^{3+} has lowest no. of unpaired electrons hence lowest paramagnetic behaviour is shown by $[Co(H_2O)_6]^{2+}$

Question276

The correct order of ligands in the spectrochemical series is
[Online May 19,2012]

Options:

A. $Cl^- > en > CN^- > NCS^-$

B. $CN^- > en > NCS^- > Cl^-$

C. $en > CN^- > Cl^- > NCS$

D. $NCS^- > CN^- > Cl^- > en$

Answer: B

Solution:

Solution:
Correct order is : $CN^- > en > NCS^- > Cl$

Question277

Which of the following paramagnetic ions would exhibit a magnetic moment (spin only) of the order of 5 BM?
(At. Nos. Mn = 25, Cr = 24, V = 23, Ti = 22)
[Online May 19,2012]

Options:

A. Mn^{2+}

B. Ti^{2+}

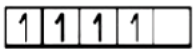
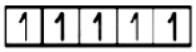
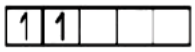
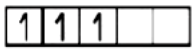
C. V^{2+}

D. Cr^{2+}

Answer: D

Solution:

Solution:

Ion	Electronic Configuration	Magnetic moment ($\sqrt{n(n+2)}$)
$Cr^{2+}d^4$		$\sqrt{4(4+2)} = 4.9$
$Mn^{2+}d^5$		$\sqrt{5(5+2)} = 5.8$
$Ti^{2+}d^2$		$\sqrt{2(2+2)} = 2.4$
$V^{2+}d^3$		$\sqrt{3(3+2)} = 3.8$

Hence Cr^{2+} has magnetic moment of the order of $5B.M.$

Question278

Square-planar geometry is shown by [Online May 19,2012]

Options:

A. $[PtCl_2(NH_3)_2]$

B. $[NiCl_4]^{2-}$

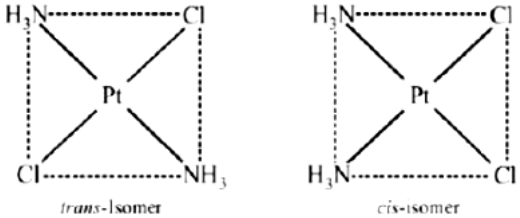
C. MnO_4^-

D. CrO_4^{2-}

Answer: A

Solution:

Solution:
$[PtCl_2(NH_3)_2]$ has square planar geometry.



Question279

Which one of the following complex ions has geometrical isomers? [2011]

Options:

A. $[Ni(NH_3)_5Br]$

B. $[Co(NH_3)_2(en)_2]^{3+}$

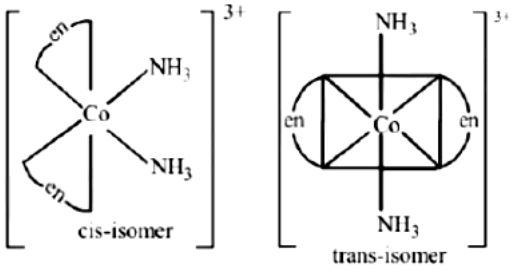
C. $[Cr(NH_3)_4(en)_2]^{3+}$

D. $[Co(en)_3]^{3+}$ (en-cthylenediamine)

Answer: B

Solution:

Solution:



Question280

A solution containing 2.675g of $CoCl_3.6NH_3$ (molar mass $= 267.5 g mol^{-1}$) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.78g of AgCl (molar mass $= 143.5 g\ mol^{-1}$). The formula of the complex is
(At. mass of Ag = 108u)
[2010]

Options:

A. $[Co(NH_3)_6]Cl_3$

B. $[CoCl_2(NH_3)_4]Cl$

C. $[CoCl_3(NH_3)_3]$

D. $[CoCl(NH_3)_5]Cl_2$

Answer: A

Solution:

Solution:

$$\underset{2.675g}{CoCl_3 \cdot 6NH_3} \longrightarrow xCl^-$$

$$xCl^- + AgNO_3 \longrightarrow \underset{4.78g}{xAgCl\downarrow}$$

Number of moles of the complex

$$= \frac{2.675}{267.5} = 0.01 mol$$

Number of moles of AgCl obtained

$$= \frac{4.78}{143.5} = 0.03 mol$$

$\therefore$ No. of moles of AgCl obtained

$= 3 \times$ No. of moles of complex

$$\therefore n = \frac{0.03}{0.01} = 3$$

Question281

Which one of the following has an optical isomer? [2010]

Options:

A. $[Zn(en)(NH_3)_2]^{2+}$

B. $[Co(en)_3]^{3+}$

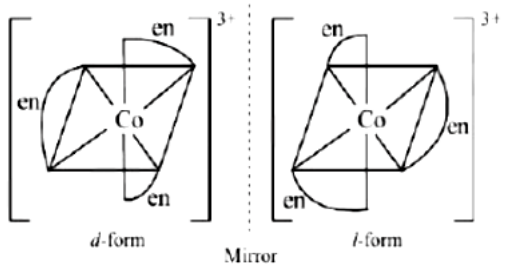
C. $[Co(H_2O)_4(en)]^{3+}$

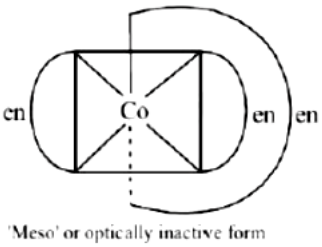
D. $[Zn(en)_2]^{2+}$ (en = ethylenediamine)

Answer: B

Solution:

Solution:
For a substance to be optical isomer following conditions should be fulfiled
(a) A coordination compound which can rotate the plan of polarised light is said to be optically active.
(b) When the coordination compounds have sam formula but differ in their abilities to rotate directions o the plane of polarised light are said to exhibit optica isomerism and the molecules are optical isomers. Th optical isomers are pair of molecules which are non superimposable mirror images of each other.
(c) This is due to the absence of elements of symmetry in the complex.
(d) Optica isomerism is expected in tetrahedral complex of the type Mabod.
Based an this only option (b) shows optical isomerism $[Co(en)_3]^{3+}$





Complexes of Zn^{2+} cannot show optical isomerism as they are tetrahedral complexes with plane of symmetry.
$[CO(H_2O)_4(en)]^{3+}$ has two planes of symmetry hence it is also optically inactive.

Question282

Which of the following shows optical isomerism [2009]

Options:

A. $[Co(en)(NH_3)_2]^{2+}$

B. $[Co(H_2O)_4(en)]^{3+}$

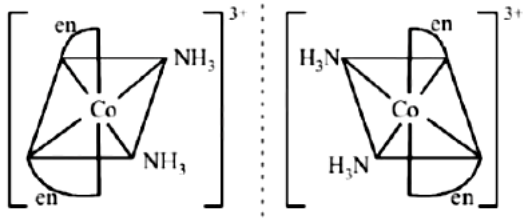
C. $[Co(en)_2(NH_3)_2]^{3+}$

D. $[Co(NH_3)_3Cl]^{+}$

Answer: C

Solution:

Solution:



Enantiomers of cis- $[Co(en)_2(NH_3)_2]^{3+}$

Question283

Which of the following pairs represent linkage isomers? [2009]

Options:

A. $[Pd(PPh_3)_2(NCS)_2]$ and $[Pd(PPh_3)_2(SCN)_2]$

B. $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$

C. $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$

D. $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$

Answer: A

Solution:

Solution:
The SCN - ion can coordinate through S or N atom giving rise to linkage isomerism
M ← SCN thiocyanato
M ← NCS isothiocyanato.

Question284

The coordination number and the oxidation state of the element ' E ' in the complex $[E(en)_2(C_2O_4)]NO_2$(where (en) is ethylene diamine) are, respectively, [2008]

Options:

A. 6 and 2

B. 4 and 2

C. 4 and 3

D. 6 and 3

Answer: D

Solution:

Solution:
In the given complex we have two bidentate ligands (i.e en and C_2O_4), so coordination number of E is 6
$(2 \times 2 + 1 \times 2 = 6)$
Let the oxidation state of E in complex be x, then
$[x + (-2) = 1]$ or $x - 2 = 1$ or $x = +3$, so its oxidation state is +3
Thus option (d) is correct.

Question285

In which of the following complexes of the Co(at. no. 27), will the magnitude of Δ_0 be the highest? [2008]

Options:

A. $[Co(CN)_6]^3$

B. $[Co(C_2O_4)_3]^3$

C. $[Co(H_2O)_6]^{3+}$

D. $[Co(NH_3)_6]^{3+}$

Answer: A

Solution:

Solution:
In octahedral complex, the magnitude of Δ_0 will be highest in a complex having strongest ligand. Of the given ligands, CN^- is strongest, so Δ_0 will be highest for $[Co(CN)_6]^{3-}$. Thus option (a) is correct.

Question286

Which of the following has a square planar geometry? (At. nos: Fe = 26, Co = 27, Ni = 28, Pt = 78) [2007]

Options:

A. $[PtCl_4]^{2-}$

B. $[CoCl_4]^{2-}$

C. $[FeCl_4]^{2-}$

D. $[NiCl_4]^{2-}$

Answer: A

Solution:

Solution:
Complexes with dsp^2 hybridisation are square planar. All the complexes of Pt^{2+} are square planar including those with weak field ligand such as halide ions. Thus option (a) is correct.

Question287

The IUPAC name for the complex $[Co(NO_2)(NH_3)_5]Cl_2$ is : [2006]

Options:

A. pentaammine nitrito-N-cobalt(II) chloride [2006]

B. pentaammine nitrito-N-cobalt(III) chloride

C. nitrito-N-pentaamminecobalt(III) chloride

D. nitrito-N-pentaamminecobalt(II) chloride

Answer: B

Solution:

Solution:
$[Co(NO)_2(NH_3)_5]Cl_2$
Pentaamminenitrito-N-cobalt (III) chloride

Question288

Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively:
[2006]

Options:

A. one, square planar

B. two, square planar

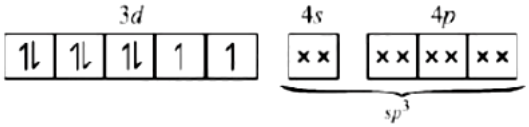
C. one, tetrahedral

D. two, tetrahedral

Answer: D

Solution:

Solution:
$[NiX_4]^{2-}$, the electronic configuration of Ni^{2+} is



It contains two unpaired electrons and the hybridisation is sp^3 (tetrahedral).

Question289

How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion?
[2006]

Options:

A. One

B. Two

C. Six

D. Three

Answer: A

Solution:

Solution:
EDTA is hexadentate, four donor O atoms and 2 donor Natoms, and for the formation of octahedral complex one molecule is required.

Question290

In $Fe(CO)_5$, the $Fe-C$ bond possesses
[2006]

Options:

A. ionic character

B. σ -character only

C. π -character

D. both σ and π characters

Answer: D

Solution:

Solution:
Due to some backbonding by sideways overlapping between d -orbitals of metal and p -orbital of carbon, the $Fe-C$ bond in $Fe(CO)_5$ has both σ and π character.

Question291

The oxidation state Cr in $[Cr(NH_3)_4Cl_2]^+$ is
[2005]

Options:

A. 0

B. +1

C. +2

D. +3

Answer: D

Solution:

Solution:
Oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$.
Let it be x, $1 \times x + 4 \times 0 + 2(-1) = 1$
Therefore $x = 3$.

Question292

The IUPAC name of the coordination compound $K_3[Fe(CN)_6]$ is
[2005]

Options:

A. Tripotassium hexacyanoiron (II)

B. Potassium hexacyanoiron (II)

C. Potassium hexacyanoferrate(III)

D. Potassium hexacyanoferrate (II)

Answer: C

Solution:

Solution:
$K_3[Fe(CN)_6]$ is potassium hexacyanoferrate (III).

Question293

Which of the following compounds shows optical isomerism? [2005]

Options:

A. $[Co(CN)_6]^{3-}$

B. $[Cr(C_2O_4)_3]^{3-}$

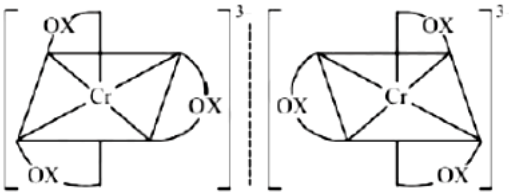
C. $[ZnCl_4]^{2-}$

D. $[Cu(NH_3)_4]^{2+}$

Answer: B

Solution:

Solution:



Non-superimposable mirror images, hence optical isomers.

Question294

Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour? (At. Nos: Cr = 24, Mn = 25, Fe = 26, Co = 27) [2005]

Options:

A. $[Co(CN)_6]^{3-}$

B. $[Fe(CN)_6]^{3-}$

C. $[Mn(CN)_6]^{3-}$

D. $[Cr(CN)_6]^{3-}$

Answer: B

Solution:

Solution:

		No. of unpaired electrons
(a)	Co^{3+}	4
(b)	Fe^{3+}	1
(c)	Mn^{3+}	4
(d)	Cr^{3+}	3

The effective magnetic moment is given by the number of unpaired electrons in a substance, the lesser the number of unpaired electrons lower is its magnetic moment in Bohr - Magneton and lower shall be its paramagnetic

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Question295

The coordination number of a central metal atom in a complex is determined by [2004]

Options:

A. the number of ligands around a metal ion bonded by sigma and pi-bonds both

B. the number of ligands around a metal ion bonded by pi-bonds

C. the number of ligands around a metal ion bonded by sigma bonds

D. the number of only anionic ligands bonded to the metal ion.

Answer: C

Solution:

Solution:
The coordination number of central metal atom in a complex is equal to number of monovalent ligands, twice the number of bidentate ligands and so on, around the metal ion bonded by coordinate bonds.
Hence coordination number = No. of σ bonds formed by metals with ligands.

Question296

Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect? [2004]

Options:

A. Cyanocobalamin is B_{12} and contains cobalt

B. Haemoglobin is the red pigment of blood and contains iron

C. Chlorophylls are green pigments in plants and contain calcium

D. Carboxypeptidase - A is an enzyme and contains zinc.

Answer: C

Solution:

Solution:
The chlorophyll molecule plays an important role in photosynthesis. It has porphyrin rings and the metal Mg, not Ca.

Question297

Which one of the following has largest number of isomers? [2004]

Options:

A. $[Ir(PR_3)_2H(CO)]^{2+}$

B. $[Co(NH_3)_5Cl]^{2+}$[2004]

C. $[Ru(NH_3)_4Cl_2]^{+}$ (R = alkyl group, en = ethylenediamine)

D. $[Co(en)_2Cl_2]^{+}$

Answer: D

Solution:

Solution:
Isomers
$[Ru(NH_3)_4Cl_2]^{+}$, $[Co(NH_3)_5Cl]^{2+}$,
cis and trans none
$[Ir(PR_3)_2H(CO)]^{2+}$, $[Co(en)_2Cl_2]^{+}$
cis and trans cis and trans and optical isomers

Question298

Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN ion towards metal species is
[2004]

Options:

A. c. a

B. b, c

C. a, b

D. a, b, c

Answer: A

Solution:

Solution:
CN ion acts good complexing as well as reducing agent.

Question299

Which one of the following complexes is an outer orbital complex?
(Atomic nos. : Mn = 25; Fe = 26; Co = 27, Ni = 28)
[2004]

Options:

A. $[Co(NH_3)_6]^{3+}$

B. $[Mn(CN)_6]^{4}$

C. $[Fe(CN)_6]^{4}$

D. $[Ni(NH_3)_6]^{2+}$

Answer: D

Solution:

Solution:

$\underset{d^2sp^3}{[Fe(CN)_6]^{4-}}, \underset{d^2sp^3}{[Mn(CN)_6]^{4-}}, \underset{d^2sp^3}{[Co(NH_3)_6;]^{3+}}, \underset{sp^3d^2}{[Ni(NH_3)_6]^{2+}}$

Hence $[Ni(NH_3)_6]^{2+}$ is outer orbital complex.

Question300

One mole of the complex compound $Co(NH_3)_5Cl_y$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of AgCl (s). The structure of the complex is
[2003]

Options:

A. $[Co(NH_3)_3Cl_3] \cdot 2NH_3$

B. $[Co(NH_3)_4Cl_2]Cl \ . NH_3$

C. $[Co(NH_3)_5Cl]Cl_2$

D. $[Co(NH_3)_4Cl]Cl_2, NH_3$

Answer: C

Solution:

Solution:

$Co(NH_3)_5Cl_3 \rightleftharpoons [Co(NH_3)_5Cl]^{+2} + 2Cl^-$

$\therefore$ Structure is $[Co(NH_3)_5Cl]Cl_2$

Now $[Co(NH_3)_5Cl]Cl_2 + 2AgNO_3 \longrightarrow [Co(NH_3)_5Cl](NO_3)_2 + 2AgCl$

Question301

In the coordination compound, $K_4[Ni(CN)_4]$, the oxidation state of nickel is
[2003]

Options:

A. 0

B. +1

C. +2

D. -1

Answer: A

Solution:

Solution:

Let the O . N , of N in $K_4[Ni(CN)_4]$ be $= x$ then

$4(+1) + x + (-1) \times 4 = 0$

$\Rightarrow \ 4 + x - 4 \ = 0$

$x = 0$

Question302

The type of isomerism present in nitropentammine chromium (III) chloride is
[2002]

Options:

A. optical

B. linkage

C. ionization

D. polymerisation.

Answer: B

Solution:

Solution:
The chemical formula of nitropentammine chromium (III) chloride is $[Cr(NH_3)_5NO_2]Cl_2$
It can exist in following two structures
$[Cr(NH_3)_5NO_2]Cl_2$ and
Nitropentamminechromium (III) chloride
$[Cr(NH_3)_5ONO]Cl_2$
Nitritopentamminechromium (III) chloride
Therefore the type of isomerism found in this compound is linkage isomerism as NO_2 group is linked through N as NO_2, or through O as $-ONO$.

Question303

A square planar complex is formed by hybridisation of which atomic orbitals? [2002]

Options:

A. s, p_x, p_y, d_{yz}

B. $s, p_x, p_y, d_{x^2-y^2}$

C. s, p_x, p_y, d_{z^2}

D. s, p_y, p_z, d_{xy}

Answer: B

Solution:

Solution:
A square planar complex is formed by hybridisation of s, p_x, p_y and $d_{x^2-y^2}$ atomic orbitals

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Question304

$CH_3-Mg-Br$ is an organo metallic compound due to [2002]

Options:

A. $Mg-Br$ bond

B. $C-Mg$ bond

C. $C-Br$ bond

D. $C-H$ bond.

Answer: B

Solution:

Solution:
Compounds that contain carbon-metal bond are known as organometallic compounds.
In $CH_3 - Mg - Br$ (Grignard's reagent), a bond is present between carbon and Mg (metal), hence it is an organometallic compound.
